

발 간 등 록 번 호

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Policy Research 2023-22-02



2023 K-Innovation Partnership Program with Azerbaijan

Policy Consultation on the Operational Programs of the HTP in Azerbaijan: Implementation and Capacity Building

(2nd Year) Improvement of the Startup and Incubation System in Azerbaijan and Development of the Startup and Incubation Program for the High Technologies Park (HTP)

2023 K-Innovation Partnership Program with Azerbaijan

Title	Policy Consultation on the Operational Programs of the HTP in Azerbaijan: Implementation and Capacity Building (2nd Year) Improvement of the Startup and Incubation System in Azerbaijan and Development of the Startup and Incubation Program for the High Technologies Park (HTP)
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Executive Summary

Azerbaijan, located at a strategic point along the coast of the Caspian Sea, is the centre of science and technology in the Eurasian region and has great importance for cooperation in the international energy market as a production and transit hub for crude oil and natural gas. Azerbaijan is hugely reliant on the oil and gas industry, which accounts for more than 90% of its total exports, and has an economic structure trade that is contingent upon three neighbouring countries, namely, Russia, Iran, and Turkey. In other words, it is a country that is economically and socially affected by unstable fluctuations in international oil prices and by changes in the circumstances of neighbouring countries.

Recently, Azerbaijan's government resolved to move away from excessive dependence on the oil industry and make the transition to a knowledge-based economy, in a bid to avoid a financial crisis caused by unstable oil prices and to create a democratic political environment. It is in this context that the government established 'Azerbaijan 2020: Look into the Future' and 'Azerbaijan 2030: National Priorities for Socio-Economic Development' with the aim of achieving sustainable economic development based on innovation in the field of science and technology. The creation of the Azerbaijan National Cooperation Strategy (2016) shows that the government is making efforts to strengthen the country's science and technology innovation capacity under the national development plan and to implement industrial diversification policies aimed at building national competitiveness. In addition, it emphasizes the need to increase investment in science and technology innovation and to improve the operation of its system by increasing the efficiency of knowledge creation, technology transfer, and the application and commercialization of technology through the 'Azerbaijan 2030 Strategic Roadmap'.

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CHAPTER 1

Project Overview



Chapter 1. Project Overview

1. Project Background

According to the definition provided by the World Intellectual Property Organization (WIPO), technology transfer plays a pivotal role in facilitating the entire life cycle of a technology, encompassing its inception, diffusion into the market, and eventual commercialization¹. Within this context, the commercialization of technologies entails a process in which an idea transitions from a knowledge base to the market, thereby creating financial value. Technology commercialization (TC) also enhances economic efficiency by offering consumers a diverse and improved range of goods and services, resulting in the generation of value-added revenue streams. Consequently, investment in technology and research and development (R&D)-related activities, along with the subsequent commercialization of their outcomes, necessitates the involvement of a wide array of institutions, programs, and mechanisms to achieve the optimal results.

In the context of Azerbaijan, technology commercialization (TC) has recently emerged as a prominent topic of discussion within its innovation ecosystem. This report is an integral part of a collaborative project between the Azerbaijan High Technologies Park (HTP) and the Science and Technology Policy Institute (STEPI) spanning 2022 to 2024. The report's primary focus is to delve into the nuances of technology transfer and commercialization that are specific to Azerbaijan. To accomplish this objective, our research will explicitly centre on the developments occurring in Azerbaijan post-July 2022, coinciding with significant institutional reforms in the aforementioned domains. We anticipate that this report will contribute significantly to enhancing the effectiveness of the STEPI-HTP partnership's operational programs.

¹ <https://www.wipo.int/technology-transfer/en/index.html>

Consistent with the WIPO's delineation, effective technology transfer requires universities and research institutions to operate within a well-structured innovation ecosystem². This ecosystem comprises an interconnected network involving governmental bodies, industry stakeholders, research institutions, and various enabling factors, including human capital, technology transfer infrastructure, and the sophistication of businesses and markets. Within such an ecosystem, these entities pool their resources and expertise to collaboratively drive innovation, ultimately serving regional and economic development objectives.

Recognizing these imperatives, Azerbaijan has established a comprehensive framework of institutions and infrastructures to regulate and bolster TC activities. For instance, the "Social-Economic Development Strategy of the Republic of Azerbaijan, 2022-2026" outlines a strategic framework, part of which is dedicated to enhancing the competitiveness of the country's human capital and positioning it as a hub for innovation. Consequently, Azerbaijan aspires to create a "Science-Industry Cluster 4.0" concept, establish the "Absheron Valley" innovation cluster, and greatly expand its research capacity and commercialization initiatives.

The K-Innovation Partnership Program, a collaboration between the Science & Technology Policy Institute (STEPI) of the Republic of Korea and the High Technologies Park (HTP) of the Republic of Azerbaijan, spanning 2022 to 2024, is primarily geared towards capacity building and fostering cooperation in the realms of effective technology transfer and commercialization. Within the ambit of this project, this document has been tasked with delivering a status report concerning TC policies, programs, and projects in Azerbaijan. To fulfill this mandate, the report will serve as a comprehensive reference, providing in-depth insights into governance mechanisms, institutional and legal frameworks, intellectual property rights, and a concise overview of Azerbaijan's national innovation ecosystem. Its overarching objective is to make a substantial contribution to the K-Innovation Partnership Program by offering a comparative analysis and timely updates on the pertinent reforms and policies in Azerbaijan.

² <https://www.wipo.int/technology-transfer/en/index.html>

2. Project Objectives

The primary objective of the 2023 initiative is to offer policy advisory services concerning the enhancement of the startup and incubation ecosystem in Azerbaijan, along with the establishment of a startup and incubation program tailored for the High Technologies Park (HTP). Additionally, the initiative aims to foster mutually advantageous economic collaboration between the two nations, thereby aiding the advancement of Science, Technology, and Innovation (STI) in Azerbaijan in accordance with the principles outlined in the 『FRAMEWORK ACT ON INTERNATIONAL DEVELOPMENT COOPERATION. 』 in the Republic of South Korea.

[Table 1-1] Framework Act on International Development Cooperation

Article 3. Basic Ideas and Objectives	
(1) The basic objectives of international development cooperation are to reduce poverty in developing nations, improve the human rights of women, children and people with disabilities, achieve gender equality, realize sustainable development and humanitarianism, promote mutually beneficial economic relationships with cooperation partners, and pursue peace and prosperity in the international community.	
(2) The objectives of international development cooperation shall lie in achieving the following matters so as to implement the basic ideas presented under the foregoing paragraph (1):	
The Basic Ideas of International Development Cooperation	Development of developing nations and improvement of the systems and conditions for such development.
	Promotion of friendly relationships and mutual exchanges with developing nations.
	Contribution to the resolution of global problems related to international development cooperation.

Source: Framework Act on International Development Cooperation, No.17302(2020), Republic of Korea.

- Project Objectives: 1st year (2022) - 3rd year (2024)
 - (2022) Improvement of the TC System in Azerbaijan and Development of the TC Program at ANAS-HTP
 - (2023) Improvement of the Startup and Incubation System in Azerbaijan and Development of the Startup and Incubation Program for HTP
 - (2024) Implementation of the Application Process with the Implementation of Capacity Building of the TC Startup and Incubation System in Azerbaijan

3. Project Framework

The 2023 project is focused on improving the technology commercialization system in Azerbaijan and developing startup and incubation programs for the High Technologies Park (HTP).

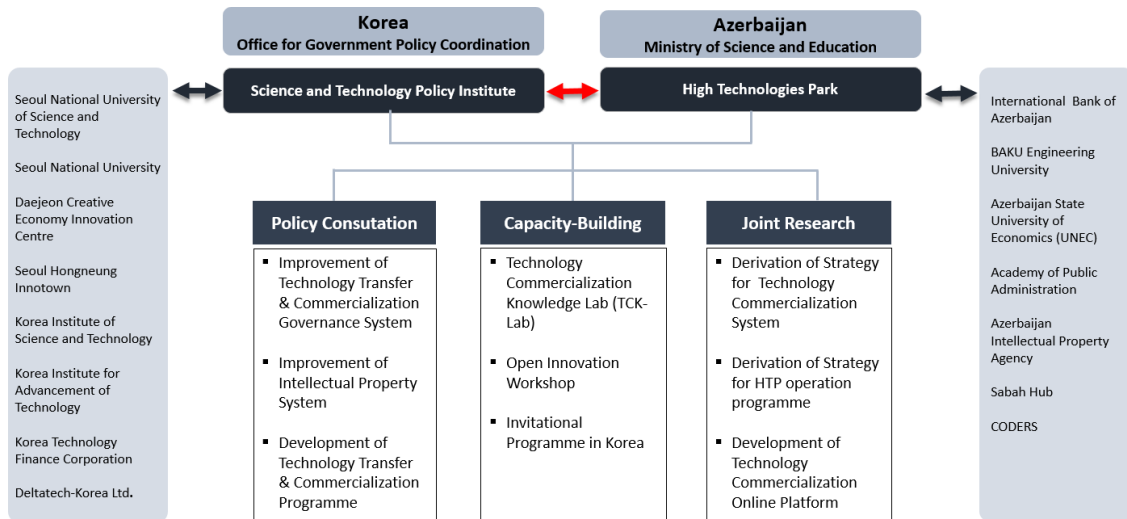
This project began with a kick-off seminar and an extensive current status analysis. During this phase, we formed a dedicated research team and outlined the scope of our research. This part of the project consisted in investigating the local landscape by conducting a thorough assessment of the existing startup and incubation regulations in Azerbaijan, while simultaneously examining the startup and incubation services offered at HTP, pinpointing areas that required further attention.

In the pursuit of enhancing capacity, a collaborative effort with diverse experts and organisations in South Korea was initiated. This collaboration included inviting Azerbaijan experts for training and organising specialized workshops on technology transfer and commercialization. These initiatives aimed to foster exchanges of knowledge and share Korea's experience and methodologies.

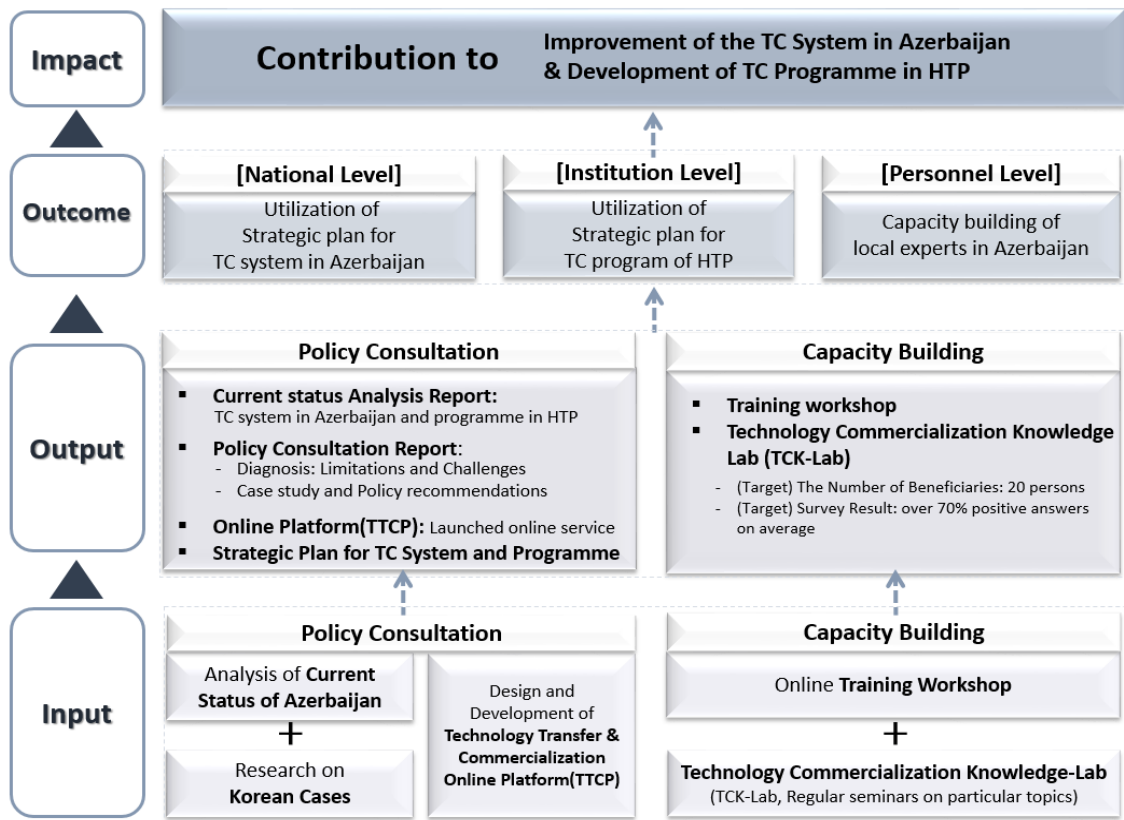
An integral aspect of the 2023 project involved the development of an online platform for technology transfer and commercialization. This platform was designed to enhance the ecosystem of technology transfer and commercialization in Azerbaijan by offering information, incorporating enhancements in content and technological functionality, and ensuring that it would serve as a valuable resource for stakeholders.

Finally, a forum has been conducted to disseminate the outcomes of the 2023 project. This forum played a pivotal role in providing policy recommendations concerning technology commercialization governance in Azerbaijan and boosting the roles of the HTP. The aim was to ensure that the insights gained from capacity-building activities were not only shared but also effectively disseminated to the relevant stakeholders, thereby contributing to the broader ecosystem of technology transfer and commercialization in Azerbaijan.

[Figure 1-1] Schematization of the 2023 Project Structure



[Figure 1-2] Detailed Framework of the 2023 Project



4. Project Team

The project team is composed of four key groups: the project management team, Korean consultants specializing in policy recommendations, Korean experts focusing on the TCK-Lab, and a team of Azerbaijan experts. The project management team oversees overall project coordination, resource allocation, and timeline management; the Korean consultants offer policy insights, aligning international standards with local needs; the Korean TCK-Lab experts facilitate knowledge transfer through seminars and workshops; and, last but not least, the Azerbaijani experts contribute local insights and conduct situational assessments to tailor the project activities. In summary, this multidisciplinary team combines international expertise with local insights to enhance Azerbaijan's technology commercialization system and foster innovation-driven growth.

[Table 1-2] Project Team

Type	Name	Position/Affiliation	Role/Area of Expertise
Project Management	Eun Joo Kim	Principal Researcher, STEPI	Project Manager
	Shinae Kang	Researcher, STEPI	Project Coordinator
Korean Consultants	Soo J. Sohn	Senior Researcher Fellow, STEPI	TT&TC governance
	Youngtack Shim	Professor, Seoul National Univ.	Intellectual property system
	Gyeongsoo Shim	Professor, Seoul National Univ. of S&T	TT&TC programs
	Youngtack Shim	Professor, Seoul National Univ.	TCK-Lab (IP Silk Road)
	Seung Ho Lee	CEO, Deltatech-Korea	Open innovation workshop
Azerbaijan Experts	Bünyamin Seyidov	Head, Incubation and Technology Transfer Centre, HTP	Project Coordinator, strategic planning for program
	Isa Gasimov	Director, Baku Engineering University and Technopark	TC platform management, strategic planning for the TC system
	Vusal Suleymanli	Innovation Consultant, ANAS	Management of TCK-Labs
	Samir Veliyev	Senior Economist, Azerbaijan International Bank	Current status research

5. Project Activities

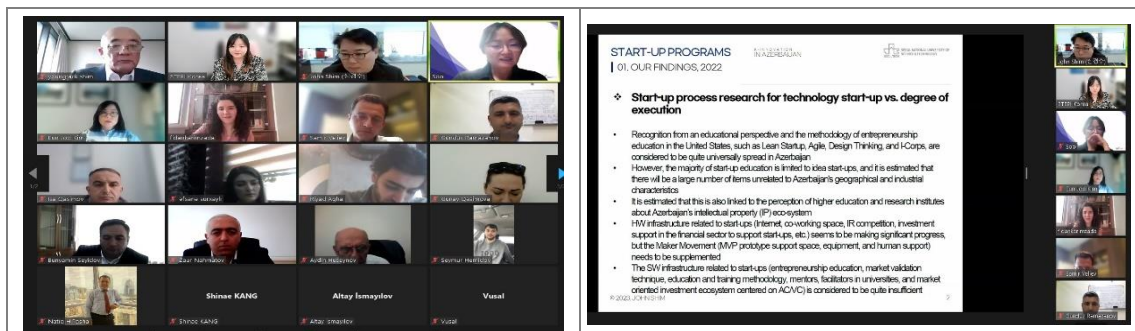
5.1 Kick-off Seminar

- Date/Venue: 15:00-17:00, 30 March 2023 /Online
- Participants: 20 persons (STEPI and HTP Project Team, Korea-Azerbaijan experts)
- Purpose: To establish a consensus and foster mutual comprehension regarding the project's subject matter, strategic blueprints, and goals
- Contents:
 - Project Objectives and Detailed Plans
 - Roles and Responsibilities of the Project Teams

[Table 1-3] Kick-off Seminar Program

Time	Contents	Presenters
15:00-15:05	Opening Remarks	Dr. Eun Joo Kim, Principal Researcher, STEPI
15:05-15:10	Welcome Remarks	Dr. Zaur Nahmatov, Executive Director, HTP
	Plan for Research on the Current Status of Azerbaijan's TT&TC Ecosystem	
15:10-15:25	Briefing on research	Samir Veliyev, Senior Economist, Azerbaijan International Bank
15:25-15:30	Feedback	2023 Project Team
	Plan for Policy Consultation	
15:30-15:40	Governance of TC	Dr. Soo J. Sohn, Senior Research Fellow, STEPI
15:40-15:50	Incubating Program	Prof. Gyeongsoo Shim, Seoul National University of S&T
15:50-16:00	Intellectual Property	Prof. Youngtack Shim, Seoul National University
16:00-16:05	Feedback	2023 Project Team
16:05-16:30	Q&A and Wrap-up	Mr. Bunyamin Seyidov, Head, Incubation and TT Centre, HTP

[Figure 1-3] Kick-off Photos



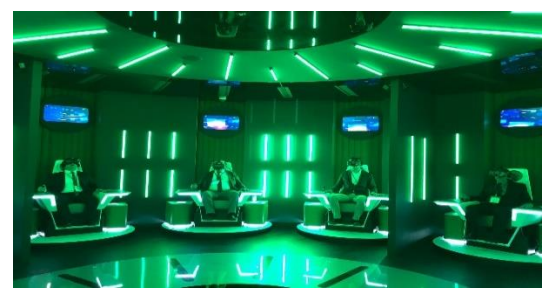
5.2 (Capacity-Building) Invitational Program in Korea

- Date/Venue: 21 – 27 March 2023 / Korea
- Purpose
 - To make study visits to major technology commercialization institutions in Korea and to enhance the capabilities of Azerbaijan's technology commercialization program
 - To hold brain-storming sessions for the Action Plan of Azerbaijan's technology commercialization program
 - To establish a network between Korean and Azerbaijani experts in technology commercialization and academic-industrial institutions
- Participants: Nine individuals, including experts related to technology commercialization, intellectual property, and technology startup ventures
- Key Contents
 - Visit prominent technology commercialization institutions in South Korea to strengthen Azerbaijan's capabilities for operating the technology commercialization programs at the High Technologies Park (HTP)
 - Hold an action plan workshop designed to strategize and plan Azerbaijan's technology commercialization programs
 - Promote networking among technology commercialization experts and academic institutions from South Korea and Azerbaijan

[Table 1-4] Invitational Program in Korea

Module	Type	Details
1. Train the Trainers	Lecture	Korean I-Corps education and training program
2. Technology Transfer & Commercialization Policy and Programs	Study Visit	Korea Institute for the Advancement of Technology (KIAT)
	Study Visit	Korea Institute of Science & Technology (KIST)
	Study Visit	Korea Technology Finance Corporation (KIBO)
	Study Visit	Daejeon Creative Economy Innovation Centre (CCEI)
	Study Visit	Seoul Hongneung InnoTown
	Study Visit	Korea University KU Startup Space
3. Action Plan Workshop	Workshop	Group work: Action Plan for the TT&TC program

[Figure 1-4] Invitational Program in Korea

	
Korea Institute for the Advancement of Technology (KIAT)	Korea Institute of Science & Technology (KIST)
	
Seoul Hongneung InnoTown	Korea Technology Finance Corporation (KIBO)
	
Daejeon Creative Economy Innovation Centre	SK T.um ICT Future Technology
	
Sejong National Research Complex	Closing Ceremony

5.3 (Capacity-Building) TCK-Lab: IP Silk Road Azerbaijan

In the current landscape of policy-making, the demand for innovative approaches to enhancing capacity-building is evident. Various learning methodologies have emerged as viable options, one of which is the “learning lab” approach. The concept of a learning lab pertains to a physical or virtual environment in which individuals engage in both academic and practical learning activities.

The inception of the Technology Commercialization Knowledge Lab (TCK-Lab) was inspired by the principles of the learning lab approach. Its primary mission was to align with the objectives of the 2023 project, which sought to bolster the operational capabilities and competitive edge of the Azerbaijan National Academy of Sciences High-Tech Park (ANAS HTP). This endeavour aimed to achieve these goals through the implementation of essential training programs designed to enhance operational expertise and foster exchanges of knowledge with local stakeholders in the realm of Intellectual Property (IP) Management.

The TCK-Lab for the year 2023 constitutes an integral component of the capacity-building initiative undertaken under the purview of the K-Innovation Partnership Program. Notably, this year's TCK-Lab featured the participation of a distinguished expert in the field of Intellectual Property (IP), who shared his invaluable knowledge and insights. The course's overarching objectives were to impart the fundamental legal principles and to illuminate the commercial dimensions of patents to a diverse audience comprising current and prospective inventors, entrepreneurs, technology commercialization and transfer experts, and Technology Transfer Organization (TTO) professionals among others.

Through the auspices of the TCK-Lab, a multifaceted approach was adopted to contribute to the capacity-building efforts of policy-makers. This approach encompassed learning, practical application, rigorous evaluation, innovative discovery, and the sharing of ideas. Consequently, the anticipated outcome of this initiative is the enhancement and advancement of Azerbaijan's institutional framework and conditions within the domain of Technology Commercialization/Technology Transfer (TC/TT), as viewed through the prism of Intellectual Property considerations.

- Technology Commercialization Knowledge Lab (TCK-Lab): IP Silk Road Azerbaijan
- Date: March to May 2023 / Online
- Lecturer: Professor Youngtack Shim, Seoul National University
- Participants: Local experts in technology commercialization and entrepreneurship, and the relevant stakeholders
- Overview: A series of seven seminars focusing on patents and intellectual property rights

[Table 1-5] 2023 TCK-Lab Curriculum

	Date	Session
1	20 March	• Inventing and Filing a Patent Application
2	5 April	• Patent Requirements
3	12 April	• Patent Infringement and Prevention
4	19 April	• How to Make a Powerful Patent
5	26 April	• Breeding Patents and Ownership
6	3 May	• Patent Licensing and Commercialization
7	10 May	• Patent Business

[Figure 1-5] 2023 TCK-Lab Participants

K-Innovation Partnership Program
Technology Commercialization Knowledge Lab (TCK-Lab)

IP SILK ROAD
AZERBAIJAN
- Intellectual Property -

DATE & TIME	SESSION
1 30 March 2023 16:00-18:00	INVENTING AND FILING A PATENT APPLICATION
2 05 April 2023 16:00-18:00	PATENT REQUIREMENTS
3 12 April 2023 16:00-18:00	PATENT INFRINGEMENT AND PREVENTION
4 19 April 2023 16:00-18:00	HOW TO MAKE A POWERFUL PATENT
5 26 April 2023 16:00-18:00	BREEDING PATENTS AND OWNERSHIP
6 03 May 2023 16:00-18:00	PATENT LICENSING AND COMMERCIALIZATION
7 10 May 2023 16:00-18:00	PATENT BUSINESS

Educator Information
Professor. SHIM, Youngtack
Professor, School of Law, Seoul National University (2009-Present)
Co-CEO, TCK-Lab
COO, The Technology Licensing Office, Seoul National University
President of Intellectual Ventures Korea
Patent Attorney, The Major U.S. Patent Law Firms

CONTACT:
Yusuf Saleymani <yusufsaleym@gmail.com>

Hosted by STEP-IH **Organized by** High Technologies Park YTP

April 5, 2023

Patent Law = Territorial Law

- **Patent law is a territorial law:**
 - Each country has its own patent law.
 - There is **NOTHING like an international patent** or else (cf. PCT?).
- **What does this mean?**
 - If I want patent protection in Korea, I must file a patent application in the Korean IP Office and get my application issued by the KIPO.
 - If I want patent protection in the US, I must file a patent application in the USPTO and get my application issued by the USPTO.
- **If I want a global patent protection?**
 - The Google search done on April 28, 2022 shows 193+2 countries in the world.
 - This means that I have to file applications and obtain patents in 193+2 countries.
 - **Doable? Cost?**
- **Thus, we need a practical strategy!**

3

5.4 (Capacity-Building) Open Innovation Platform Workshop

In light of the increasing demand for innovative policy-making capabilities, a broad spectrum of diverse learning methodologies has emerged as a viable option in contemporary contexts. One such methodology that warrants consideration is the workshop approach.

The Open Innovation Platform Workshop was conceived and executed with the overarching objective of aligning with the objectives of the 2023 project. This project's primary mission was to fortify the operational prowess and competitive standing of the HTP (High Technologies Park) through the implementation of indispensable training programs aimed at augmenting operational expertise. Additionally, it sought to facilitate the exchange and dissemination of knowledge among local stakeholders within the realm of TTCP (Technology Transfer and Commercialization Platform) Management.

The Open Innovation Platform Workshop for the year 2023 represents an integral component of the capacity-building initiative within the purview of the K-Innovation Partnership Program. The workshop's ultimate aim was to furnish the participants with foundational operational principles and illuminate the commercial dimensions of technology, specifically pertaining to the TTCP platform. The diverse participant base included individuals from various professional backgrounds, including current and prospective inventors, entrepreneurs, experts in technology commercialization and transfer, as well as professionals affiliated with Technology Transfer Organizations (TTOs), among others.

Through the strategic deployment of this workshop, it is anticipated that a multifaceted approach encompassing learning, practical application, rigorous evaluation, innovative discovery, and the exchange of ideas will contribute significantly to the capacity-building efforts of ecosystem stakeholders. As a result, this initiative is poised to catalyse improvements and advancements within Azerbaijan's Technology Transfer (TT) and Technology Commercialization (TC) landscape through the digital platform provided by the TTCP.

- Date/Venue: September to October 2023 / Online

- Lecturer: Mr. Seung Ho Lee, CEO of DeltaTech Korea
- Overview: A series of four workshops aimed at enhancing open innovation and platform management capabilities
- Participants: Targeted at the operating team of the Azerbaijan Technology Commercialization Platform

[Table 1-6] Open Innovation Platform Workshop Curriculum

	Date	Session
1	7 September	• Overview of Open Innovation Platforms
2	12 September	• Methodology for Generating Technical Information
3	26 September	• Technology Transfer and Technology Commercialization Marketing Strategies
4	10 October	• Methodology and Agreements for Technology Transfer

[Figure 1-6] Participants in the Open Innovation Platform Workshop

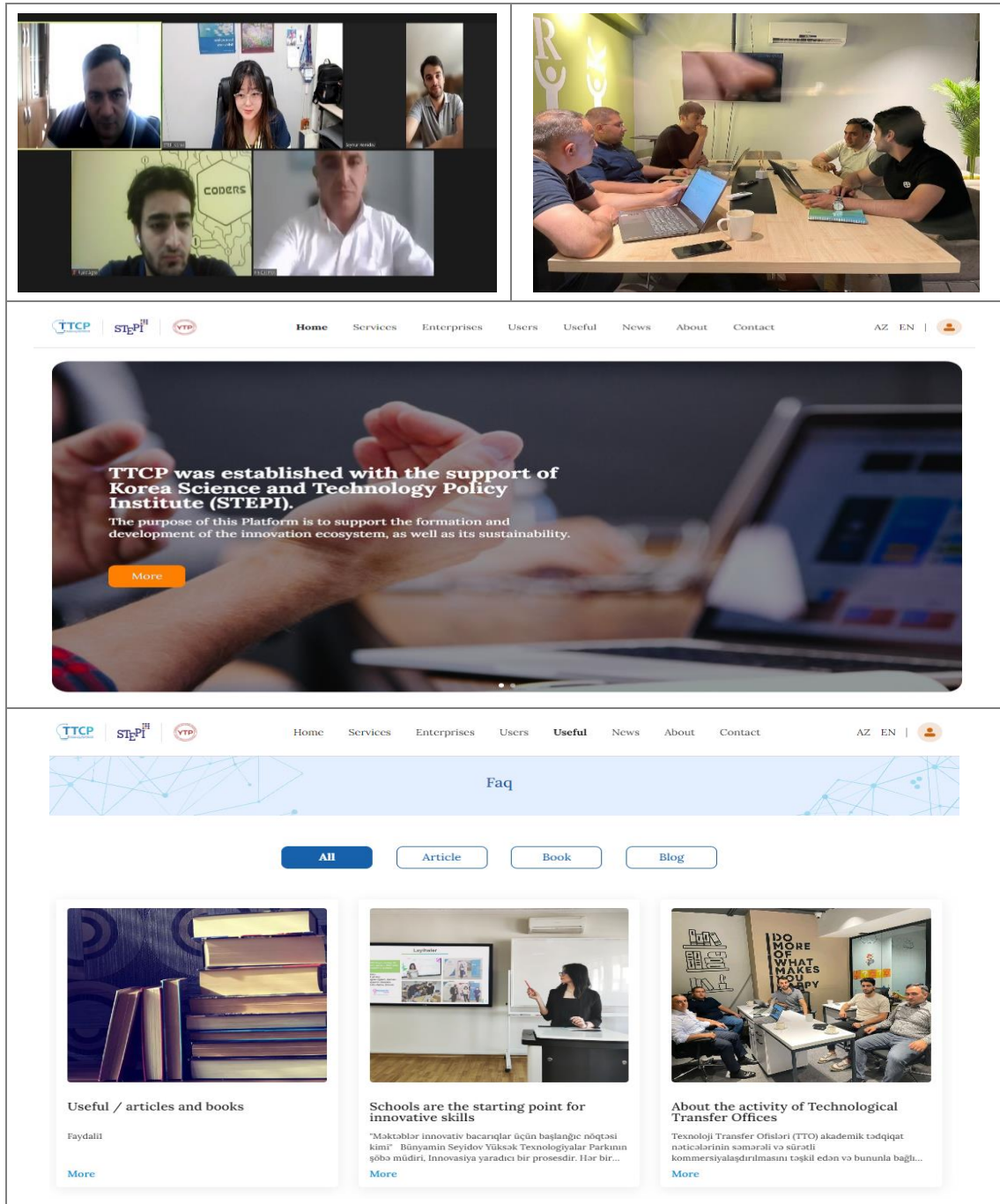


5.5 Technology Transfer & Commercialization Online Platform

- Purpose: To provide information on technology commercialization policies and programs, and to serve as a platform for activating networks and collaboration among stakeholders in Azerbaijan
- Development Process: (*Access link: <http://ttcp.az/>)
 - (May to November 2022) The development team (HTP) was formed to develop the technology commercialization platform in collaboration with CODERS, a local IT company. A comprehensive execution plan, including the platform's structure and functions, was established, and phased construction began. This included UI/UX design, user-based interface development, content database collection, and concurrent front-end and back-end development. Open beta services were launched, and user feedback was collected from technology commercialization stakeholders in Azerbaijan for the purpose of making further improvements.
 - (March to November 2023) Technical updates and platform promotional activities were carried out, including:
 - Updates of over 60 contents, including 45 patents;
 - Aggregation of 56 technology commercialization experts and 4 technology commercialization-related companies registered on the platform;
 - Hosting of 3 webinars for TTCP promotion with over 60 participants;
 - Front-end redesign to accommodate the latest standards and user-based interface upgrade; and
 - Optimization of platform performance, including simplification of data search and improved loading times.
- Expected Benefits and Future Plans
 - The technology commercialization platform aims to provide an effective space for database management, networking, and technology transfer and commercialization.
 - However, there is still a significant shortage of users and uploaded information. To enhance and activate the platform, the following activities are planned for 2024:

- Regular seminars and forums for technology commercialization experts; and
- Project-based events such as startup demo days and hackathons.

[Figure 1-7] Development of Technology Transfer & Commercialization Online Platform



5.6 Dissemination Forum

The event aimed to provide insights into various aspects of innovation ecosystems, technology commercialization, and government policies, with a focus on South Korea's experiences and expertise. The presentations were delivered by experts from different fields and institutions.

The first presentation, delivered by Dr. Soo J. Sohn, a senior research fellow at STEPI, focused on the theme of 'Innovation Ecosystems and the Role of Government.' The presentation drew attention to South Korea's approach to fostering an innovation ecosystem, which necessitates interdisciplinary collaboration and the infusion of diverse innovation resources. To achieve this, South Korea has devised the relevant support policies across a range of governmental departments. It was emphasized that such policy governance can take various forms, including centralized and decentralized approaches. The presentation underscored the importance of formulating a governance framework for commercialization policies that aligns with the nation's innovation environment.

The second presentation, delivered by Professor John Shim from Seoul National University of Science and Technology, introduced the 'Korean I-Corps Program.' The program's objective is to promote technology entrepreneurship, particularly in non-petroleum sectors. It was noted that the Azerbaijani government aspires to introduce digital government, including ICT, to fuel future growth. In this context, the experiences and expertise accumulated by South Korea through its participation in the U.S. NSF I-Corps Program since 2015 were highlighted as potentially valuable for Azerbaijan in the context of nurturing a conducive ecosystem.

The third presentation, delivered by Professor Youngtaek Shim from Seoul National University, was centered on the topic of 'Patents and Intellectual Property Rights.' Professor Shim stressed that, in order to stimulate innovation and economic growth within a country, it is imperative to establish an ecosystem that not only supports innovators but also legally protects and commercializes their innovations. South Korea's experience in this regard, particularly the roles played by the Patent Office and the Korea Invention Promotion Association, were highlighted. The presentation recommended that Azerbaijan, under the

purview of the HTP, should consider formulating and implementing policies designed to expand the scope of innovation and invention.

The final presentation, delivered by Dr. Yoon Seong Hwang, Head of the Management Planning Division at the Daejeon Creative Economy Innovation Centre, focused on 'Technology Commercialization and Startup Support Programs' for Azerbaijan's High-Tech Park (HTP). The presentation offered insights into the growth stages of startups, mission-specific support strategies, and the successful organization of the 2023 SIW (Startup Korea Investment Week). Dr. Hwang proposed future strategies for global cooperation, particularly in conjunction with SIW events in Azerbaijan.

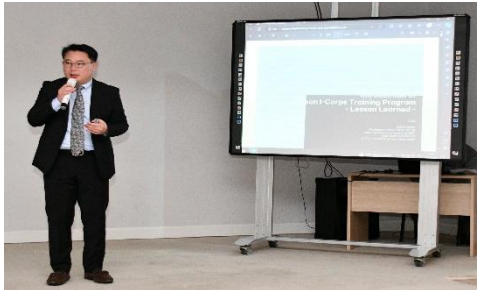
The event provided a valuable platform for knowledge exchange and highlighted South Korea's experiences and expertise in fostering innovation ecosystems, promoting technology entrepreneurship, and protecting intellectual properties. The insights presented during the event are expected to contribute to the ongoing efforts in Azerbaijan to strengthen its innovation and technology commercialization landscape. Further collaboration between South Korea and Azerbaijan in these domains is anticipated in the future.

- Date/Venue: 25 October 2023 /Baku Azerbaijan
- Participants: 50 stakeholders in Azerbaijan
- Purpose: To conduct policy advisory and capacity enhancement activities in the implementation of the K-Innovation Partnership Program for Azerbaijan's "Project to Improve the Startup and Incubation System in Azerbaijan and Develop the Startup and Incubation Program for the High Technologies Park (HTP)" and to disseminate the outcomes.

[Table 1-7] Dissemination Forum Program

Time	Contents	Presenters
	<Opening Session>	
10:00-10:20	Opening Remarks	Dr. Eun Joo Kim Principal Researcher, Division of Global Innovation Strategy Research, STEPI
	Welcome Remarks	Dr. Vasif Abbasov General Director, HTP
	Congratulatory Remarks 1	Ambassador Eun Yong Lee Korean Embassy in Azerbaijan
10:20-10:40	2023 K-Innovation Partnership with STEPI and HTP	Dr. Zaur Nahmatov, Executive Director, HTP
	Photo session	
<Session 1. Results of Korean Policy Consultation and Their Application in Azerbaijan>		
10:40-11:00	The Technology Commercialization Ecosystem and the Roles of Ministries	Dr. Soo J. Sohn Senior Research Fellow, Office of Innovation Law & Institution Research, STEPI
11:00-11:20	Ecosystem and Policy Technology Commercialization in Azerbaijan	Samir Veliyev, Local expert
11:20-11:40	I-Corp Program in Korea	Prof. John Shim, Seoul National University of S&T
11:40-12:00	I-Corp Program –its application and utilization in Azerbaijan	Dr. Isa Gasimov, Founder of "Yeni Fikir" LLC
12:00-13:00	Lunch	
13:00-13:20	Intellectual Property Management	Prof. Youngtack Shim, Seoul National University
13:20-13:40	Result of TCK-Lab and Plan for IP management in Azerbaijan	Mr. Vusal Suleymanli, Innovation consultant, local expert
13:40-14:00	Centre for Creative Economy and Innovation (CCEI) Technology Commercialization Support Program	Dr. Yoon Sung Hwang, Chief, Division of Strategic Planning & Administration, Daejeon Centre for Creative Economy and Innovation (CCEI)
<Session 2. Promoting the TC Ecosystem in Azerbaijan via TTCP >		
14:00-14:20	Renewal and Update of the TT&TC Platform	Mr. Seymur Hamidov, Founder, CODERS
14:20-14:40	Coffee-break	
14:40-15:00	Utilization and Optimization of the TT&TC Platform	Mr. Bunyamin Seyidov, Head, Incubation and Technology Transfer Centre, HTP
15:00-15:30	Open Discussion	Chair. Dr. Vüqar Babayev, Senior Lecturer, Academy of Public Administration

[Figure 1-8] Dissemination Forum

	
Dr. Eun Joo Kim, Principal Researcher, STEPI	Dr. Vasif Abbasov, General Director, HTP
	
Ambassador Eun Yong Lee, Embassy of Korea	Dr. Soo J. Sohn, Senior Research Fellow, STEPI
	
Professor Youngtack Shim, Seoul Nat'l Univ.	Professor Gyeongsoo Shim, Seoul Nat'l Univ. of S&T
	
Attendees of the Dissemination Forum	Attendees of the Dissemination Forum

6. Project Schedule

[Table 1-8] Project Schedule

Activity	2023												2024
	1	2	3	4	5	6	7	8	9	10	11	12	1
Project Planning													
Design Project Framework													
Organizing Project Team													
Kick-off Seminar													
TCK-Lab													
Invitational Program													
Open Innovation Workshop													
Development of TC Platform													
Analysis of Current Status													
Diagnosis of Current Status													
Policy Consultation													
Establishment of Strategic Plans													
Dissemination Forum													
Submission of Final Report													
Presentation of Final Report													
Evaluation of Final Report													
Publication													



CHAPTER 2

Current Status of Technology Commercialization in Azerbaijan

Bünyamin Seyidov

Head, Incubation and Technology Transfer Centre, HTP

Isa Gasimov

Director, Technopark of Baku Engineering University

Samir Veliev

Senior Economist, International Bank of Azerbaijan

Vusal Suleymanli

Innovation Consultant, Azerbaijan National Academy of Sciences

Chapter 2.

Current Status of Technology Commercialization in Azerbaijan

1. Research Framework

A systematic review of existing academic literature on technology commercialization can demonstrate the ambiguous role of the capabilities of technology transfer offices (TTO) in transferring technologies from a university/research institute to industry. For example, Lee and Jung (2021) identified three distinctive capabilities of technology commercialization (TC) while analysing longitudinal data from South Korea for the period 2009-2015. These capabilities include academic research capability, applied research capability, and TTO operation capability. Their analysis of the effects of these capabilities on TC performance reveals that, unlike the other two capabilities, the TTO operation capability has no significant effect on TC performance. Therefore, establishing an effective operational capability alone for a high-technology park cannot solve the challenges associated with TC activities. Min, Kim and Vonortas (2020) also analysed the conditions for TC and commercial exploitation of public technologies in Korea. Their analysis of 514 cases of TC by major public research institutes and universities underline that market competition is a key determinant of successful TC, and that it also affects business growth in general.

Moreover, regardless of the market situation, effective partnership is critical for successful TC. Recently, Belitzki, Aginskaya and Marozau (2019) also identified the role of TTOs and direct industrial funding in TC in the transition economies of Azerbaijan, Belarus and Kazakhstan during 2015-2017. Interestingly, their research reveals that successful research commercialization is neither associated with the existence and awareness of TTOs, nor with their contracts with partners. However, direct industrial funding of research is a significant factor.

1.1 Macro-level analysis of the current status: a view of the TC ecosystem in Azerbaijan

■ Policy-making issues

A successful innovation policy for a developing country requires careful consideration of a wide range of activities. To achieve better TC results, policy makers should develop a comprehensive innovation policy and then consider certain key issues when implementing that policy. Such issues may include, but are not limited to, the following: the state of innovation; strategic priorities; access to finance; intellectual property protection; capacity building; and collaboration and cooperation.

■ Institutions

TC is a process by which necessary measures are undertaken in order to pass a given research output through the relevant procedures and make it beneficial for commercialization in various applications. For TC to act effectively, several types of institutions play important roles: government agencies; research centres and universities; business incubators and accelerators; venture capital firms; and associations and consortia.

■ Funding mechanisms

TC requires several funding mechanisms to be available. The range of such funding schemes may vary depending on the stage of technology development, or on the goals of the stakeholders who seek to commercialize such technologies. Various major funding mechanisms also play an important role, including government grants, equity financing, debt financing, corporate partnerships, and crowdfunding.

■ Regulatory frameworks

Complying with legislation and regulations is a must-do requirement for TC. Therefore, it is critical to assess regulatory frameworks that may impact it. There are several regulatory frameworks that are particularly important for TC, and such frameworks - encompassing intellectual property, taxation, international trade, consumer protection, data privacy, and environmental, health and safety regulations among others - may vary depending on the specific industry and technology involved.

1.2 Micro-level analysis of the current status of the HTP

■ Operational issues

Improving the operational issues related to TC, particularly in high-technology parks, involves different strategies and actions. By improving them, stakeholders can achieve a better assessment of the status quo, develop appropriate and effective strategies and plans, and monitor their implementation and results in later stages. There are certain operational issues that are critical to TC, i.e. needs assessment, improvement plans, infrastructure and services, collaboration and cooperation, and monitoring and evaluation.

■ Other issues

Along with the abovementioned elements of a rational framework of analysis for this report, the following other issues are also important to consider when aiming for successful TC: market potential, business model, timing, team, etc.

Based on the research framework described above, this report will analyse the current status of TC in Azerbaijan.

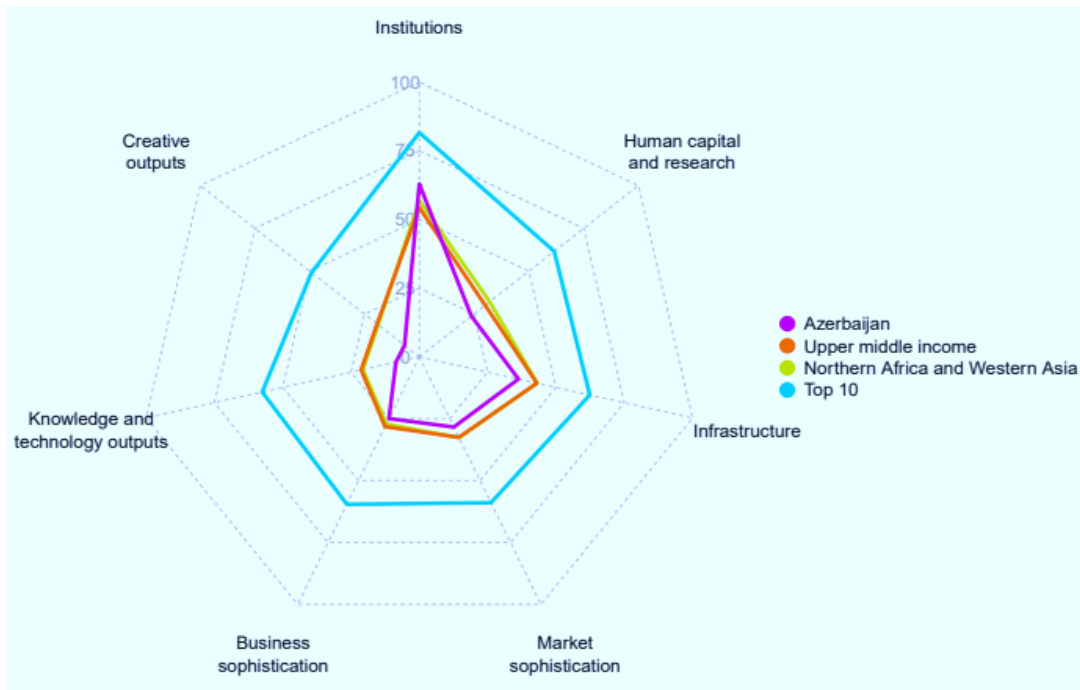
2. Macro-Level Analysis of Technology Commercialization in Azerbaijan

A successful innovation policy for a developing country requires careful consideration of a wide range of activities. In order to achieve better TC results, policy makers should develop a comprehensive innovation policy and then consider certain key issues when implementing that policy. Gaining added value through TC plays a key role in economic development. The rapid development of technology and increasing competition are among the main factors that justify the commercialization of science, as well as the development of innovative entrepreneurship. There are complex infrastructural factors in Azerbaijan to regulate the process of TC. At the same time, there are factors that restrict this process and partially hinder it, such as legislation, the weakness of university-industry cooperation, incomplete formation of the intellectual property ecosystem, and other factors. Also, the lack of research that meets the needs of the industry can be considered as a factor that slows down the commercialization process.

Regarding the state of innovation in Azerbaijan, the country generally performs below expectations for its level of development, relative to GDP. According to the 2023 Global Innovation Index, it ranked 93rd among the 132 economies featured in the GII 2022, a poorer ranking than it achieved in the previous two years, i.e. 80th in 2021 and 82nd in 2020. Compared to its level of investment in innovation, it produces fewer innovation outputs. Furthermore, it performs relatively poorly in various domains.³ For example, its best performing sub-index is “Institutions” while its weakest performance is in “Knowledge and technology outputs”. Particularly, it underperforms significantly in the “creative outputs” and “knowledge and technology outputs” categories. As regards the “knowledge and technology output” indicator, Azerbaijan ranks 107th in knowledge creation, 120th in knowledge diffusion, and 106th in knowledge impact. These indicators give a clear picture of Azerbaijan’s innovation output.

³ https://www.wipo.int/edocs/pubdocs/en/wipo_pub_2000_2022/az.pdf (accessed date: 10 May 2023)

[Figure 2-1] The Seven GII Pillar Scores for Azerbaijan



Source: World Intellectual Property Organization (2021), p,21.

Other underperforming indicators include IP payments as a percentage of total trade (122nd); IP receipts as a percentage of total trade (113th); production and export complexity (114th); expenditure on education as a percentage of GDP (114th); etc. Again, the GII report underlines that Azerbaijan exported (total value: USD 40 million) high-technology products, and this indicator is equivalent to the 106th position in the ranking.

These indicators clearly demonstrate that Azerbaijan needs to develop certain key areas in order to improve its level of innovation. Regarding strategic priorities, prioritizing certain policy targets over others may have its own pitfalls. However, if done correctly, some strategic priorities can bring about high growth and success in the selected areas.

Usually, the President's office initiates the preparation of national development plans and assigns leadership of their implementation to the relevant ministries. Ministries working on issues like economic development, digital development, social protection, and agriculture, etc. proposes their initial drafts, and then develop the plan further until it is officially approved and adopted by the government.

In this regard, “Azerbaijan 2030: National Priorities for Socio-Economic Development” was an ambitious development plan until the national economy encountered fluctuations (the The Second Karabagh War, Covid-19 pandemic, etc.). Therefore, the government preferred to develop another five-year national development plan, namely the “Socio-Economic Development Strategy 2022-2026”. Hence, “Azerbaijan-2030” is no longer being pursued as a national development plan.

Therefore, national development plans such as the “Azerbaijan 2030: National Priorities for Socio-Economic Development” prioritize certain strategic targets to be pursued. For example, the “Azerbaijan 2030” concept considered “A Creative and Innovative Society” and underlined that: “In the future, the digital society, which supports the latest technologies and innovations, will become a leading force in development. An ecosystem that stimulates creativity and innovation in society must be built to keep the country competitive. To this end, the development of the processing industry should be broadened, and the development of science-intensive and high-income medium and high-tech industries should be ensured. In addition, competitive funding mechanisms should be established to promote results-oriented research”.⁴

Moreover, the Social-Economic Development Strategy of the Republic of Azerbaijan, 2022-2026,⁵ adopted on 22 July 2022, underlines the essential points regarding the development of innovations and technological development. As the strategy highlights, since Azerbaijan is entering a new phase of strategic development in the post-pandemic and post-conflict era, its strategic aims include, but are not limited to, the following:

- Mobilizing modern and sustainable sources of economic growth;
- Strengthening innovation-orientation and the financial sustainability of the national economy;
- Effectively integrating the national economy into global value chains.

⁴ <http://www.e-qanun.az/framework/25029> (accessed date: 10 May 2023)

⁵ <https://president.az/az/articles/view/56725> (accessed date: 10 May 2023)

One of the five strategic frameworks of the 2022-2026 strategy concerns “competitive human capital and the destination of modern innovations”. It underlines that:

- Commercialization of scientific-research and technical-construction projects (in other words, R&D projects) and creation of a favourable environment for the transformation of research results into products will be supported, obstacles to scientific research will be removed, and scientific research will be facilitated. It is also planned to transfer innovation to markets and increase research and development spending to 0.1% of GDP. The Government supports research and development projects through the Science Fund and SOCAR Science Funds, as well as various types of grant competitions. Indeed, in terms of creating scientific infrastructure and funding scientific research, these may not be considered as full support tools. However, progress in this direction is observed every year. In 2022, the expenses from the state budget for science in Azerbaijan amounted to 168 million manats, and the funds allocated to research and development projects were 202 million manats.

By increasing the share of non-oil goods and services in the national economy, Azerbaijan aims to diversify and boost the volume of its exports. It will be accompanied by incentives for FDI in those industries. The Strategy 2022-2026 also aims to form the “Science-Industry Cluster 4.0”. Moreover, the government of Azerbaijan also aims to establish the “Absheron Valley” Innovation Cluster to form an innovation agenda, prioritizing innovations in priority fields. Overall, “Absheron Valley” aims to host 300 startups that could attract USD 13 million of investment. By utilizing the benefits of private-public partnerships, it will expand incubation and acceleration programs at universities and private entities. For example, “Absheron Valley” will also host an innovation centre specializing in agricultural technologies.⁶

As outlined in the Socio-Economic Development Strategy 2022-2026, “Absheron Valley” will be an innovation cluster that receives governmental support to achieve its agenda. The long-term goals of the cluster include attracting 300 new startups that would obtain funding

⁶ <https://president.az/az/articles/view/56725> (accessed date: 10 May 2023)

to the tune of USD 13 million. However, it is too early to launch such an ambitious initiative. The relevant government authority should decide on its location, establish the necessary legal framework, and build the infrastructure required to start working. Therefore, there is no established relationship as yet between “Absheron Valley” and the HTP. Considering ambitions within the “Absheron Valley” initiative, the HTP may cooperate on issues such as offering the necessary infrastructure for its startups, expanding the incubation and acceleration programs within public-private partnership projects, and strengthening the power of the tech diaspora, etc.

Regarding access to finance⁷, gross expenditure on R&D as a percentage of GDP was just 0.2 (ranked 93rd) according to the GII 2023⁸. This is extremely low compared to advanced economies such as Korea and Israel. The state budget of the Republic of Azerbaijan (total: AZN 3.8 billion) allocates AZN 228.5 million (approximately, USD 134.4 million) for science-related expenditures. Azerbaijan’s Strategy 2022-2026 document also highlights that researches in various fields of science will be expanded, which will be made possible by implementing certain measures, including the establishment of a mechanism for competitive and purposeful funding of scientific research. Moreover, a funding mechanism will be established for each stage of startup growth in order to develop a system of purposeful fiscal incentives for startups and investors.⁹

There is a limited number of private funds (although their assets and investment sizes are not clearly communicated) in Azerbaijan. For example, Khazar Ventures has a longer experience in the market. INNOLAND is in fact operated by the same professionals who founded Khazar Ventures. Meanwhile, “Technovate Investments” was established as recently as April 2022 by one of the successful founders of “GoldenPay” and “eMobile”. “Technovate Investments”, a venture studio whose founder currently resides in the USA, aims to bring startups from Azerbaijan and Georgia to Silicon Valley. Its portfolio includes five startups, four of which have moved to the USA.

The same team cooperates with the Innovation and Digital Development Agency (IDDA)

⁷ https://www.wipo.int/edocs/pubdocs/en/wipo_pub_2000_2022/az.pdf (accessed date: 10 May 2023)

⁸ https://www.wipo.int/edocs/pubdocs/en/wipo_pub_2000_2022/az.pdf (accessed date: 10 May 2023)

⁹ <https://president.az/az/articles/view/56725> (accessed date: 10 May 2023)

under the Ministry of Digital Development and Transportation, and one of the biggest private holdings in the country ("Pasha Holding" LLC). As a result, the three parties, together with other local investors, have recently established a venture capital fund, called Caucasus Ventures LLC, with equity capital of USD 11.3 million (IDDA: AZN 5 million; Pasha Holding: AZN 5 million; and individual entrepreneurs: AZN 1.3 million). "Caucasus Ventures" was established on November 29, 2022 with the participation of Innovation and Digital Development Agency, "PASHA Holding" Group of Companies and individual investors. It is the first such fund in Azerbaijan. Representatives of companies representing the public and private sectors, investors, responsible employees of incubation and acceleration centers, foreign diplomats, heads of international organizations and media representatives participated in the presentation ceremony held in connection with the creation of the fund.

The main direction of the fund's activity is to invest in post-seed growth and "A" stage startups. The authorized capital of the fund is 11,300,000 Azerbaijani manats. "Pasha Holding" Group of Companies agreed to allocate 5 million manats to the fund and individual entrepreneurs 1.3 million manats. At the same time, the Innovation and Digital Development Agency supported the initiative of private partners and allocated an additional 5 million manats to "Caucasus Ventures". "Caucasus Ventures" fund is planned to invest in 50 startups from Azerbaijan, Georgia, Turkey, Central Asian and Eastern European countries by 2028. More than 60% of the fund's budget is planned to support Azerbaijani projects, and the rest will be directed abroad. With the consent of the shareholders, the period covered by the fund can be extended until 2030. The size of investment tickets is expected to be between 50,000-250,000 USD, depending on the stage of development of the startup. Some of the verticals that the fund will focus on are travel technology, SaaS solutions and financial technology.

There are no crowd-funding schemes in Azerbaijan. Recently, corporate partnerships have been evolving, especially led by Pasha Holding and the International Bank of Azerbaijan OJSC, which develop their own corporate innovation programs. The only viable and feasible funding scheme from which startups can obtain financing is Caucasus Ventures, as described in the Status report previously. This will finance an applicant's equity capital through a joint fund held by Pasha Holding, IDDA, and some private investors. Moreover,

the government runs smaller grant schemes (usually up to AZN10,000) in different platforms.

Unfortunately, underfunding is only one side of the coin. Usually, funding agencies complain about the quality of applicant projects when it comes to selecting promising ones. Please note that there is only one VC-like funding agency, namely Caucasus Ventures, which is described in the Status report. Recently, it announced that it had started investing startups in February 2023. So far, only three startups (TuranAPP, TestedWeb, and MagicPort) have received funding from this VC.¹⁰ As a rule, residents of HTP attract investment themselves. However, there may be certain start-ups in which HTP will provide seed investment.

Most startups are funded at various stages of their growth by incubation and acceleration centres. However, such centres usually provide mentoring and development support rather than funding in significant amounts. For example, INNOLAND supports startups in their preparations for pitching and transforming their business ideas into prototypes or full projects. However, their support is limited as well. Moreover, corporate innovation centres such as Pasha Holding LLC and the ABB Innovation Centre, or units under private corporations, aim to support startups from idea to product, although their views cannot be considered a national program.

Certain government grants deigned to stimulate innovation are available within entities under the Ministry Science and Education. For example, there is a grant project to finance educational development and innovations adopted since 2016¹¹. The Cabinet of Ministers approved the procedures and other relevant guidance for such grants. The ultimate goal of these grants is to stimulate healthy competition and innovation activities among education and research entities. Once a year the Ministry of Science and Education announces the call for the project proposals in two categories, namely, general education entities and individual projects. Applications are assessed based on technical and content expertise. Grants for general education entities can be up to AZN 100,000 (approximately, USD 59,000), whereas grants for individual projects can be up to AZN 10,000 (approximately, USD 5,900).

¹⁰ <https://tech.az/posts/caucasus-ventures-3-startapa-yatirim-edib-2895> (date accessed: 10 May 2023)

¹¹ <https://e-qanun.az/framework/35883> (accessed date: 10 May 2023)

HTP has sent a draft of the commercialization-related proposal to the Agency, and a project proposal for strengthening university-industry cooperation has been prepared and presented.

Regarding regulatory measures, especially in the field of IP protection, Azerbaijan ranked 30th out of 141 countries in the latest Global Competitiveness Index published by the World Economic Forum. Although there is no specific, single legal act on IP protection, there is a strong legal base for such measures¹², as listed below:

- The Law of the Republic of Azerbaijan on Patents, and the Law of the Republic of Azerbaijan on Trademarks and Geographical Indications.
- The Law of the Republic of Azerbaijan on Legal Protection for the Topographies of Integrated Circuits.
- The Law of the Republic of Azerbaijan on Copyright and Related Rights.
- The Law of the Republic of Azerbaijan on Legal Protection for Compilations of Data.

Azerbaijan, which has successfully entered the international intellectual property system, is a party to the Berne Convention for the Protection of Literary and Artistic Works held under the auspices of the World Intellectual Property Organization (WIPO), Conventions on the Protection of the Rights of Performers, Phonogram Producers, and Broadcasting Organizations, the WIPO Internet Contracts on Copyright, and the Records of Performances and Phonograms. It has also signed the UNESCO "World Authorship Agreement" and the "Convention for the Safeguarding of the Intangible Cultural Heritage".

From a broader view, the following - but not limited to - legislative acts may be regarded as legislative acts on the partial commercialization of technologies:

- Decree of the President of the Republic of Azerbaijan on ensuring coordination in the field of innovative development in the Republic of Azerbaijan.
- Law of the Republic of Azerbaijan on Science.

¹² <https://www.ip-coster.com/IPGuides/c/intellectual-property-azerbaijan> (accessed date: 10 May 2023)

- Rules for maintaining the registration of residents of the technology business incubator, as well as the registers of the residents of technology parks and those of technology business incubators.
- Form of registration certificates of technology parks and technology business incubators.
- Resolution of the Cabinet of Ministers of the Republic of Azerbaijan on approval of the “Application form for registration as a resident of the Technology Park and the List of Attached Documents.”
- Resolution of the Cabinet of Ministers of the Republic of Azerbaijan on approval of the “Criteria for determining startups.”
- Azerbaijan 2030: National Priorities for Socio-Economic Development.
- Decree of the President of the Republic of Azerbaijan on ensuring the activities of the Innovation Agency (current name: Agency for Innovation and Digital Development) under the Ministry of Transport, Communications and High Technologies of the Republic of Azerbaijan (Ministry of Digital Development and Transport).
- Social-Economic Development Strategy of the Republic of Azerbaijan: 2022-2026.

Azerbaijan has not yet adopted draft laws on “Innovation Activity”, “Venture Capital” and the “Competition Code”. Moreover, the draft of the “Innovation Strategy” developed by global consultants has not yet been approved officially.

Within the “Azerbaijan 2020: Strategic Roadmaps”, one of the policy priorities (Priority 5.1) was dedicated to improvement of the national innovation system. Under this Priority, the government underlined that although there is no specific legislative base for innovation activities in Azerbaijan, there are certain acts that support and stimulate innovation policies (e.g. Patent Law, Science Law, etc.). Therefore, it aimed to improve the legislative framework for innovation activities by considering the enactment of a law on innovation activities. With the participation of the Ministries of Economy, Justice, and Communication and High Technologies, the government drafted the law and intended to enact it by 2020.

Unfortunately, as few of the priorities within the Strategic Roadmap initiatives, has not yet been adopted by Azerbaijan. The draft law would consider improving legal frameworks for the establishment of high technology parks as a critical element of the innovation infrastructure. Moreover, it would also cover funding and implementation mechanisms for innovation-related projects, including the establishment of venture capital centres.

The draft law on innovation activities, along with the draft strategy for innovation development in Azerbaijan, touch upon such issues as the definitions, roles, rights and responsibilities of each stakeholder (especially, the government), as well as human resource development, international cooperation, evaluation systems, incentives and penalties.

Since innovation activities are considered vital within the Socio-Economic Development Strategy 2022-2026, those draft acts and documents should be updated and enacted according to the action plan of the government. So far, TC related legislation is underdeveloped in Azerbaijan.

Regarding institutions, TC is a process by necessary measures are carried out in order to ensure that a given research output passes through the relevant procedures and to make them beneficial for commercialization in diverse applications. However, there is no central organization in Azerbaijan that is engaged in technology commercialization. Usually, all ministries include their right to conduct scientific-technical projects within the scope of their statutes as a standard clause. However, the activities of the Ministries of Digital Development, Transport, Economy, and Science and Education, as well as the Intellectual Property Agency, partially coincide with this area.

There is no single ministry dedicated to technology commercialization in Azerbaijan. Therefore, each related authority has its own TC approach. However, the Centre for Commercialization and Transfer of Technologies (established in 2021) under the IP Agency is the only dedicated institution, although it is not a central body tasked with carrying out country-level TC policies. It has no authority to lead the TC policies by itself, nor order other ministries to conduct TC functions as it sees fit. In order to achieve its objectives, this centre will support the provision of incentives for invention activities, and the integration of science and industry, as well as supporting “innovation economy” policies by promoting IP activities.

With equity capital of AZN 100,000 (approx. USD58,500), it will also support the development of expertise capable of assessing the commercialization potential of scientific research outputs. Other than this centre, there is no ministry or agency dedicated to the formulation and implementation of TC policies.

The previous status report within the K-Innovation Partnership Program, submitted in 2022, highlighted the institutional settings before the date of July 2022 because of the significant changes that happened after July 28, 2022. According to the Decree of the President of the Republic of Azerbaijan on “certain measures related to the improvement of governance in science and education”¹³, the Ministry of Education has been renamed as the Ministry of Science and Education. With the new reforms, the Ministry acquired the authority to implement state policy and regulations in science, and to coordinate and direct the activities of scientific institutions and organizations. Accordingly, the following two new agencies have been established under the Ministry:

- State Agency on Science and Higher Education; and
- State Agency on Pre-school and General Education.

Perhaps the most significant change was related to the Azerbaijan National Academy of Sciences (ANAS), whose authority and scope have been considerably weakened. In addition, a significant number of research institutes of ANAS have been transferred to the Ministry of Science and Education, as well as to the Ministry of Culture:

¹³ Presidential Decree on certain measures related to the improvement of governance in science and education in the Republic of Azerbaijan; <https://e-qanun.az/framework/50052> (accessed date: 20 May 2023)

From ANAS to the Ministry of Science and Education:

Research institutes

- Institute of Physics
- Institute of Mathematics and Mechanics
- Institute of Control Systems
- Institute of Radiation Problems
- Institute of Information Technology
- Institute of Biophysics
- Shamakhi Astrophysical Observatory (named after Nasiraddin Tusi)
- Institute of Petrochemical Processes (named after the academician Yusif Mammadaliyev)
- Institute of Catalysis and Inorganic Chemistry (named after the academician Murtuza Naghiyev)
- Institute of Chemistry of Additives (named after the academician Ali Guliyev)
- Institute of Polymer Materials
- Institute of Geology and Geophysics
- Institute of Geography (named after the academician Hasan Aliyev)
- Museum of the History of Nature
- Institute of Oil and Gas
- Institute of Botany
- Institute of Zoology
- Institute of Physiology (named after the academician Abdulla Garayev)
- Institute of Microbiology
- Institute of Dendrology
- Institute of Molecular Biology and Biotechnologies
- Institute of Genetic Resources
- Institute of Soil Science and Agro-Chemistry
- Institute of Economy

Regional divisions

- Nakhchivan Division
(Institute of Natural Resources; Institute of Bio-resources; Batabat Astrophysics Observatory)
- Ganja Division
(Institute of Ecology and Natural Resources; Institute of Bioresources; Institute of Agrarian Problems; Botanical Garden)

Scientific service organizations

- Experimental-Industrial Plant
- Cybernetics, a scientific-production association
- Nafta, a scientific-production association
- Ashgar, a scientific-production association
- "Composite, a scientific-production association

High Technologies LLC

From ANAS to the Ministry of Culture:

- National Museum of the History of Azerbaijan
- National Museum of the Literature of Azerbaijan (named after Nizami Ganjavi)
- Huseyn Javid Home-Museum.

The following entities have been kept under the authority of the ANAS:

- Institute of Manuscripts (named after Mahammad Fuzuli)
- Institute of Architecture and Art
- Institute of Folklore
- Institute of Literature (named after Nizami Ganjavi)
- Institute of Linguistics (named after Nasimi)
- Institute of Archaeology, Ethnography and Anthropology
- Institute of Philosophy and Sociology
- Institute of the Caucasian Studies
- Institute of History (named after Abbasgulu Agha Bakikhanov)

- Institute of Oriental Studies (named after Academician Ziya Bunyadov)
- Republican Seismic Survey Centre
- Central Botanical Garden
- Sheki Regional Scientific Centre
- Lankaran Regional Scientific Centre
- Guba Regional Scientific Centre
- Research Centre of “Azerbaijan National Encyclopaedia”
- Central Scientific Library
- Virtual Electronic Action Society (VEAS)
- "Science" publishing house

Note that the transfer of all these institutes and entities from ANAS to the ministries has not yet been completed (according to the Decree, it had to be completed within three months). Therefore, the fate of many institutes and the HTP is uncertain and somewhat chaotic nowadays.

Considering the most recent institutional setting, ANAS has lost its authority as a “centre of science” in Azerbaijan and henceforth will mostly deal with humanitarian and social sciences. Therefore, innovation activities, particularly technological innovations, will no longer be relevant to ANAS. Henceforward, the activity in the K-Innovation Partnership Program will help HTP to become one of the active organizations of the Ministry of Science and Education.

State agencies and state-owned companies do not yet have clearly defined responsibilities with regard to their work on technology commercialization. The Azerbaijan National Academy of Sciences (ANAS), Science Development Fund, State Agency for Public Service and Social Innovations, Space Agency (Azercosmos), Centre for Analysis and Coordination of the 4IR, Economic Zone Development Agency, Centre for Analysis of Economic Reforms and Communication, and Intellectual Property Agency are the most active public authorities when it comes to working coordinated and together on innovation-related activities. Some other entities have also become increasingly interested in collaboration (or at least, an imitation of collaboration) with incubation and technology transfer units. Azergold, SOCAR,

and a few other companies support startup programs with the aim of finding new solutions to their relevant industrial upgrades.

In particular, the Intellectual Property Agency of the Republic of Azerbaijan was established to improve management in the field of consumption market control, standardization, metrology, and the protection of intellectual property rights.” The Patent and Trademark Examination Centre¹⁴ and the Technology Commercialization and Transfer Centre” were established under the IP Agency¹⁵. The latter provides a link between science and industry, mutual order and supply relations, technology transfer, and the commercialization of intellectual property results. The ANAS, Science Development Fund, Agency for Innovation and Digital Development (AIDD), and certain other entities also carry out a number of activities aimed at regulating and ensuring the process of scientific research, technology transfer, and commercialization. With no clearly defined legal mandate as yet, AIDD was established to organize and implement (including control and regulation) activities related to digital transformation. Moreover, it promotes and funds scientific research and innovation projects (including startups), as well as innovative entrepreneurship.¹⁶

The table below highlights the role of the most relevant ministries, as their charters suggest, in TC activities.

¹⁴ <https://www.copat.gov.az/en/index> (accessed date: 20 May 2023)

¹⁵ <https://e-qanun.az/framework/47305> (accessed date: 20 May 2023)

¹⁶ <https://president.az/az/articles/view/53407> (accessed date: 20 May 2023)

[Table 2-1] The Role of other Ministries in TC-related Activities

Ministry	Role of technology commercialization	Institutions/ Program for technology commercialization : e.g. incubating, matching technologies, demonstration, making prototypes, funding, etc.	Similar to the HTP	Differentiation from the HTP
Economy	The Ministry's roles include the formulation of innovation and scientific-technical policy in the national economy together with other relevant public entities. The ministry provides recommendations on investing in education and science for human capital development, with the aim of increasing South Korea's international competitiveness. It is authorized to conduct scientific-technical and research projects, and to implement their findings in its operations. One of the most important duties of the Ministry is to prepare recommendations for incentivizing, supporting and promoting innovation activities to increase competitive final products in new industrial sectors with high added-value (together with other relevant public entities).	The Ministry of Economy does not directly engage in TC activities. However, certain agencies under the Ministry of Economy have particular roles in innovation-related activities. For example, KOBIA (an SME-support agency) provides a "startup certificate" (94 startups by May 2023) which grants incentives from the government. The Economic Zones Development Agency (EZDA) under the Ministry of Economy manages six industrial parks, which host residents with modern technologies for manufacturing innovative, export-oriented and competitive products.	The Startup Certificate requires startups to develop commercially viable, innovative, technology-intensive, and competitive products. The awarded startups are exempted from income tax on revenues from their innovation activities for three years EZDA awards its residents in industrial parks with tax breaks for 10 years (property, land, profit, custom duties and VAT while importing technologies).	Technopark supports science-intensive activities and provides necessary conditions and privileges to residents operating in its territory. Residents are exempted from property, land and profit tax for 10 years. In addition, residents are granted exemptions from paying customs duties and value added tax (VAT) for 7 years on imported machinery, technological equipment and devices.

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Digital Development and Transport	As one of the key government authorities, the Ministry prepares and implements scientific-technical and innovation policies for high technologies. It works on fostering high technologies sectors, competitive and scientific goods and services, as well as promoting innovative entrepreneurship. The Ministry is authorized to conduct all necessary measures for the implementation of scientific-technical achievements, and to coordinate innovation activities for the analysis, development, implementation and improvement of high technologies. Moreover, it supports research projects on high technologies, improving their efficiency and implementing their findings in the relevant industries (nuclear industry, space industry, etc.). It prepares recommendations on implementing high technologies and scientific products in various manufacturing and industrial sectors. According to the recent amendments to the Statute, the Ministry governs the promotion of innovation activities, and assists local businesses in obtaining modern			
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	technologies and technological solutions. Moreover, it manages technology transfer, supports innovative scientific research projects, funds innovative projects (including startups), and promotes innovative entrepreneurship via the Agency for Innovation and Digital Development.			
Agriculture	The Ministry participates in advanced research projects, and implements their technical and technological achievements in the relevant fields. It manages the implementation of scientific-research projects, as well as efficient inventions and discoveries in the agricultural sector. It coordinates the activities of its own scientific institutions, and supports the development of the agricultural sciences and implements their findings.	The Agriculture Innovation Centre (AIC) under the Ministry of Agriculture aims to improve innovative agricultural solutions and boost access to innovations, promote technological advancements and methods, and support the implementation of innovative technologies. It also has a special division, the Technology Transfer and Innovation Division.	The scope of its activities and organizational structure allows it to work on TC activities.	AIC organizes various events and projects to support agriculture innovations, such as “Green innovation”, “AgroX” startup contest (with INNOLAND), and “AgroHackathon”. The winners get awards, but it is not long-term, well-design support program.
Finance	The Ministry participates in the development of investment and innovation programs funded by the state budget. It ensures the efficient use of resources dedicated to these purposes.	Not available	Not available	Not available
Culture	The Ministry is authorized to conduct scientific-research and experimental-construction projects in the fields of culture, archaeology,	Not available	Not available	Not available

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	etc. and it considers the opinion of ANAS in certain projects. The Ministry also conducts and supports projects related to the creative economy (“CREATIVE AZERBAIJAN”).			
Foreign Affairs	One of the Ministry’s duties is to coordinate and provide diplomatic support for Azerbaijan and its public entities in order to establish international and scientific-technical relationships. The Ministry also has the right to involve experts and scientists, as well as research institutions, in advisory services.	Not available (The Ministry of Foreign Affairs and embassies abroad may support “tech diaspora” initiatives, but such cases are rare and inconsistent.	Not available	Not available
Health	The Ministry authorizes and controls scientific-research projects in the field of healthcare. It has a right to develop scientific-research projects and implement their outputs. The Ministry also conducts R&D projects jointly with ANAS and directions.	Not available	Not available	Not available

Source: Authors’ elaboration.

By the Decree of the President of the Republic of Azerbaijan, the Technology Commercialization and Transfer Centre was established on 6 April 2021 under the Intellectual Property Agency.¹⁷ The Centre provides a link between science and industry, mutual order and supply relations, technology transfer, and the commercialization of intellectual property results in order to increase the quality and application of inventive activities, to support innovative development, including startup projects, and the commercialization of research results. It also provides services for the transfer of technologies and projects to target markets, and coordinates activities in this area, with the aim of transferring them for non-commercial uses, as well as generating income from these results.

The main tasks of the Centre are as follows: to assess the innovation potential of intellectual properties and assist with choosing ways of commercializing them; to create a database of innovative projects and science-based technologies; to obtain and transfer their rights for the acquisition and commercialization of intellectual properties; to monitor the results of research and innovation activities in research institutions; to provide advisory support in the creation, development and financing of small innovation enterprises (startups); to study market needs with regard to the use of intellectual properties; and, finally, to promote international cooperation in the field of transfer, mastering, and commercialization of innovative technologies, etc.

According to the decree the President of the Republic of Azerbaijan dated 11 October 2021, the Innovation and Digital Development Agency (IDDA) was established under the Ministry of Transport and Digital Development. The newly established agency is a successor to the National Nuclear Research Centre, Innovations Agency, and Research Centre of High Technologies¹⁸. Furthermore, since November 2022, the Ministry of Transport and Digital Development has seen a further increase in its authority with the transfer of the “E-Gov. Development Centre” from the State Agency for Public Service and Social Innovations to its IDDA unit. Furthermore, the Ministry is now authorized to implement state policy and regulation, as well as to organize and coordinate the relevant activities for e-government in

¹⁷ <https://e-qanun.az/framework/47176> (accessed date: 22 May 2023)

¹⁸ <https://e-qanun.az/framework/48367> (accessed date: 22 May 2023)

Azerbaijan¹⁹. According to its charter, the IDDA is in charge of organizing, coordinating and implementing *digital development*; formulating and implementing complex measures for *high technologies*, nuclear science, nuclear technologies, and nuclear energy; and carrying out *innovation activities* to support the development of modern technologies and technological solutions, as well as to promote and fund innovation-oriented academic research innovative projects (including startups) and innovative entrepreneurship.

More recently, the President of the Republic of Azerbaijan signed several documents that consider certain measures relating to the ICT sector. Accordingly, the measures include conditions pertaining to the residents of a techno park. Since January 1, 2023, the IDDA has implemented the process covering applications for, and the assessment and granting of residential status at, its techno park. Residents performing system integration, software preparation and development activities operating outside the technology park will be eligible for the following privileges:

- Exemption from certain taxes (income and profit tax, dividend tax, property tax, and land tax).
- Income tax incentives (0% on income up to AZN 8000 and 5% on income exceeding AZN 8000; from 2026, 5% on any income for the next seven years).
- Exemption from work permit conditions for foreign specialists.
- Depending on the employee's choice, payment of mandatory state social insurance equivalent to an amount that is four times the minimum wage, or income from a paid job for the next ten years.²⁰

With this reform, the residents of techno parks, regardless of their location (outside or inside), will receive the exemptions and incentives provided by the government.

Industrial and technology parks in Azerbaijan are usually granted fiscal and administrative incentives to manufacture high-technology goods. Most of them are carried out within specific government programs in dedicated industries and, typically, they are not interested

¹⁹ <https://e-qanun.az/framework/52788> (accessed date: 22 May 2023)

²⁰ <https://www.idda.az/az/projects/our-projects/tecnopark> (accessed date: 22 May 2023)

in commercializing new research outputs *per se*. Such projects might be knowledge-intensive, but they mostly originate from market-leading firms that agree to sell their know-how to local partners to produce goods in the industrial parks.

The HTP is the most fit institution for this purpose and might be considered the most representative TC program in Azerbaijan.

Generally, the residents of techno parks and industrial parks in Azerbaijan obtain fiscal incentives such as tax exemptions.

Another useful project of the IDDA is the “Ecosystem Map”, an information portal about all the participants in the innovation ecosystem of Azerbaijan. The portal serves to disseminate information about innovative developments in Azerbaijan not only for the domestic audience but also for interested entities beyond its borders.

The IDDA also supports relocation programs, including Technest, which provides education on ICT skills, and Technion, which provides education on cybersecurity, etc. It will also cooperate with Nokia on improving digital literacy and establishing a lab and network in the capital, Baku. Nokia will also work together with Azerbaijan Technical University to establish an R&D Centre.²¹

Under a cooperation initiative with local universities, the IDDA and Azerbaijan State University of Economics (UNEC) recently announced an appeal for applications for the “UNEC Startup Cup” which will award AZN 20,000 to the winning startup.²²

Furthermore, the IDDA organizes the “IDDA Awards” under the nine categories listed below for the purpose of promoting and supporting the most promising innovators, stimulating competition among stakeholders, and raising awareness:

- Startup of the year: *Push30*
- ICT entrepreneur of the year: *Ms. Gunay Imanzade*

²¹ <https://www.bakupost.az/nokia-bakida-laboratoriya-acir> (accessed date: 22 May 2023)

²² <https://inc.unec.edu.az/unec-startup-cup-az/> (accessed date: 22 May 2023)

- Invention of the year: *AYVA*
- Locomotive of the ecosystem: *Mr. Tariyel Aghazade*
- Ecosystem player of the year: *Innoland*
- Digital project of the year: *Labour and Employment Sub-system*
- Technology media of the year: *Technote*
- Investor of the year: *n/a*
- Capability power of the year: *Innovations Centre LLC ("Skills for the Future")*

So far, ninety-four startups have received the Startup Certificate from the Ministry of Economy. However, the certificate does not guarantee the receipt of any funding as such: Rather, it aims to grant fiscal incentives in cases where an applicant succeeds in obtaining funding from another source by itself. Two of the granted startups (namely: Push30 and SmarTap) recently had their Startup Certificate deactivated due to their successful growth, and both of them immediately became regular taxpayers under the existing legislation.²³

The "StartUp School" project has been created to turn the ideas of more young people into reality and support them along the way through cooperation with the "Enterprise Azerbaijan" portal of the Centre for the Analysis of Economic Reforms and Communication and the Ozmen Centre for Entrepreneurship of the University of Nevada, USA. The main goals of the project include, but are not limited to, supporting startups with academic knowledge and fostering a spirit and culture of entrepreneurship. It also has the mission of "assisting startups with presenting their projects to investors, enabling them to enter the Global Startup Ecosystem, and attracting alternative sources of funding to the country's economy". Moreover, Sabah.Lab provides a 15-day training program in conjunction with the project. Sponsored by the IDDA, the project participants may obtain the following benefits:

- An opportunity to participate in I2B training in the USA (for the 1st place winner).

²³ <https://xeberler.az/new/details/startap-sehadetnamesi-alib-dovriyyeni-artiran-startaplar-derhal-vergiye-celb-edilib--30079.htm> (accessed date: 22 May 2023)

- Inventory support from the State Employment Agency for the 1st, 2nd and 3rd placed winners and "The Best Idea" category.
- An opportunity to participate in the acceleration program of SABAH.HUB for the 1st, 2nd and 3rd placed winners.
- Provision of accelerated support by KOBIA for applications for a "Startup Certificate" for the ten best projects.
- The team selected under the "Fintech" category received an opportunity to have an internship at PASHA Bank.
- Gifts were presented by DIV Academy.

Universities, business incubators and accelerators are regarded as some of the key drivers of TC activities. Currently, various universities are actively running startup incubation activities. A few years ago, Azerbaijan State University of Oil and Industry and Baku Engineering University (formerly, Qafqaz University) were pioneers in hosting techno parks and incubation centres. However, they became relatively more sluggish in this respect due to certain internal factors. In addition, it is possible that the COVID-19 pandemic also had a negative effect on such programs at universities.

As mentioned in the previous report in 2022, there is no TLO at the higher education institutions operating in Azerbaijan. However, some universities have departments that conduct small-scale (primary activities) activities, such as a technology park, technology transfer office, or business incubation centre. It has been observed that the patenting of inventions created at universities is mainly carried out on the personal initiative and efforts of the inventor, although in some cases the university administration is closely involved in the process. Azerbaijan State University of Economics operates the UNEC INC startup unit, which has just begun cooperating with various private and public organizations, including the IDDA. ADA University is also enthusiastic in this field and plans to empower its Innovation Centre soon, while Azerbaijan State University of Oil and Industry and Azerbaijan Technical University have shown strong technological and scientific potential.

After the COVID-19 pandemic, universities once again reinforced such activities, with the following universities running ahead of the others in comparison:

- Students of the French-Azerbaijani University are very active in the startup ecosystem. This university regularly organizes innovation-related competitions and its students have received awards in various local and regional contests, such as the “TEKNOFEST”.
- The “E-azi” Startup Incubation Centre at Azerbaijan State University of Oil and Industry has adopted an incubation program from the “Skills for the Future” initiative (described below). They are eager to attract private sector partners to this process.
- Azerbaijan University of Architecture and Construction as established a functional dedicated department and incubation centre. Its students also participate in various contests in the fields of engineering, technology, etc.
- Azerbaijan Technical University is one of the most dynamic universities and has a strong mentor base. Its students also participate in global competitions, especially those in the fields of robo-technics, transport, design, etc.

An initiative called “Skills for the Future,” created under the Innovations Centre LLC, has been working with local universities to train mentors for innovation-related activities at each partnering university. The initiative has two main components: ²⁴

- Training university instructors to become mentors in idea-to-business processes through eighteen modules (basic skills) and a three-day boot camp (interactive specialized programs). Once trained, these mentors go on to help students on incubation programs currently being operated at thirteen local universities.
- The “Tusi (2C)” incubation program supports universities in their efforts to prepare students in startup development. It has three tools – training (18 modules), execution (2-3 months for prototyping), and a pitch-day for funding.

So far, approximately 500 students have participated in the training programs on “Skills for the Future”. The initiative has attracted 135 university instructors from 13 universities, who currently mentor their students on the “Tusi 2C” program. Overall, more than 500 training

²⁴ <https://www.facebook.com/gelecekuchunbacariqlar> (accessed date: 22 June 2023)

sessions have been carried out under this initiative.

INNOLAND has been established under the State Agency for Public Service and Social Innovations as an incubation and acceleration centre. It offers the following four services:²⁵

- “Incubation” (3-6 months) aims to support startups in the early stages in order to bring their ideas into MVP. Currently, INNOLAND does not accept applications.
- “Accelerator” (3 month) aims to help startups grow (business development, pitching, etc.) locally and then expand to the global level.
- Innoland has modern and comfortable coworking spaces available 24/7 with office equipment, lockers, high speed Wi-Fi and additional benefits of unlimited tea/coffee.

Virtual Residents of “INNOLAND” will get free access to all events of “INNOLAND”, opportunity to participate at “ASAN Learning” sessions, free e-OFFICE for financial reports and will get mentoring support for their startups.

- The “IT Training and Education Centre” is a specialized unit that teaches programming and bootcamp courses.

SABAH.Lab is a startup acceleration centre supported by the Ministry of Science and Education as a part of the Education Institute (under the Ministry itself). The Centre has become active in attracting the young generation of startups and collaborates with large private enterprises. Its main objective consists in creating and supporting early-stage companies through the following learning programs: ²⁶

- “Accelerate” (5 months) aims to accelerate teams through the MVP stage and prepare them for investors.
- “Build” (6 months) aims to help teams through ideation, analysis, and MVP creation.
- “SABAH.incubation” (3 months) aims to help people form a startup idea and team and then proceed to the next stages.

²⁵ <http://innoland.az/en> (accessed date: 22 May 2023)

²⁶ <https://www.sabahlab.edu.az/> (accessed date: 22 May 2023)



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The Centre designs its educational programs in collaboration with the IE Business School, Europe's number one school of innovation and entrepreneurship. The curriculum includes the participation of several professors and industry leaders from local and international universities. Since its establishment in 2021, SABAH.lab programs has supported around ten startup teams for each program.

3. Micro-Level Analysis of Operational and Managerial Issues at the HTP

According to the Road Map of the HTP, its ultimate goals are to conduct scientific research and experimental projects and to provide an advantageous environment and support for the residents in their efforts to commercialize the outputs of those research and experimental projects in various industries. In order to achieve its goals, the development concept/improvement plan of the HTP adopted some of the following, but not limited to, targets:

- Provide an advantageous environment and support the related entrepreneurship promote competitive and science-intensive development based on innovations and high technologies.
- Promote and implement the commercialization of innovation, scientific research, experimental works, inventions and discoveries, and technologies in Azerbaijan.
- Commercialize and promote residents' inventions and startup and business ideas with sales potential.
- Support the development of the innovation ecosystem in Azerbaijan.
- Support SMEs in high technology sectors.
- Attract science-intensive high technologies and technological products to the market.
- Increase the share of science-intensive products in the economy and the transfer of high technologies.
- Develop competitive, export-oriented and import-substituting high technologies, and create a better environment for their transfer to customers in commercial terms.
- Forge partnerships with the private and public sectors, universities and research institutes in order to develop joint projects.
- Support the creation of cutting-edge technologies and IP objects, and promote their implementation, etc.

There exist a number of conditions that are necessary for the HTP to achieve the abovementioned targets. They include, but are not limited to, the following:

- Identify a broader set of incentives through legislation.
- Improve the territorial, road-transportation, management and social infrastructure of the HTP.
- Improve the consistent and sustainable provision of the basic utility infrastructure for the HTP.
- Build modern office areas for startups, business incubators, and residents.
- Create a better environment in which large companies can develop innovation projects for a better TC.
- Expand access to international markets.
- Learn from the development experience of advanced countries and use it in the growth strategy of the HTP.
- Promote exchanges of experience with different countries to improve the nationwide ecosystem.
- Support residents in their efforts to establish and grow their SMEs.
- Establish an advantageous business environment that attracts investment at various stages of growth.
- Align TC activities - along with the implementation of innovation projects, experimental works and prototypes - with international cooperation and grant projects, including attracting potential donors.
- Establish an advantageous environment and organizational arrangements for funding, investment, and the promotion of innovation projects.
- Develop the potential of specialized human resources and a technology incubator to train entrepreneurs in innovation management and innovation marketing topics, and improve their legal, economic and managerial expertise.

- Establish a scheme for public procurement of the products manufactured at the HTP with private-public partnership frameworks.
- Strengthen collaboration with higher education and research institutes, etc.

If these necessary conditions are not put in place, it will be challenging for techno parks to achieve their targets, and they may underperform, even making it impossible in worst-case scenarios.

As mentioned in the previous report in 2022, certain privileges are available for the Managing Organization and the residents of the HTP, and these privileges create the conditions for increasing the efficiency of interaction, attracting the interest of investors, and inducing the production of competitive products.

Residents of the HTP can take advantage and exemptions listed below:

- Income Tax
- Value-added tax
- Property tax
- Land tax
- Customs fee

Such incentives will allow the HTP to offer beneficial conditions to its residents and earn a certain amount of revenue from their operations. Residents can decrease their costs and enjoy these benefits for seven years after registration.

The HTP and its TC activities were supported by the Azerbaijan National Academy of Sciences (ANAS) until it underwent structural reforms in 2022. Currently, the HTP is operating under the Ministry of Science and Education, and therefore is supported by this ministry. The HTP could seek to cooperate closely with the Ministry of Economy in startup and SME related projects, and with the Ministry of Transport and Digital Development in connection with innovation and new venture projects.

The HTP also cooperates with public, private and international partners in creating better

conditions for TC activities. Stakeholders include research institutes, universities, techno parks, companies, innovation centres, etc. For example, the HTP presented a project proposal to the Erasmus+ program framework together with several public universities and private companies. Within the project to “Implement project-based training methods at universities in Azerbaijan: promotion of innovation and entrepreneurship among students and scholars” (coordinated by Leipzig University, Germany), the HTP participates with other local and foreign universities. TTCP is also a part of this project. The main purpose of the project is to improve the practical aspects of higher education in Azerbaijan by learning from European experiences. It also helps to develop university-industry relations and raise awareness of innovation and entrepreneurship among students. The HTP also participates in grant projects and programs (e.g. Horizon Europe) established for techno parks and incubation centres.

The HTP also cooperates with a variety of Turkish partners, including Cyberpark at Bilkent University, METU Technopark at Middle East Technical University, Teknokent at Istanbul Technical University, Özyegin University, Istanbul Koc University, Yildiz Technical University, Cappadocia Technopark, and Bilishim Valley, as well as the Turkish Aerospace Industry Company and the Digital Transformation Office of the President Administration of Turkey. Such partnerships include practical learning from Turkish experience of operational mechanisms for techno parks, TT and TC processes, business incubation, IP management, and academia-industry cooperation, as well as participating in international projects and programs together. For example, HTP, SABAH.lab and Cappadocia Techno Park have been working on a joint incubation program with the purpose of developing an innovation environment that responds to local needs and local content in the region of Azerbaijan, including efforts to promote cooperation among researchers, startups, investors, industrialists, and regional universities and to increase the number of startup projects. The HTP and SABAH.lab jointly initiated the organization of an international conference on “Innovation culture in Turkic-speaking countries” with the aim of boosting cooperation among these countries, identifying priorities, introducing techno parks and innovation centres, and supporting their national innovation strategies.

Currently, the HTP is not engaged in any active relations with these organizations; however,

it is envisaged that such relations will eventually be established. Of course, the HTP considers this kind of cooperation to be real and values it as an important element in the development of the innovation ecosystem.

Currently, negotiations are ongoing with the Cappadocia Techno Park regarding the development of an incubation program. The HTP intends to implement such a program with the participation of Turkish trainers. In the framework with Cappadocia University HTP will apply for Turkish and Azerbaijan finance recourses.

Sabah.Lab is a centre within the Ministry of Science and Education, and the HTP will most likely run an incubation program with them. A meeting has already been held in the Techno Park with the participation of managers. Sabah.Lab is funded by the state, so it is likely that this collaboration will be funded by the Ministry of Science and Education.

Within the framework of the Erasmus+ project, the project for the "Application of project-based learning methods in Azerbaijani universities: promotion of innovation and entrepreneurship among students and teachers" has already been presented. Three European universities and ten Azerbaijani universities, as well as the HTP and private organizations, are represented in the project. The aim of this project is to improve the quality of teaching and research methods with integration-based, project-based learning methods and European higher education experience and apply them to the relevant Azerbaijani universities. The Erasmus Project will be supported by the European Commission.

The HTP is currently working on re-establishing its membership of the International Association of Science Parks (IASP) as this will potentially increase opportunities for further cooperation. The HTP also represents Azerbaijan at the "Regional Forum on Sustainable Development in the UNECE Region" organized by the United Nations.

Although it can be considered the most representative TC program, the HTP is an underdeveloped and underperforming institution, making it unsuitable to conduct TC programs alone. Since it is responsible for implementing such programs, it faces strong challenges in delivering its goals due to certain shortcomings, such as limited authority, frequently changing leadership, and organizational restrictions.

Perhaps the most important partnership for the HTP is its long-term cooperation with the STEPI within the K-Innovation Partnership Program 2022-2024. Under this program, the Technology Transfer and Commercialization Platform (ttcp.az) has been established to promote and support TT and TC activities in the innovation ecosystem of Azerbaijan, and to develop it in accordance with global and local demands. It also aims to achieve consistent and sustainable development of intellectual capital and high technologies. Currently, the platform is in its last phase of development before release. So far, various enterprises, researchers, patent attorneys, startups, and participants from other categories have been registered in the portal and have submitted their list of services, solutions and projects.

Ongoing discussions and meetings regarding TTCP continue, and some additional changes are being made in terms of content. Currently, the process of placing patents continues. There will be changes in the design of the platform. The project experts regularly meet with the CODERS team to discuss the optimization of TTCP.

Moreover, under the project titled “The Operational Program for the HTP: implementation and capacity building” conducted jointly by the HTP and STEPI, it has established partnerships with several private and public organizations. Representatives from the Intellectual Property Agency, Agency for Development of Free Economic Zones, Education Institute, Baku State University, Azerbaijan State University of Oil and Industry, Azerbaijan Technical University, Baku Engineering University, Academy of Public Administration, “New Ideas” LLC, Institute of Control Systems, and many other organizations have attended online and offline training programs conducted by Korean experts.

■ Improvement of operational and managerial issues

Since its establishment in 2016, the HTP has gone through various stages of development and experienced fluctuations in its fortunes. Fortunately, it provides strong research opportunities and fiscal incentives for its residents. On the other hand, the frequent changes in leadership, sluggish and old-fashioned image of ANAS, inappropriate location in a factory zone outside Baku, and a lack of vision for development can be regarded as shortcomings in the path of the HTP. Additionally, the HTP is one of the most passive members of the innovation ecosystem of Azerbaijan in terms of visibility and marketing.

Considering the status reported above, as well as the real challenges lying ahead, the HTP must improve its operational and managerial capabilities by focusing on the following critical dimensions:

■ Needs assessment

Ever since its inception, there have been continuous discussions about the necessity for the HTP to succeed in the ecosystem and achieve its intended targets. However, a detailed needs assessment, ideally conducted by an independent third-party, has never been conducted to grasp the real picture and calculate the appropriate budget and allocation of resources. A needs assessment would enable identification of specific operational issues that need to be improved to ensure for a better TC. A comprehensive needs assessment may focus on infrastructure, services, management, and other areas. A needs assessment should also be interpreted from the viewpoint of the HTP's strategic vision.

■ Improvement plan

Although the HTP has prepared a Road Map for future growth, this document consists of a list of targets that might seem rather vague and ambitious while lacking a proper initial assessment of its needs. Therefore, based on the results of a needs assessment, a detailed improvement plan that outlines the specific actions required to address each issue is critical for successful TC. Implementation plans also allocate tasks and set timelines and milestones for their completion.

■ Infrastructure and services

Successful TC requires state-of-the-art infrastructure to support the needs of stakeholders. Such an infrastructure would include facilities, equipment, manufacture spaces, and work spaces, etc. Moreover, services such as networking events, mentorship programs, and access to legal and financial advice are also crucial for TC. From this point of view, the HTP faces huge challenges. For example, its location is less interesting for private firms and individuals; and although its infrastructure is sufficient for experimental-production works, it is not attractive as a modern business incubation and acceleration centre; nor does it have a reputation for strong mentorship and networking capabilities. Therefore, depending on

its improvement plans, the HTP should consider how to become an attractive hub equipped with modern infrastructures and services.

■ **Monitoring and evaluation**

Assessing needs and then developing implementation programs is not a sufficient way to proceed. Therefore, regular monitoring of the progress of the improvement plan and an evaluation of its effectiveness is critical for TC, as this would allow stakeholders to adjust their plans as necessary to ensure that the intended outcomes are achieved. The HTP needs to establish an effective monitoring and evaluation system for its own scope and, perhaps, for its residents as well. This would allow it to become more flexible and agile and to improve its growth strategy in a timely manner.

■ **Timing**

The HTP has already been in operation since 2016, and this period has witnessed massive changes in both industries and technologies. Therefore, in order not to miss critical opportunities, timing is a critical factor for the purposes of TC. It is important to assess the market trends and the readiness of technologies before entering the market. Time is pressing upon the HTP, and it may also affect its political credibility as a reliable and efficient public entity with a broad range of fiscal incentives. Therefore, it should immediately assess the market potential locally and globally, connect with potential customers, and compete against other providers of TC.

■ **Business model**

A good business model that evaluates the market, competition and other factors is essential for the successful commercialization of technologies. Because of the abovementioned factors that the HTP needs to improve, there is no clear business model for its TC activities. The founding purpose of the ANAS HTP (currently, the total budget amounts to about USD 300 000) was to gradually decrease its dependence on the state budget so that it could become a self-financing, not-for-profit entity in the long term. To achieve that goal, however, it needs to build diverse and sustainable revenue streams.

■ Team

A strong team armed with the right skills and expertise is crucial for the successful commercialization of technology. Frequent leadership changes at ANAS itself, as well as at the HTP (currently 50 employees), have led to a misallocation of human resources and a high turnover, which in turn has negative effects on the performance and continuation of strategies, action plans, and projects. HTP involves improving the activities of people with relatively lower wages and qualifications by involving them in specialized capacity building programs.

■ Cooperation and collaboration

Cooperation and collaboration are critical elements for promoting and developing TC activities among government, academia, and industry. Policy makers should aim to encourage international partnerships (e.g. overseas development assistance, etc.) and to develop a culture of collaboration and knowledge sharing. As far as the HTP is concerned, this would provide opportunities for knowledge-sharing and networking benefits, raise awareness in broader communities, and improve TC if considered and implemented attentively. It could also result in joint R&D projects, shared marketing, a stronger sense of community, and professional development and capacity-building programs, etc. The HTP could also consider joining different associations, networks and consortia that are currently working on similar technologies or operating in inter-related industries.

4. Conclusion

This Report on the Status of the K-Innovation Partnership Program 2022-2024 between STEPI and the HTP can be considered a follow-up to the previous status report submitted in 2022. This report mainly concentrates on the updates that have occurred since July 2022, while considerable institutional reforms have been implemented within ANAS. Numerous institutions, mostly in the field of natural and technical sciences, have been transferred within the structure of the Ministry of Science and Education. As for ANAS, it mostly manages institutes in humanitarian and social sciences. Most importantly for this project, the ANAS HTP is no longer a unit of ANAS, although the interagency transferring process is still ongoing and needs to be finalized.

Another significant event entails strengthening the authority of the Ministry of Transport and Digital Development as a central power to govern the innovation ecosystem, digital transformation, and e-governance. All these responsibilities are currently assigned to the Innovation and Digital Development Agency. The HTP would benefit from closer collaboration with the IDDA and active participation in the innovation ecosystem.

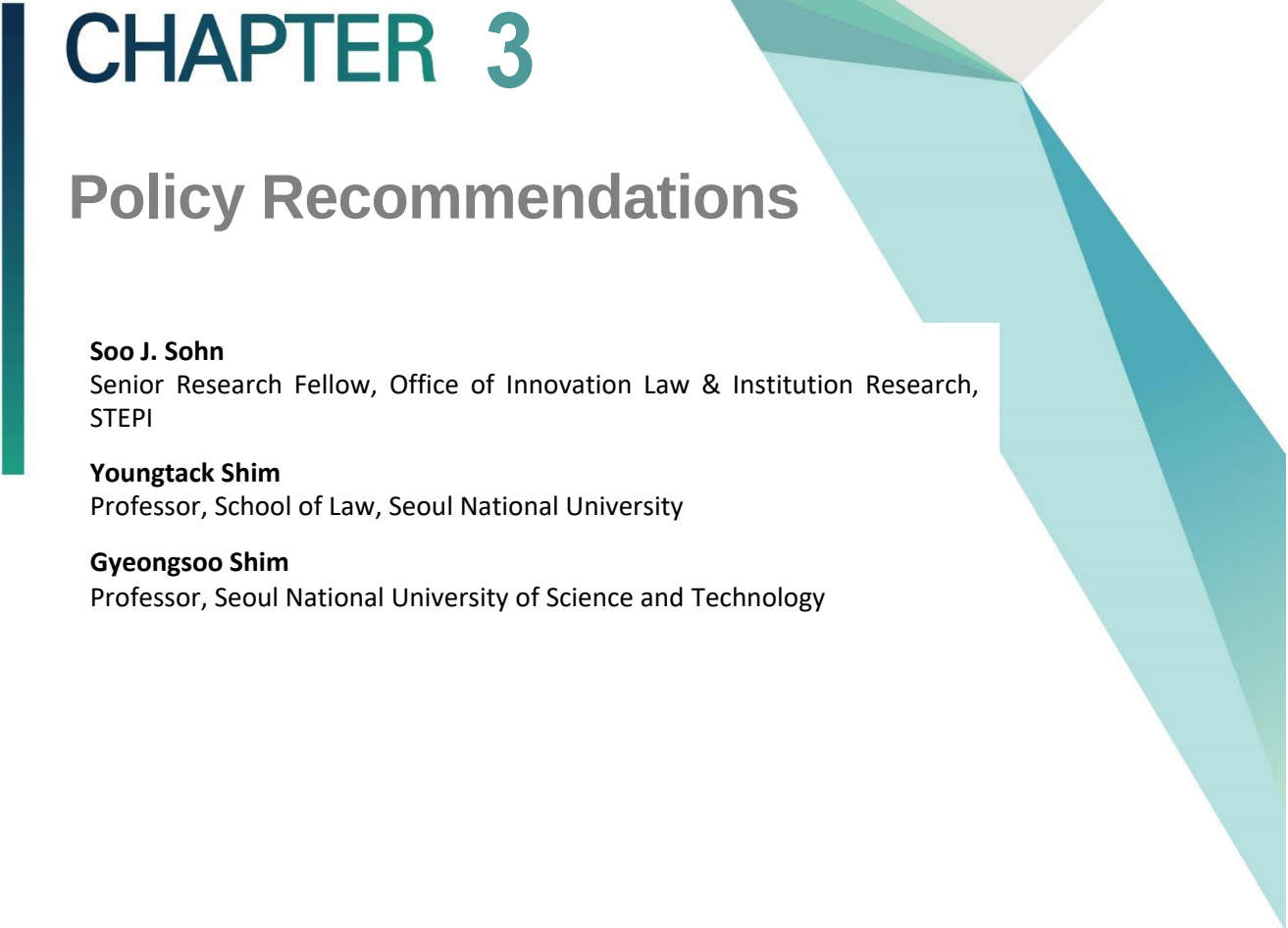
So far, Azerbaijan has not introduced any legal acts on competition, venture capital, etc. Moreover, a draft of the national innovation strategy needs to be approved in order not to miss the right timing or lose out on the market potential. Equity and debt financing, and crowdfunding are underperforming forms of funding mechanisms in the country. Furthermore, the unavailability of venture capital firms in the domestic market makes the creation of more dynamic markets and financing an even more challenging task.

We believe that the process of re-formation of HTP will take place after it becomes part of the Ministry of Science and Education. The visits of the minister and other management staff to the HTP, and their respective ideas, indicate that the HTP will once again be formed into a modern technology park.

Regarding the need to improve the HTP's operational and managerial issues, it will soon conduct a needs assessment and develop improvement plans based on the findings of the

assessment and its own strategic aspirations. Deploying a solid business model will surely have a positive impact on the sustainability and profitability of its operations. Moreover, it is important for the HTP to strengthen its team capacity, monitoring and evaluation system, and participate in the innovation and TC ecosystem more actively.

The K-Innovation Partnership Program is a valuable opportunity for Azerbaijan, and for the HTP in particular, to develop a far better and more conducive environment for technology commercialization.

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CHAPTER 3

Policy Recommendations

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Chapter 3. Policy Recommendations

1. Technology Commercialization Ecosystem

1.1 Governance for Technology Commercialization

1.1.1 Structure

The bridge between technologies and markets (value added) is a major engine of economic growth. Therefore, to design the path for technology commercialization is necessary strategy for having national compatibility.

To acquire national growth engines, the Ministry of Science and Education should establish the vision and policy for technology commercialization. The HTP, which is charged with that mission, should assign the size (manpower, budget, etc. and the role of division of technology commercialization (hereinafter referred to as “Division”), which is in charge of commercialization. Division is basically needed three parts-infrastructure, startup, collaboration- at least. (i) In the part of infrastructure, it has a role on IPRs, education, information(TTCP), etc. Especially, it should establish national guidelines for the creation, protection, and use of IPRs. Furthermore, it needs to operate manpower training programs and certify experts (e.g. traders, evaluators, etc.) in the field of commercialization. The most important thing is to upgrade and activate the data platform, TTCP. (ii) Startup support must provide hardware and software for startups. In this regard, hardware refers mainly to facilities, equipment, and spaces, while software refers to startup support programs. (iii) Recently, the issue of cooperation has become prominent. In particular, as the shocks caused by innovation such as digitalization, climate change, and infectious diseases have increased, it has been realized that innovation is impossible without cooperation. Therefore, a part for inbound and outbound collaboration has become more important. Above all,

collaboration with the related ministries is important to ensure the efficiency and effectiveness of the policy. And to encourage and design the national policy direction should set a basic national plan with relevant ministries. In addition, cooperation with neighbouring countries to share resources and form markets is necessary. For example, neighbouring countries could also build TTCPs and increase the amount of available IPR information through connections between TTCPs.

Above all, to achieve these missions, the Division must secure experts and be able to carry out sustainable programs.

1.1.2 Role

1.1.2.1 Technology Infrastructure Part

In the field of technology commercialization, the main role of the government is focused on establishing the basic infrastructures. Diverse factors or components can be included within the scope of infrastructure, such as laws, regulations, institutions, hardware (facilities, spaces, etc.), manpower, information networks, and so forth. Therefore, the infrastructure part must be able to set focus areas by year.

The most important infrastructure in its current state is the information network, so activating TTCP is important. In addition, it is important to secure professional personnel for the planning, implementation, and management of technology commercialization. Such personnel can accumulate knowledge, know-how and expertise through learning and experience. Thus, the government must provide specialized training courses for them. In reality, the most important area in which the government and academia should cooperate is R&D aimed at creating good IPRs. To that end, it is necessary to establish an IPR system-education, monitor and provide consulting for inventors, provide incentives, evaluate the technology quality level, and establish an international IPR database, etc. Furthermore, long-term planning and huge investments in the construction of the system will be required.

1.1.2.2 Technology Startup Part

Startups are a major driving force in encouraging economic growth based on innovation. Therefore, finding and matching startup ideas, providing education on entrepreneurship, financing seeds and early stages are the most important policies for creating jobs and value-added.

Therefore, the main role of the part is constructing hardware, such as maker spaces, incubation centres, and facilities, etc. and managing the related programs, such as academies (I-corps) and events for entrepreneurs, etc.

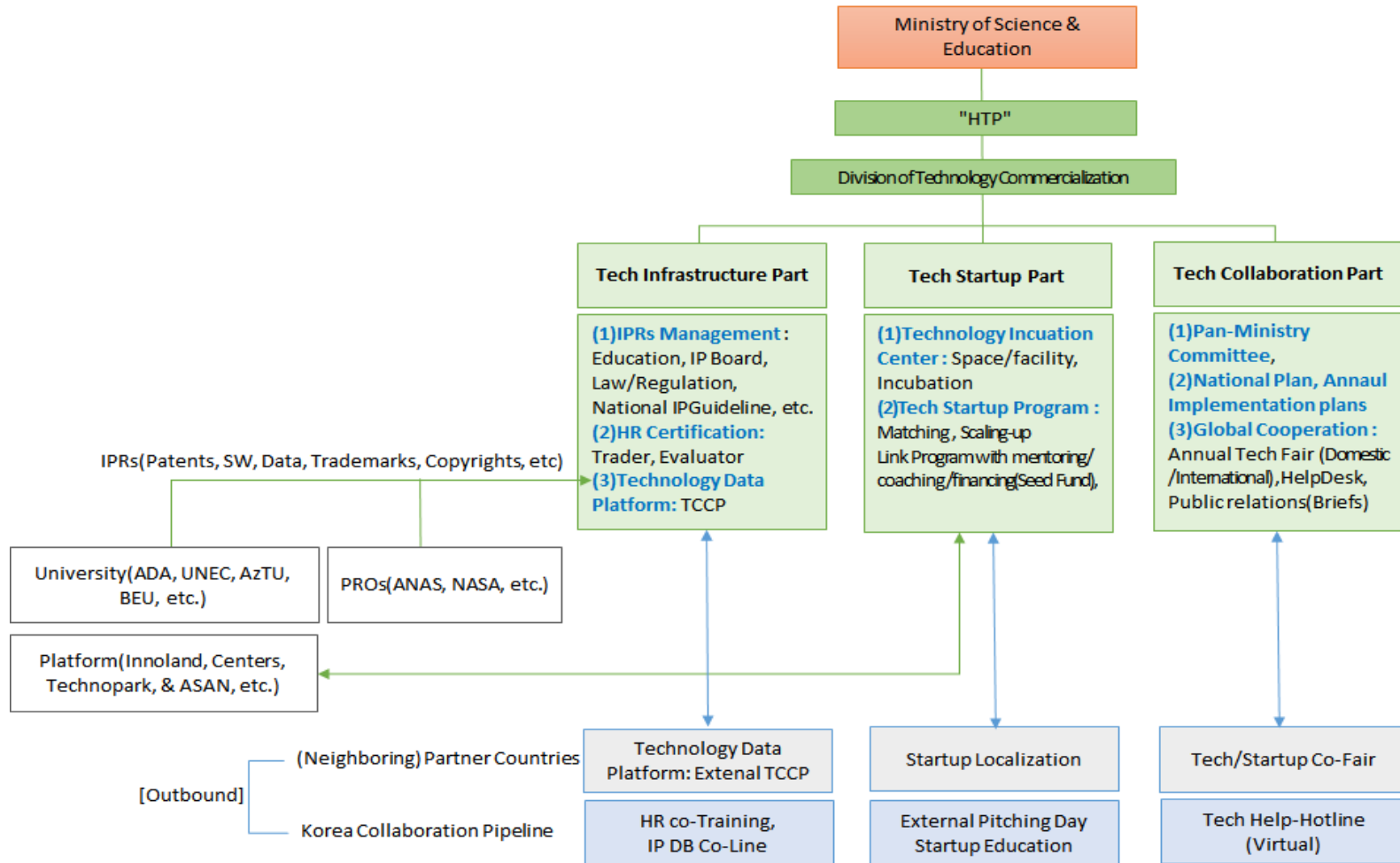
1.1.2.3 Technology Collaboration Part

The era in which innovation is accomplished by one entity in one field is over. Nowadays, many actors must cooperate regardless of the sector, department, or national boundaries. Therefore, the main role of this part is to provide programs designed to encourage various cooperative activities among innovation entities both domestically and overseas.

The Division should lead the design of a basic plan for pan-ministerial technology commercialization every three years. To do this, first, a committee should be formed with the relevant ministries and agencies or people in charge. This committee should be given authority and responsibility in establishing national commercialization strategies, and should appoint a group of experts to consider a wide range of factors - including the innovation environment, digital transformation, sectors, startups, technology financing, IPRs, incubation, innovative clusters, facilities, and other commercialization-related factors - at the national level.

The Division also presents the Ministry of Science & Education's annual implementation plan. And this part should monitor its process, performances, and limitations.

[Figure 3-1] Governance of Technology Commercialization Programs



Source: Author's elaboration.

1.2 Institutions

1.2.1 Infrastructure: IPRs Data platform, TTCP

1.2.1.1 TTCP promotions

The HTP provides a variety of programs designed to inform researchers and companies about the use process and expected benefits of TTCP in an easy and convenient way. Basically, it is expanding active methods, such as video production, access via apps and information mailing services, for patent applicants.

By accessing TTCP, users can check the process of finding a desired technology, the transfer process of a given technology, the nurturing process of the transferred technology, and the expected effects. In addition, TTCP pays users points based on usage, and allows the points to be used for related education, information or service acquisition, etc.

These overall activities are not limited solely to domestic users, but also include potential users from neighbouring countries.

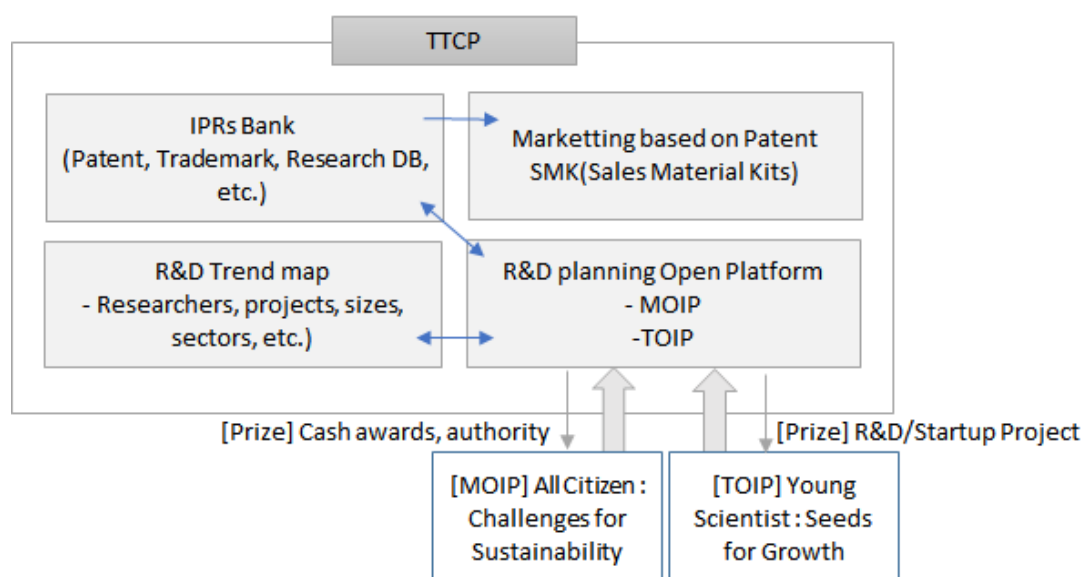
1.2.1.2 TTCP functions and TTCP as a R&D planning open platform

TTCP must be upgraded to perform not only the functions of an information accumulation platform and IPR bank, but to serve as both an information hub on R&D trends and patents and as an R&D planning open platform. In order to provide patent information, SMKs (Sales Material Kits) for major patents should be provided to increase consumer understanding.

R&D planning open platform can be approached in one of two ways: MOIP (Mission-oriented IP) or TOIP (Target-oriented IP). First, MOIP is a way to frame the social issues facing Azerbaijan as a 'mission' and to invite all citizens to submit research topics aimed at solving them. A contest is held on social issues that the general public needs, such as the pollution of the residential environment or the Caspian Sea. Then, the selected topics can be planned as public R&D projects. The research team is encouraged to participate as a joint consortium of multiple entities rather than as a single team. In cases where the project based on the selected topic is designated, the proposer may choose between either a cash award or a share of the research results as a prize. Then, a virtual room for TTCP will be opened for the project; and innovators, investors, and accelerators who are interested in the topic can

enter the room and share information with each other, while the proposer of the idea can monitor the process of the project. Second, TOIP is for the topic of technological development to create value-added from Azerbaijan's natural resources. Examples include petroleum resource-based applications, food crop utilization, and such like. Therefore, the selected topic is linked to entrepreneurship programs as well as R&D projects, and focuses on young scientists equipped with knowledge and awareness of new technologies. In addition, an expert committee selects areas with high impact from a market and job-based perspective, and supports the related IP pooling and market planning. As regards IP pooling, the possibility of licensing or receiving (technical assistance) necessary technology from outside can also be considered.

[Figure 3-2] Expected Functions of TTCP



Source: Author's elaboration.

1.2.1.3 T-Day

"T-Day" is a networking event for TTCP members and prospective members. Each year, a specific date is designated as 'T-Day' and used as a festival for innovation entities. On the day itself, awards are presented to those who have achieved various activities through TTCP that year.

1.2.2 Infrastructure: Manpower Training and Certification System

1.2.2.1 Manpower Training

The process needs to be run in one of two ways, i.e. targeting existing personnel or targeting new (prospective) personnel.

First, training for existing personnel is mainly aimed at TLO personnel. Among the public research institutes and universities under the Ministry of Science and Education, some institutions with R&D awareness and the related resources have established dedicated organizations such as the TLO. These organizations have dedicated personnel, but the scope of knowledge or networks with which they must be familiar is extremely wide, making continuous and regular education and training necessary. As such, regular programs must be organised to provide education on technology nurturing, technology finance, legislation and international laws and treaties, regulations, contracts, and market planning, etc. Additionally, the formation of networks of TLO personnel can be expected through these programs.

As another program, the pre-expert training course may be understood as an internship course. It provides short-term courses on TLO work for students majoring in engineering, economics/management, and law, etc. This course, which runs for one semester or the vacation period, will be available not only domestically but also in other countries upon agreement. In particular, cooperation programs involving countries with similar cultures or innovation environments are also important because this may make it possible to exchange human resources and promote interactions between them in the future. Through this process, the position of ‘technology commercialization expert’ can be presented as a career path for highly skilled personnel from Azerbaijan’s universities (graduate schools). Technology commercialization experts generally include ‘technology traders’(who perform specialized work such as technology commercialization consultation, advice, guidance, brokerage, mediation, research analysis, exploration, and discovery, etc.) and ‘technology evaluators’(who perform specialized work such as technology valuation, technology-based finance, etc.). Such a position can also be extended to include accelerators and company builders.

1.2.2.2 Manpower Certification System

South Korea has a 'technology trader' system that certifies qualifications in the relevant field for experts with specialized knowledge and experience. Since 2001, some 5,000 technology traders have been registered in Korea. Personnel who meet the appropriate conditions can apply for a certificate and are granted the qualification after undergoing a screening process, but they must complete training provided by a dedicated organization on a regular basis. These personnel play a professional role when universities, research institutes, companies, etc. participate in government projects for technology commercialization, and they also participate in government project planning to convey the needs of the field, as well as participating in arbitration, negotiation, consultation, and contracting for technology transactions conducted in the private sector.

The HTP is in charge of certification and manages the relevant certifications, while the Ministry of Science and Education is responsible for certifying the qualifications. To do this, a national model must be designed, including the underlying laws, application qualification criteria, certification status, examination procedures, curriculum, and management plans.

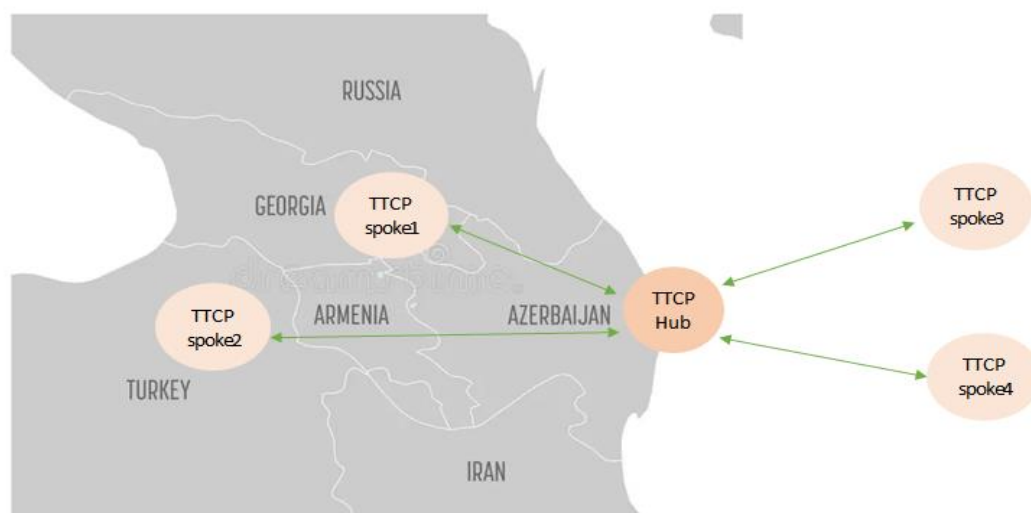
1.3 Cooperation with Outbound

1.3.1 Neighbouring Countries

There are basically three models that can be used for promoting cooperation with neighbouring countries, i.e. the data, startup, and co-fair models.

First, there is the data cooperation model, which can be viewed from the perspective of extending TTCP. This is a method of supporting neighbouring countries' efforts to create their own sub-TTCP (including the transfer of experience, know-how, technical models, etc.) and then connecting this to the HTP's TTCP. In other words, it is a hub-spoke linkage model in which the HTP's TTCP becomes the main IPR data hub in the area centred on the Black Sea and the Caribbean, and each country's TTCP becomes a spoke in this model.

[Figure 3-3] Hub & Spokes of TTCP



Source: Author's elaboration.

Second, in the startup model, geographical boundaries can be lowered through mutual cooperation so as to increase the number of startups and SMEs in the surrounding areas. The HTP supports prospective entrepreneurs from neighbouring countries to move into the HTP's incubation centres, while partner countries are introducing cooperation programs to lower the legal and institutional obstacles facing startups from Azerbaijan in order to help them enter neighbouring countries. Through these activities, each country can expect the localization of startups to become more active.

The third model is a technology or startup joint exhibition, or co-fair. The countries involved promote startups and support them in their efforts to find sales channels by holding joint exhibitions in the form of startup expos.

1.3.2 South Korea

There are basically three models that can be approached for cooperation with South Korea: HR co-training, entrepreneurship education, and consulting.

In the case of human resources (HR), cooperation is expected in the training of technology traders. South Korea already has experience and actual programs related to training

‘technology traders’, which means a cooperative channel for nurturing human resources will be possible. In this way, cooperation between the two countries will make it possible to foster commercialization experts in various fields, such as IP, TLO, negotiations, valuation, etc.

As regards entrepreneurship education aimed at revitalizing startups, it will create a way to share the experience of Korea’s various programs (I-Corps, Startup academy, etc.). In particular, a win-win situation can be expected in terms of revitalizing startups in both countries through cooperation between Korea’s startup promotion agency and the HTP.

Regarding technological cooperation, a virtual ‘Tech-help hotline’ can be opened between the two countries. Such a hotline would serve as a channel through which both countries could secure the solutions they need by connecting Azerbaijan’s raw materials and natural resources with Korea’s technology and know-how, etc. Furthermore, it could be used as a route for advice on the technologies needed for the MOIP or TOIP mentioned above. Such a constant communication channel will also be helpful in promoting innovation in both countries.

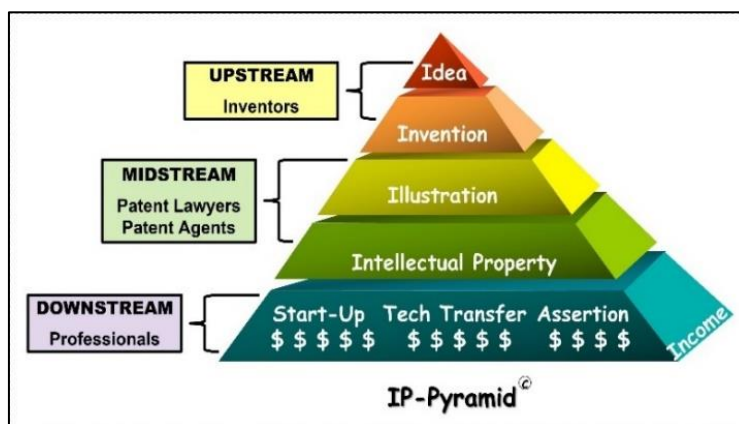
2. Intellectual Property Management

To promote the technology transfer and commercialization of home-grown technologies and their intellectual properties (IPs), such as patents, the Azerbaijan government needs to set up and fine-tune its IP management policies. In doing so, however, it is strongly recommended that the government take account of how to develop and nurture the human resources required for technology transfer and commercialization.

It is appreciated that technological innovations do not spring from nowhere and cannot be automatically protected under the IP (intellectual property) laws. Therefore, various professionals equipped with diverse expertise have to be involved in creating technological innovations, protecting such innovations under the various IP laws, and then transferring or commercializing the technologies.

It is also appreciated that technological innovations, inventions, patents, and their transfer or commercialization require a complex IP ecosystem. Such an IP ecosystem must include an upstream, a midstream, and a downstream, all of which must be interrelated with each other in both a forward direction and a backward direction. More importantly, each stream of the IP ecosystem requires professionals from different backgrounds and equipped with diverse expertise.

[Figure 3-4] Graphical Representation of an IP Ecosystem



Source: Shim (2016), p. 887.

[Figure 3-4] is a graphical representation of an exemplary IP (intellectual property) ecosystem, which may be referred to as an “IP-Pyramid.” As shown in the figure, the ecosystem typically includes (1) an upstream, (2) a midstream, and (3) a downstream segment, each of which is connected to the others in a series mode, in both an upward direction and a downward direction.

As will be explained in greater detail below, the upstream segment requires a pool of talented innovators or inventors, while the midstream segment requires patent professionals well versed in not only domestic patent law but also in foreign patent systems. Meanwhile, the downstream element requires technical or business experts capable of transferring innovations, inventions and patents to industry or commercializing them.

It should not be forgotten that it is the innovators who can create innovations and inventions; that it is the legal professionals who can protect such innovations and inventions as patents; and that it is the technological or business experts who can transfer or commercialize such innovations, inventions or patents. Accordingly, Azerbaijan’s central government should establish IP management policies which take account how to recruit a pool of such home-grown professionals and then develop them properly.

To this end, this Chapter explains:

- ① The professionals required by Azerbaijan to set up a domestic IP ecosystem;
- ② The basic knowledge which such professionals must be equipped with; and
- ③ Exemplary educational programs or materials that can provide such knowledge.

2.1 Intellectual Property (IP) Ecosystem

Innovations, inventions, and new technologies do not appear from nowhere. That is, innovations, inventions, and new technologies are neither created from scratch nor obtained at no cost. On the contrary, they come from an ecosystem comprising several layers.

It has been noted that different people who are equipped with different skill sets usually

play a major role in each such layer. In addition, these people tend to work in a sequence so that, for instance, people in the second layer cannot start to perform their tasks until the people in the first layer have completed their tasks, while people in the third layer cannot start to perform their tasks until people in the second layer have finished their work, and so on. To reflect these characteristics, one can assume a hypothetical IP ecosystem, and then use that IP ecosystem to analyze the characteristics of each layer as well as the various interactions among the different layers.

2.1.1 The Upstream Segment of the IP Ecosystem Requires Innovators and Inventors

The upstream of the IP ecosystem is characterized by the creation of innovations by innovators or the conception of inventions by inventors. More particularly, the upstream starts when an innovator or an inventor comes up with an idea or an innovative technology.

The innovator or inventor can then modify his or her idea and can also end up with lots of related ideas or ideas for improvement. Eventually, the innovator or inventor can sublimate the best idea into an invention (i.e., the inventor conceives the invention). Thus, there is no doubt that the inventor or innovator is the main player in the upstream of the IP ecosystem. When the midstream and downstream of the IP ecosystem of a certain country are fully developed, innovators or inventors may only need to master their technological expertise. However, when the midstream or downstream is not fully developed, it may be worthwhile to educate innovators and inventors with knowledge related to patents, technology transfer, and commercialization. Once equipped with such knowledge, innovators and inventors may be able to pursue their R&D, while manipulating their R&D in the direction of increasing the chances for potential technology transfer or commercialization.

2.1.2 The Midstream Segment Requires Patent Law Professionals

The midstream element of the IP ecosystem consists in filing a patent application and getting the patent application issued as a patent. More particularly, an inventor hires a patent lawyer (or a patent agent), who then prepares and files a patent application to a

patent office in the country (or another country) of the inventor.

After filing the patent application, the patent lawyer deals with a patent examiner at the patent office, and assists his client in getting the patent application issued as a patent. Therefore, the patent lawyer or the patent agent is the main character in the midstream.

When the upstream and downstream segments of the IP ecosystem are well developed, legal professionals may only need to know the patent law itself. If the midstream or downstream segment is not fully developed, however, it may be worthwhile to educate legal professionals in the various fields of knowledge related to R&D, the invention process, and technology transfer or commercialization.

Equipped with such knowledge, legal professionals can better understand and efficiently communicate with innovators and inventors, and may then draft patent applications while increasing the chances for potential technology transfer or commercialization.

2.1.3 The Downstream Segment Requires Various Technological or Business Professionals

Finally, the downstream of the IP ecosystem is characterized by different modalities of monetizing or commercializing a patent. Accordingly, the inventor (or patent owner) can resort to at least three different modalities to reap benefits from his invention.

2.1.3.1 1st Modality of Commercialization ~ Self-Practice

The 1st modality of commercialization is self-practice, i.e. the inventor applies his invention and makes his own product using his or others' manufacturing facilities. To this end, the inventor may have to set up a startup. It would be even better, however, if the inventor himself owned his own factory.

In this first modality of commercialization, the inventor can generate sales revenue by selling his product in the market, and then take a portion of the sales revenue as his profit. Of course, the commercialization may fail if the inventor cannot sell his product in the market or if the balance stays in the red even after the inventor has generated sales revenues.

2.1.3.2 2nd Modality of Commercialization ~ Technology Transfer

The second modality of commercialization is that the inventor may transfer his technology and patent to a licensee for profit. Scientific or technological innovators are not usually entrepreneurial, preferring to work in the lab rather than on the manufacturing front.

It is not surprising to find that most of these innovators tend to be constantly on the lookout for a potential licensee-manufacturer. When an inventor finds and negotiates with a licensee-manufacturer and signs a licensing contract, he or she can earn royalties in one or more of the following ways: (1) an upfront royalty, (2) a running royalty, or (3) a milestone royalty. The upfront royalty generally refers to the royalty which an inventor receives upon signing a contract for technology transfer. Conversely, the running royalty refers to a portion of the licensee's revenue.

In other words, as the licensee generates sales revenue, the licensee shares a certain portion of the revenue with the inventor, and that portion corresponds to the running royalty. The milestone royalty is the one which the licensee pays to the inventor when a certain condition is met or when a certain event happens. It should be noted that the various types of royalty discussed above reflect different features of technology transfer and technology commercialization.

For example, a technology transfer which includes the upfront royalty, but not the running royalty, means that the real fruit of technology commercialization is kept solely by the licensee. Accordingly, once the technology transfer contract is signed, there will be very little (if any) incentive for the inventor to assist the licensee in carrying out the actual commercialization.

Conversely, a technology transfer that includes the upfront royalty as well as the running royalty means that the inventor and the licensee agree to share the actual fruits of technology commercialization. Therefore, compared with the upfront-only licensing outlined in the above paragraph, the latter technology transfer can provide plenty of incentives for an inventor to assist the licensee in carrying out and successfully completing the actual commercialization.

2.1.3.3 3rd Modality of Commercialization ~ Patent Assertion

Circumstances may force a naive inventor to go to court to file a suit against a competitor for patent infringement. For example, there have been many cases in which an inventor has self-practiced his patent, manufactured the products, and sold them in the market, but where a competitor has made a dead-copy product and sold this counterfeit in the market. And, due to the lack of patent protection in the inventor's country, the inventor has lost his market.

In some cases, although an inventor has proposed to license the patent, the competitor refuses. In this case, the inventor's only remaining option is to take the case to court, file a suit against the copying competitor, win the lawsuit, and obtain damages from the copying manufacturer.

In other cases, however, an inventor may not be interested in self-practicing his own patent at all. Instead, the inventor meets lots of manufacturers, advertises his new technology and patent, and proposes a licensing deal with those manufacturers.

However, the manufacturer may still choose not to sign the contract and instead copy the patented technology of the inventor. Similar to the above example, the inventor's only remaining option in such a case is to take the case to court, sue the copying competitor, win in the court, and then obtain damages from the copying manufacturer.

Because the only right awarded to the patent owner is the right to exclusive competitors from practicing his patent, asset the patent in a court is the only right owned by the patent owner. There is nothing wrong in suing the infringer.

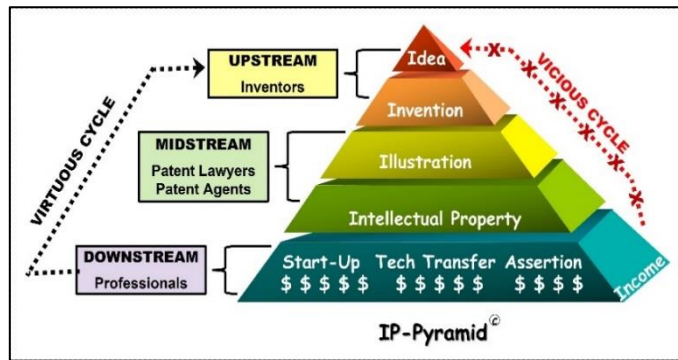
2.1.4 Virtuous IP Ecosystem vs. Vicious IP Ecosystem

As shown in Figure 1-1, an invention is transformed into a patent through a series of processes, wherein different people with different types of expertise play their roles in each of the upstream, midstream and downstream elements of the IP ecosystem.

Some IP ecosystems flourish, while others perish. The main difference between flourishing and failing ecosystems usually lies in their efficiency or productivity in generating profits and

in distributing such profits to the different players in its different layers.

[Figure 3-5] Graphical Representation of a Virtuous IP Ecosystem vs.a Vicious IP Ecosystem



Source: Shim (2016), p. 898.

[Figure 3-5] is a graphical representation of an IP ecosystem that flourishes in a virtuous cycle (see the left-hand side of the figure), and another IP ecosystem that operates in a vicious cycle and perishes (see the right-hand side of the figure).

As depicted in the figure, new technologies and their patents are commercialized (or monetized) by various professionals in the downstream element of the IP ecosystem. When a technology or its patent is successfully commercialized, the entrepreneur-inventor (of the 1st modality) can make profits from the sales revenue of the product which embodies the patented technology.

A licensor-inventor (of the 2nd modality) can make profits from the royalty he receives from the licensee-manufacturer, while the plaintiff-inventor (of the 3rd modality) can make profits from the damages which a jury or judge awards to the inventor.

According to the incentive theory of the patent system and, also, our common sense, once an inventor receives his share of profits, this financially rewarded and emotionally charged inventor will go on to make the next generation of innovations, inventions, and patents.

In short, an IP ecosystem operating according to a virtuous cycle always has channels in which profits generated in the downstream can be funneled back to the upstream. As a result, inventors are motivated to make inventions, to obtain patent protection, and to

commercialize their technologies and patents.

Because an IP ecosystem operating in such a virtuous cycle has a larger pool of serious and qualified inventors, the patent attorneys or agents serving those inventors can also make profits while assisting them in getting their inventions issued into patents. As this virtuous cycle goes on, all players in each layer of the IP ecosystem can make profits. In addition, such players can invest at least a portion of their profits back into the IP ecosystem. Accordingly, the IP ecosystem flourishes while acquiring additional width and depth.

Conversely, in an IP ecosystem which operates in a vicious cycle, everything goes in exactly the opposite direction. Without knowing about the “vicious feature” of the system, an inventor invests her time, effort, and money in coming up with a decent invention. Then, the inventor spends more money in filing a patent application and acquiring a patent.

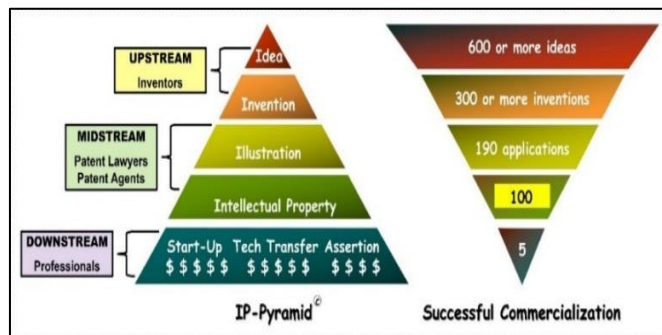
However, the inventor cannot make any profits through commercialization due to, for example, a lack of (or lax) patent protection. Some persistent inventors may go through this vicious cycle a few more times, but it will not take long for an inventor to realize that generating inventions and innovations and obtaining patents only helps his copycat competitors. The inventor will then determine to join the pool of copiers and, ultimately, there will be no one in the pool of inventors and innovators. As a result, the system will have far fewer innovations and inventions, and the public will not be able to reap any benefit from technology.

2.1.5 A virtuous IP Ecosystem requires the right players equipped with the right knowledge

Even within an IP ecosystem operating according to a virtuous cycle, not all inventions can be commercialized. In fact, only a small percentage of inventions are ever transferred to a manufacturer and, in turn, only a small portion of such transferred inventions are successfully commercialized. Furthermore, only a small portion of the commercialized inventions will find a way to occupy a meaningful market share. As mentioned above, less than 5% of the technologies developed in a university are “converted into an actual

commercial product that generates revenue for both the inventor and the university.”²⁷

[Figure 3-6] Chances of Successful Technology Commercialization



Source: Shim (2016), p. 887., author's elaboration.

Assuming that successfully commercialized university technologies are also protected by patents, [Figure 3-6] represents a graphical representation of the chances of successful technology commercialization, where the number of ideas, of inventions, and of issued patents required to reach the 5% level of successful commercialization are also shown.

Looking at [Figure 3-5], it may be reasonable to assume that about 190 patent applications are needed to obtain 100 issued patents. This assumption seems reasonable in that about 55% of all patent applications are issued into patents in the U.S.

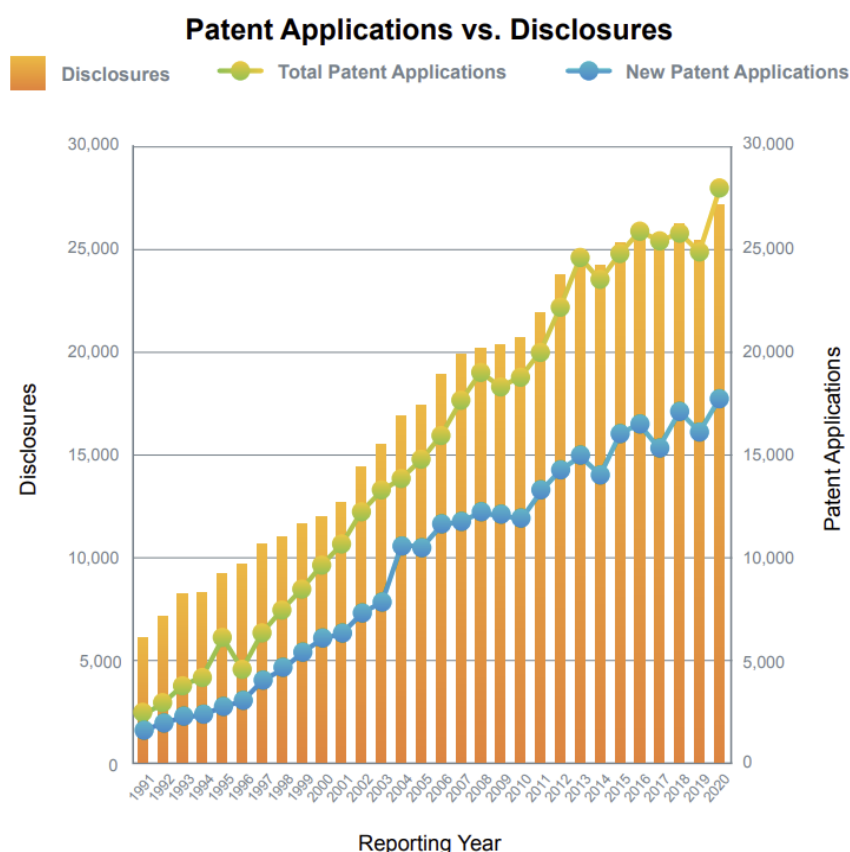
The conclusion which can be inferred from Figure 1-3 is that many inventions may lack technical merit or commercial potential, or may not be issued into a patent. Therefore, to promote technology transfer or commercialization, a country may have to establish an environment in which more inventions can be generated. This reflects the fact that such a society would need more innovators or inventors to build a useful IP ecosystem. However, if a country does not have an established IP ecosystem, its government may have to start educating potential innovators and inventors and to equip them with the appropriate background knowledge about technology transfer and commercialization.

[Figure 3-6] shows a comparison of the number of invention disclosures received by many

²⁷ <https://www.afr.com/work-and-careers/education/commercialising-new-technologies-needs-a-two-way-partnership-20211108-p5973j> (date accessed: 10 October 2023)

technology transfer offices (TTOs) of representative US universities with the number of patent applications filed by such universities.

[Figure 3-7] Comparison of the Number of Invention Disclosures Received with the Number of Patent Applications Filed by Representative U.S. Universities



Source: AUTM(2020), p. 10.²⁸

In this figure, it may be reasonable to assume that about 300 invention disclosures are needed to get 190 patent applications, as U.S. universities file about 60% to 65% of the invention disclosures they receive from professors, researchers, graduate students, and others working for the representative U.S. universities.

It may also be assumed that about 600 or more crude ideas would be needed in order to

²⁸ <https://ipwatchdog.com/2021/08/31/pandemic-cant-stop-bayh-dole-politicians-might/id=137235/> (Date accessed: 10 October 2023)

generate about 300 patent-eligible inventions. Even though there are no statistical data to support this assumption, this 2:1 conversion ratio may actually be too optimistic, given that most inventors are “serial and addicted” inventors. Most inventors might even think up about 1,000 or more ideas before filing 300 invention disclosures with the TTOs of their universities.

Whatever the actual conversion rates of crude ideas, inventions, patent applications, and issued patents may be, the bottom line seems to be that the most important task of the government in promoting technology transfer and commercialization is to build a pool of:

- ① Innovators or inventors who are equipped with the knowledge required to bring in lots of innovations or inventions;
- ② Patent law professionals who are equipped with knowledge of domestic as well as foreign patent systems; and
- ③ Technology and business professionals who are equipped with knowledge that is crucial in collaborating with innovators, inventors, patent law professionals and, more importantly, manufacturers or service providers who may be the final consumers of the innovations, inventions or patents.

2.1.6 Shortcuts to establishing an IP Ecosystem in Azerbaijan

According to Proclus, Euclid responded to King Ptolemy’s request for an easier way of learning mathematics by saying that: “There is no Royal Road to geometry.” A similar expression is also attributed to Menaechmus, the ancient Greek mathematician, in his reply to Alexander the Great.

The same may apply to building a virtuous IP ecosystem in a country. However, it is very reasonable to assume that some roads may be shorter than others, and that some roads may need fewer workers than others, among other such considerations.

2.1.6.1 R&D-IP strategy of KISTA

The Korean government has been very active in finding such shortcuts. For example, KISTA

(Korea Intellectual Property Strategy Agency), one of many umbrella agencies of KIPO (Korean Intellectual Property Office), has educated Korean inventors, universities, research institutes, and industry about the R&D-IP strategy or the IP-R&D strategy.

According to KISTA, the R&D-IP strategy is suitable for countries which aim to follow the leading countries, whereas the IP-R&D strategy may be suitable for acquiring pioneer patents, standard and essential patents (SEPs), and such like.

2.1.6.2 KIPA's 3T (Teaching-The-Teacher) programs

In addition, KIPA (Korea Invention Promotion Association), another umbrella agency of KIPO), has offered the 3T (Teaching The Teacher) programs to various institutions. For example, KIPA has been training high school teachers, college professors, and even professionals interested in technology transfer or commercialization.

After about fifteen years of successful operation, KIPA recently stopped this 3T Program, having found that many high-schools and universities had voluntarily continued and maintained the 3T Programs, and having realized that they no longer need its intervention or assistance.

2.2 Knowledge with which major players in the various layers of the IP Ecosystem need to be equipped

As shown in Figure 1.1, the various players in the various layers of the IP system have their own rules. For example, the inventors in the upstream segment may focus on conceiving many technologically supreme and commercially valuable inventions. The inventors may also have to strive to come up with improvement inventions and include them in the disclosure of the patent applications.

Meanwhile, the legal professionals in the midstream segment can focus on generating patent applications and drafting solid claims of broad scope. To do so, the legal professionals collaborate with the inventors and efficiently convert the ideas and examples filling the brains of the inventors into various embodiments included in the patent applications. The legal professionals continue to work with the inventors in acquiring subsequent patents by,

for example, filing continuation patent applications or divisional patent applications.

Finally, the technical or business professionals in the downstream segment of the IP ecosystem can focus on self-practicing the patents, or they can transfer or commercialize the technology to or with industry. To this end, these technical or business professionals can either come from the industry or develop personal and professional relationships with the industry.

In addition, it would be helpful if the technical or business professionals in the downstream segment can acquire basic knowledge of the logistics of the R&D processes as well as the mindset of the researchers.

2.2.1 Need for a national strategy to effectively develop the professionals required for each layer of the IP ecosystem

It has been noted that the strategy explained in the foregoing paragraph applies to countries in which an IP ecosystem has already been set up, and where the various players in the various layers of the IP ecosystem can hone their expertise and successfully convert the IP ecosystem into a virtuous IP ecosystem.

However, it may be a totally different story in countries where an IP ecosystem has not been set up, where some or all of the layers of the IP ecosystem lack professionals, and where the domestic environment may not be ready to provide such professionals.

Of course, a government could wait for such professionals to educate themselves, build each layer of the IP ecosystem, and establish a virtuous IP ecosystem. However, it would take a long time for the professionals to teach themselves, accumulate experience, and actively participate in technology transfer or commercialization. Moreover, it would be extremely difficult to encourage such professionals and to start up the engines of technology transfer and commercialization.

Therefore, Azerbaijan's central government is strongly recommended to come up with a plan by which it can accelerate the education of the professional people in each layer of the IP ecosystem and give them sufficient financial and moral incentives to actively participate in laying the foundations of the virtuous IP ecosystem.

The following sections 2.2.2, 2.2.3, and 2.2.4 summarize the various kinds of knowledge that will be helpful in educating and nurturing the main players in each layer of the IP ecosystem.

2.2.2 Extra knowledge required for innovators and inventors in the upstream segment of the IP ecosystem

Inventions can be conceived in many fields of life. More often than not, inventors in the fields of science or engineering perform research and conceive inventions. As a result, inventions are very closely related to various areas of science, such as chemistry, physics, biology or geology, or various branches of engineering, such as chemical engineering, biochemical engineering, electrical or electronic engineering, mechanical engineering, computer science, or material science, etc. Of course, the inventors who conceive such inventions obtain the required background knowledge through collegiate or graduate studies.

Although not as common as inventions in the fields of science and engineering, inventions can also be conceived in non-scientific or non-engineering fields. Typical examples include inventions related to methods of doing business. Such inventions can be conceived not only by scientists and engineers but also by ordinary people.

Inventors typically obtain proper background knowledge through collegiate studies or graduate education. Typically, a government may or may not do extra work to prepare such inventors to generate more inventions.

However, the Azerbaijan government needs to train and equip such inventors with extra knowledge about the dynamics of the IP ecosystem and the basic nature of the patent system. For example, inventors need to know all about (1) the difference between R&D (and academic papers) and inventing (and patents); (2) inventorship (e.g. who the sole inventor is or who the joint inventors are); (3) ownership of the patents (this is crucially important in conducting co-R&D with other domestic or foreign partners); and (4) how competitors can detect weaknesses in inventors' inventions or patents, etc.

It has been noted that the R&D-IP strategy or the IP-R&D strategy, as described in Section

1-6 above, may also be good candidates for educating and training the inventors to manipulate the direction of their inventions more strategically. With such strategies in mind, inventors can align their R&D activities appropriately with inventions in real time.

Doing so will not only accelerate the education and nurturing of inventors but also help them to communicate more effectively with legal professionals, entrepreneurs and, more importantly, those people in diverse industries.

2.2.3 Extra knowledge required for patent law professionals in the midstream segment of the IP ecosystem

Patent law professionals are those persons who are well versed in the domestic patent laws and systems. In most countries, they have to pass a domestic patent bar examination to become a patent agent or patent attorney. Accordingly, there will be no need for the Azerbaijan government to do extra work to improve the legal skills of such patent law professionals.

However, the Azerbaijan government may need to assist such legal professionals in becoming better versed in the patent systems of strategically important countries. Because the patent law is a territorial law and because Azerbaijan (like Korea) is a small country, it would be far better if the valuable inventions created by Azerbaijan's inventors were to be patented in such strategically important countries as the US, certain EU nations including the United Kingdom, France and Germany, and China and Japan among others.

Azerbaijan's patent law professionals could upgrade their skills by themselves and acquire the capability to upgrade the quality of Azerbaijan's patent applications, which in turn could be translated into English, German, French, Chinese or Japanese, and then converted into US, British, German, French, Chinese or Japanese patents.

In practice, however, it would cost much time and money for Azerbaijan's patent law professionals to learn about the patent systems of diverse foreign countries by themselves. In addition, even if these professionals were prepared to do all this, they would not be able to recoup their time and resources unless Azerbaijan's inventors were interested in filing patent applications in those countries.

Accordingly, the government of Azerbaijan (e.g. the Azerbaijan Patent Office) could take the initiative in globalizing Azerbaijan's legal professionals. For example, the Patent Office could follow up with major changes in the patent laws of strategically important countries or summarize the court decisions of important cases in those countries. Meanwhile, the Patent Office could offer seminars or short courses aimed at disseminating such information to domestic patent law professionals.

Doing so would not only improve the skills and knowledge of legal professionals but also help inventors, as Azerbaijan's patent law professionals would then be able to tailor their patent applications such that, when translated into the languages of strategically important countries, the translated patents would remain valid in those countries and could be successfully protected against infringers.

2.2.4 Extra knowledge required for technology transfer and commercialization experts in the downstream segment of the IP Ecosystem

In general, one can choose from three modes of technology transfer and commercialization, namely, self-practice, technology or patent licensing, and assertion of patent rights in the courts. Because the third mode is the exclusive remit of patent lawyers, this section will focus on the knowledge required for the first and second modes of technology transfer and commercialization.

In the first mode of tech transfer and commercialization, entrepreneurs are the key players. It can be assumed that most entrepreneurs have adequate background knowledge in science and engineering, in business, or in both. Accordingly, the government may not need to do any extra work to equip entrepreneurs with further knowledge in these fields.

However, entrepreneurs usually have only limited (if any) knowledge of inventions or patents, so they may need to be educated about the matters referred to in Section 2-2 above from time to time. Entrepreneurs may purchase others' patents, license other's patents or license out their own patents. In this context, entrepreneurs may need to grasp the basics of patent assignment or licensing.

In the second mode of technology transfer and commercialization, the major players are patent brokers or patent law professionals. Considering that a technical background and industry connections are required for more efficient technology transfer and commercialization, it may be advantageous to cultivate a pool of patent brokers who can communicate with domestic as well as foreign industries. In addition, the Azerbaijan government may come up with efficient plans for nurturing patent brokers who will invest much of their time and resources in connecting inventors with industry, or in channeling industry feedback to academia. If brokers have the basic knowledge described in Section 2-2 above, they will be able to communicate with inventors. In addition, they will also be able to come up with a proper R&D-IP or IP-R&D strategy that enables inventors and patent brokers to build up a patent portfolio and attract more attention from industry.

2.3 TCK-Lab Sessions of 2023

STEPI conducted seven two-hour IP lectures starting on March 30, 2023 and ending on May 3, 2023. This author conducted all seven lectures, with the audience of each one ranging from 12 to 15 people.

Based on the knowledge and information discussed in Section 2 above, this author has selected topics which will be beneficial to the major players in the upstream, midstream, and downstream segments of the IP ecosystem. The following [Table 3-1] presents a list of the topics covered in each IP lecture.

[Table 3-1] Details of the 2023 TCK-Lab Curriculum

(1) IP Lecture 1 (March 30, 2023)	• Introducing the lecturer • Topics of inventing projects • How to invent? Working example vs. prophetic example • Patent-eligible inventions • Parts of a patent • Essence of the patent system • Does the patent system promote or prevent innovation? Two examples • Patents are about money • When can you file a patent application? • What happens after you file a patent application?
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(2) IP Lecture 2 (April 5, 2023)	• In which country should you file a patent application? • What does a patent protect? • Patentability preview - Novelty requirement • Patentability preview - Non-obviousness requirement • Pitfalls in prior art search • Issued patents can be invalidated • Patent infringement preview - Literal infringement • Avoiding literal infringement - an example • The “all elements” rule • Patent infringement preview - Infringement under the doctrine of equivalents
(3) IP Lecture 3 (April 12, 2023)	• TRIZ applied to inventing • Patent rights are fairly limited • Pros and cons of an improvement patent
(4) IP Lecture 4 (April 19, 2023)	• Examples of good patents with a wide scope • Claim structure (1) - Preamble • Case study: Jeffrey Griffin and Claudia Griffin v. Michael Marrin (C.D.CA 2008) • Claim structure (2) - Transitional phrase • Open-ended claim • Closed claim • Partially-closed claim • Claim structure (3) - Body • Types of claim: • Machine claim • Article of manufacture claim • Composition of matter claim • Process/method claim
(5) IP Lecture 5 (April 26, 2023)	• Example of a teaching material from KIPA • Patents and their parts • Claim construction • Canon of claim construction • 1st canon - A claim must be interpreted such that its elements or limitations are presumed to have an ordinary and customary meaning. • Case study: G.E. Lighting Solutions v. AgiLight, 750 F.3d 1304 (Fed. Cir. 2014) • 2nd canon - A patent applicant can be his or her own lexicographer. • Case study: X2Y Attenuators v. ITC and Intel, 757 F.3d 1358 (Fed. Cir. 2014)

	• 3rd canon -A claim must be interpreted to have a consistent meaning. • Case study: Markem-Imaje Corp. v. Zipher Ltd., 657 F.3d 1293 (Fed. Cir. 2011) • 4th canon -A claim must be interpreted in such a way that its scope includes the applicant's preferred embodiment. • Case study: Trading Tech. Int'l v. eSpeed, Inc., 598 F.3d 1340 (Fed. Cir. 2010) • Doctrine of claim differentiation • 5th canon -A claim must be interpreted in view of a patent specification, but any claim limitation cannot be imported into the claim. • Case study: Unique Concepts, Inc. v. Brown, 939 F.2d 158 (Fed. Cir. 1991) • Lessons from the canons of claim construction
(6) IP Lecture 1 (May 3, 2023)	• Claims are important, but specifications may be more important. • Inventor strategy - Keep one patent application pending • Continuation applications or divisional applications • Examples of the importance of continuation applications • Continuation applications vs. Divisional applications • Inventorship • Murphy's Law • Conception of an invention • Joint inventorship • Employee invention • Nature of Technology Transfer Offices • Evolution of Technology Transfer Offices • German solution
(7) IP Lecture 7 (May 7, 2023)	• Team presentations of inventing projects • Discussion

It is appreciated that the above IP lectures were focused on those participants who had collaborated for more than one year since 2022. Accordingly, some contents of the IP lectures may be suitable for those who have some background in technology transfer or commercialization, but they may not be suitable to train beginners in such fields. Accordingly, this author has selected the topics and related background information which are recommended for inclusion in the training materials for beginners in the area of technology transfer and commercialization from various backgrounds. Accordingly, some

topics may look too basic to some professionals with a certain background, while other topics may look some-what advanced. However, this author thinks that including most of the basic information will be helpful in preparing such training materials.

2.3.1 Selected Topics from IP Lecture 1

2.3.1.1 How to Invent? Working vs. Hypothetical Examples

When an inventor conceives an invention, he or she can prove that invention by constructing a working example. The working example may include a mock-up model or a physical proto-type, and so on. Based thereon, the inventor can hire a patent attorney who will then prepare a patent application and file the application to a patent office. Of course, the patent attorney includes the working example in the patent application. However, the inventor may take an alternative route. That is, instead of constructing a working example, the inventor can purely rely on his imagination, and come up with an example which he or she does not actually test in the real world. This example is referred to as a prophetic example. Because it has not been proven, the prophetic example may not be properly included in an academic paper. However, because the patent aims for a purpose (profit generation) which differs from the purpose of the academic article (i.e. the pursuit of a universal truth), the prophetic example can be used in a patent application. For example, the MPEP (Manual of the Patent Examining Procedure) of USPTO allows an applicant for a patent to use a prophetic example in order to improve the enablement of the invention. As a result, the patent applicant should not be shy of using such prophetic examples.

2.3.1.2 Which is More Important? The Claim or the Specification?

It is the patent claim itself which is analyzed against various requirements of the patent law, for example, in determining its novelty or non-obviousness. Accordingly, the patent claim must be written to survive these requirements. In addition, the scope of the claim also determines the scope of infringement. Therefore, a competitor may easily infringe a claim with a broader scope, or may easily get around and escape infringement when a claim has a narrower scope. In this regard, the claim is a very important part of a patent. However, when a competitor finds a hole in the scope of a patent, it can attempt to escape the scope of the claim. A patent owner (or a patentee) may file a continuation or divisional application

while claiming the earlier filing date of the parent patent, and may trap the competitor. To do this, however, all elements and limitations of the claims of the continuation or divisional application must have been included in the specification of the parent patent. In this context, the specification of the patent may be deemed to be more important than the claim itself. In other words, the claim of the patent may be regarded as “a soldier who goes to the battlefield and fight against the enemy”. Contrary to this, the specification of the patent may then be deemed “an arsenal” from which the soldier can come and pick the proper weapon of his or her choice.

2.3.1.3 The Patent System Promotes Innovation

Some scholars argue that the patent system hinders or even prevents innovation. The major argument of these scholars is based on the fact that the patent only guarantees a negative right. When a patent is issued, the patent system guarantees a patent owner (i.e., a patentee) the right to exclude others from practicing the patented invention without the consent of the patentee. However, the patent system does not guarantee the patent owner to practice his own patent. The main reason for this is that the patent system allows a later improver (such as an inventor of an improvement invention) to obtain an improvement patent whose scope may be only a part of the pioneer patent of an earlier pioneer (such as an inventor of a pioneering invention). That is, all patentees, regardless of whether their inventions are pioneer inventions or improvement inventions, have the (negative) right to exclude others from practicing their patented inventions, but the patentees of the improvement inventions do NOT have the right to practice their own patents (positive rights).

Anti-patent scholars focus on these problems and refer to the pioneer patents and the improvement patents as “blocking patents.” They further argue that, due to the blocking nature of patents, the more patents there are, the fewer innovations there will be. However, this blocking patent conspiracy theory is counterintuitive at best. Just like patents, other properties provoke conflicts whenever their property rights overlap. Just like other properties, conflicts surrounding patents can be resolved as well. Let us suppose that Howard has a nice house with a beautiful lawn, and that Tom has a nice tent. Tom yearns

to sleep for a night on Howard's lawn while gazing up at the Milky Way. But Howard does not allow Tom to do so. Accordingly, one night, Tom brings his tent to Howard's lawn, erects his tent on the lawn, and hops into it. Howard finds Tom's tent, and calls the police. The police officer kicks Tom off Howard's lawn, and Tom has no means to stay on Howard's lawn. In the middle of this mess, however, Howard finds that Tom's tent is awesome and decides he wants to keep it. But the police officer orders Howard to return the tent to Tom. As one can see, this dispute is attributed to the overlapping property rights of Howard and Tom. However, our capitalistic society offers different ways to solve this dispute. One such way is that Tom buys Howard's house and lawn or that Howard buys Tom's tent. Another way is that Howard receives money from Tom and allows him to stay on his lawn for one night.

Similarly, the owner of a pioneer patent (i.e. a pioneer patentee) and the owner of an improvement patent (i.e., an improvement patentee) can also resolve their dispute in various ways. The first such way is that the pioneer (or improvement) patentee can sell his patent to the improvement (or pioneer) patentee. The second way is that the improvement (or pioneer) patentee can license his patent to the pioneer (or improvement) patentee. The third way is that the pioneer patentee and the improvement patentee can collaborate and set up a joint venture company. The fourth - but not the last - way is that the pioneer patentee and the improvement patentee can together sell their patents to a third party.

In other words, the disputes surrounding patents are no different from the disputes surrounding other personal properties recognized in the capitalistic society. As a result, the blocking nature of patents is no different from the same blocking nature of other personal properties of the capitalistic society.

The patent law belongs to the intellectual property law, which in turn belongs to the intangible property law, which in turn belongs to the property law. The property law of the capitalistic society defines which goods have economic value, and how such goods can be generated, kept, and exchanged. In short, patents are all about money. Accordingly, just like other financial incentives which lure numerous entrepreneurs into the business marketplace, the patent system can successfully lure numerous inventors and innovators into creating more inventions and innovations.

2.3.1.4 When to File a Patent Application? Grace Period

Who can obtain a patent when two different inventors conceive the same invention at different times? Two systems used to exist: the first-to-invent system and the first-to-file system.

The first-to-invent system awards the first inventor and allows him or her to obtain a patent, but it does not give a patent to the second inventor. This system makes a lot of sense as it is better that the patent be given to the first, earlier inventor. However, the main problem with the first-to-invent system is that some inventors are inclined to forge a document to prove the earlier date of their invention. Accordingly, the courts have often had to rule on who is the first inventor in a dispute, while validating each and every piece of evidence related to the date of invention. Accordingly, lawsuits surrounding the date of invention can be costly in terms of both resources and time.

Conversely, the first-to-file system awards the inventor who files a patent application first, and allows this inventor to obtain a patent, but it does not give a patent to another inventor who files a patent application later on, regardless of whether this late-filing inventor is in fact the first to conceive the invention. This system may seem not to make much sense, for it seems more reasonable to give the patent to the first and earlier inventor of the invention. However, the principal merit of the first-to-file system lies in the fact that the court does not have to rule who is the first inventor in the case of disputes. The reason for this is that the court under the first-to-file system does not have to review any documents about the date of invention, and requires nothing more than the date of filing, which is well documented by the patent office.

Despite all the inefficiencies attributed to the validating of the date of an invention, the US maintained the first-to-invention system up until 2013, and then adopted the first-to-file system, just like the rest of the world. Therefore, as far as we know, the entire world is now running the first-to-file system.

It is appreciated, however, that the first-to-file system may not be beneficial to the public in general. For example, when an inventor conceives an invention, he has to rush to file a patent application, as otherwise he may lose the chance to get a patent if a competitor files

a patent application for the same invention at an earlier date. As a result, the inventor may not have enough time to hone his invention or to find more valuable improvements of his original invention, and the public may not be able to reap more benefits from the same invention. In addition, the first-to-file system may place unwary inventors at great risk, as it does not allow an inventor to disclose an invention publicly before filing a patent application. However, many scientists who are not versed in patent law may publish an academic paper and disclose their inventions publicly, well before filing a patent application. In such a case, the inventor may lose the right to get a patent, as his own publication may serve as a prior art, negating the novelty and non-obviousness of his invention.

To alleviate this problem, the patent law provides unwary inventors with a certain period of grace, for example, 6 months or 12 months. In other words, even if an inventor happens to disclose his invention publicly before filing a patent application, as long as the inventor files a patent application within 6 (or 12) months of its disclosure, he will obtain a patent provided that his patent application satisfies various requirements.

Most countries now provide a grace period of 6 or 12 month. It has been noted, however, that China and many European countries previously adopted the absolute novelty system, in which no grace period was offered. Under such a regime, naïve inventors who accidentally disclosed their precious inventions in academic papers often lost the opportunity to convert the inventions into patents. However, those countries have started providing a grace period of 6 to 12 months. Accordingly, whatever the grace period may be, inventors have many reasons to file a patent application before disclosing their inventions publicly.

2.3.2 Selected Topics from IP Lecture 2

2.3.2.1 Where to File a Patent Application? In Which Countries Should One Get Patents?

As we know, the world is composed of 193 (+ 2) countries. Because the patent law is a territorial law, theoretically speaking, an inventor needs to file 193 (+ 2) patent applications in order to protect his invention.

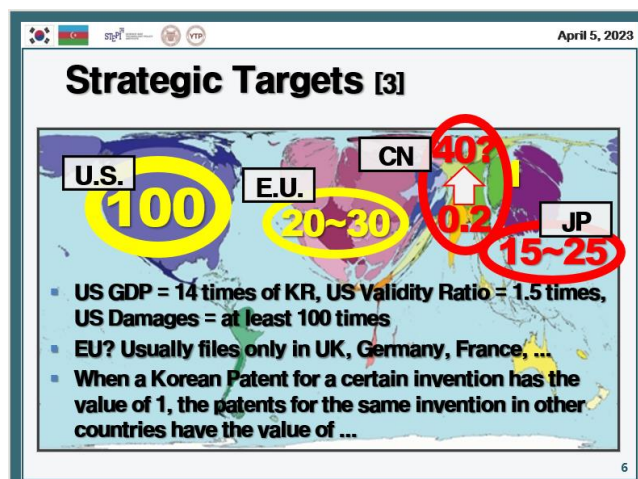
As described above, the patent is all about money (or profit). Accordingly, inventors are advised to perform a cost-benefit analysis to determine in which countries they should file their patent applications. The answer is rather straightforward for a patent owner who is also a manufacturer. In such a case, the patent owner would want to protect his patented product in the countries in which he can sell more products or services. Other things being equal, those countries would correspond to countries with a high GDP . In addition, the patent owner would also want to suppress copiers from the market, for he cannot sell more products in a country in which patent protection is weak. In such a country, copiers would literally copy the patentee's product, and sell the counterfeit products at half price. And because patent protection is very weak, the patent owner would not be able to exclude those copiers from the market.

For a patent owner who does not practice his own patents (i.e. a non-manufacturing patent owner), the answer is also straightforward. Because he does not practice his own patent, the non-manufacturing patent owner would want to sell his patent or to license the patent to a manufacturer. In this case as well, the manufacturer would prefer those patents to be issued in countries where the GDP is greater, where the courts protect patents more fiercely and, therefore, where the patent value will be higher.

In summary, the inventor would want to file patent applications in those countries with a greater GDP and stronger patent protection. Currently, such countries would include the following:

- ① The US
- ② The EU and the UK
- ③ China
- ④ Japan

[Figure 3-8] Relative Values of Patents which Claim the Same Invention in Strategic Countries



Source: Author's elaboration.

At this point, it is reasonable to raise another question. How valuable is the US patent compared to another patent in another country? [Figure 3-8] shows the exemplary (relative) values of patents in strategic countries, where each patent in those countries would claim the same invention. Let's assume that a Korean patent which claims the invention 'K' has a relative value of "1" for a prospective buyer in Korea or in any other country of the world. What, then, would be the relative value of a US patent which also claims the invention 'K'? Because the US's GDP is about fourteen times greater than that of Korea, an entrepreneur who manufactures the patented product would sell fourteen times more products in the US than in Korea. Accordingly, it seems safe to say that the US patent claiming the invention 'K' would be at least fourteen times more valuable.²⁹

In addition, if we assume that the invalidity ratio in the US courts is about 50%, while that in the Korean courts is about 70%, the US patent would be 1.4 times more valuable than the Korean patent.³⁰ Furthermore, a US patent owner can earn at least US\$ 5 million in damages upon winning a patent infringement suit in a US court, whereas a Korean patent owner would earn less than US \$100,000 upon winning the same suit in a Korean court. In other

²⁹ <https://www.populationu.com/gen/countries-by-gdp> (Date accessed: 10 October 2023)

³⁰ <https://www.nber.org/papers/w21769> (Date accessed: 10 October 2023)

words, a US patent owner can earn fifty times more patent damages in the US compared to a Korean patent owner.³¹

If one multiplies all those factors, i.e. $14 \times 1.4 \times 50$, it turns out that a US patent is 980 times more valuable than a Korean patent. Even considering errors in estimating the invalidity ratio, US patents are still far more valuable than Korean patents.

The European Union (EU) has twenty-seven member countries. The respective GDPs of the US and the EU account for more than 40% of the global nominal GDP, with the US's GDP being about 1.3 times greater than the EU's GDP.³² Accordingly, it may seem that an EU patent is as valuable as a US patent.

However, there is a “big catch.” The EU runs an European Patent Office (EPO) which is in charge of receiving and examining patent applications for all of its twenty-seven member countries. Once the EPO decides to grant a patent to a patent applicant, the applicant may choose the EU member countries in which he or she wishes to obtain a patent. Common sense dictates that the applicant might try to gain the maximum protection and earn the biggest profits, while minimizing the patent cost. Assuming that the courts of the EU countries would protect the patent similarly, it then follows that it would be advantageous to patent applicants to issue their patents in the countries with a large GDP.

In practice, too, most applicants choose to obtain their patents in the UK, Germany or France. When a patent applicant owns a manufacturing facility or distributing center in another EU country, the applicant may also obtain patents in that country as well. In summary, it is true that the EU, with its twenty-seven member countries, has a GDP comparable to that of the US. Because an inventor generally obtains patents in a smaller number of countries, however, the EU patent for the same invention ‘K’ has a lower value than that of the US patent.

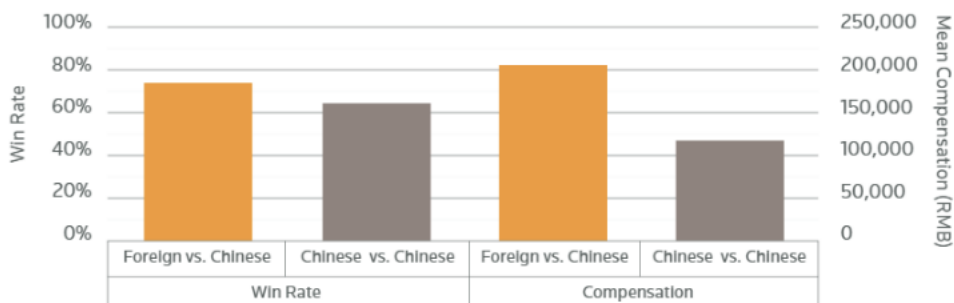
³¹ <https://www.analysisgroup.com/globalassets/content/insights/publishing/patent-damages-in-us-courts-analysis-group.pdf>. (Date accessed: 10 October 2023)

³² <https://statisticstimes.com/economy/united-states-vs-eu-economy.php>. (Date accessed: 10 October 2023)

[Figure 3-9] Patent Owner Winning Rates and Damages in China (2006-2014)

Chart 8

WIN RATE AND MEAN STATUTORY COMPENSATION COMPARISON—FOREIGN AND DOMESTIC PLAINTIFFS



Source: CIELA by Rouse

Source: Thomson Reuters (2015), p. 11.

In the period 2006-2014, the Chinese government adopted a policy of strong patent protection and, more importantly, the Chinese courts fiercely protected patents owned by its citizens as well as other patents held by foreign nationals. The Chinese courts also endeavored to prove to the world that they actually treated foreigners who held Chinese patents better than they did domestic citizens who held Chinese patents. Figure 1-6 shows this trend. As a result, the value of Chinese patents increased steadily year after year.

Recently, however, the global political situation has seen China become isolated from the US, the EU, and many other nations, all of which are trying to detach themselves - or “derisk” themselves - from current Chinese politics. Accordingly, the actual value of a Chinese patent may not be as high as it used to be. In addition, it is very difficult to predict whether or not the Chinese courts will treat foreign patent holders in as friendly a way as they used to do.

This being the case, another question naturally arises. What about the other 190 countries around the world? Should Azerbaijan’s inventors forego the excellent opportunity of obtaining patents in those 190 countries or, at the very least, in a few of them?

Of course, India, Brazil, and Argentina are also large countries. However, the situation in those countries may not bode well either in terms of GDP or patent protection, or both. When an Azerbaijan company intends to build a manufacturing facility in one or more such

countries, the company may have sufficient reason to obtain patents in them. Otherwise, a wiser policy would be to concentrate on the aforementioned, strategically important countries.

2.3.2.2 Novelty Requirement

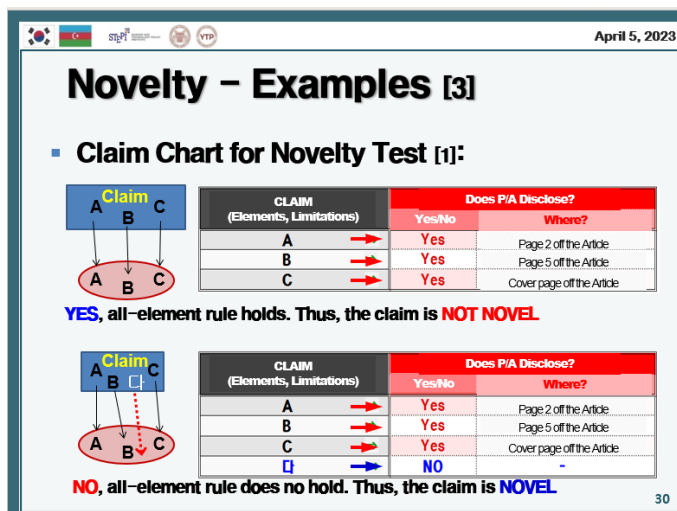
In order to obtain a patent, an invention which is recited in a patent claim must be novel, i.e. new compared to all prior art throughout the world. If the invention recited in the claim is not new, i.e. the same invention has existed before the inventor's patent, the claim is not novel. As a result, the claim included in the patent application will be rejected, or the claim included in the issued patent will be invalidated for not being novel over the prior art.

Why does the patent system require novelty? The answer is simple. Society does not have to protect an invention which has already become known publicly. If the patent system protects such a non-novel invention, then that non-novel invention, which has been in the public domain (i.e. the invention which has been publicly known and which the public has been able to use free of charge), suddenly becomes the personal property of the inventor, and the public cannot practice that invention without the consent of the inventor.

In this context, protecting an inventor's invention that is not novel does not provide any benefit to the public and society. Accordingly, such a non-novel invention must not be protected by a patent.

Because all countries have adopted the first-to-file system, "prior art" is defined as anything which is known to the public before the date of filing a patent application (i.e. the filing date). Such prior art may include a printed publication, others' published patent applications, others' issued patents (whether expired or not), merchandise, contents on the web, and such like.

[Figure 3-10] Example of the “All-elements-rule” and “Claim Charts” Applied to Determine the Novelty of a Patent Claim with Respect to a Single Prior Art



Source: Author's elaboration.

[Figure 3-10] illustrates an example of the “all-elements-rule” and “claim charts” when applied to determine the novelty of a patent claim with respect to a single prior art. When applied to determine novelty, the all-element rule states that a claim is not novel if each and every (claim) element and each (claim) limitation are disclosed by a single prior art.

A claim chart is usually a two-column table that is employed to assist comparison of a claim with respect to the prior art. To this end, all (claim) elements and (claim) limitations are listed in the left-hand column of the claim chart, whereas the right-hand column of the claim chart show whether the (claim) elements and the (claim) limitations of each row of that column are disclosed by the prior art. Accordingly, when every element and every limitation of a certain claim are disclosed by a single prior art, that claim is not novel. Conversely, when at least one element or one limitation of the claim is not disclosed by a single prior art, that claim is then novel. It is appreciated that the determination of novelty using the claim chart is to compare from left to right, rather than from right to left, when the claim is open-ended.³³ When the claim is closed,³⁴ however, the direction of the comparison for

³³ <https://www.lsd.law/define/open-ended-claim> (Date accessed: 10 October 2023)

³⁴ <https://www.lawinsider.com/dictionary/closed-claim> (Date accessed: 10 October 2023)

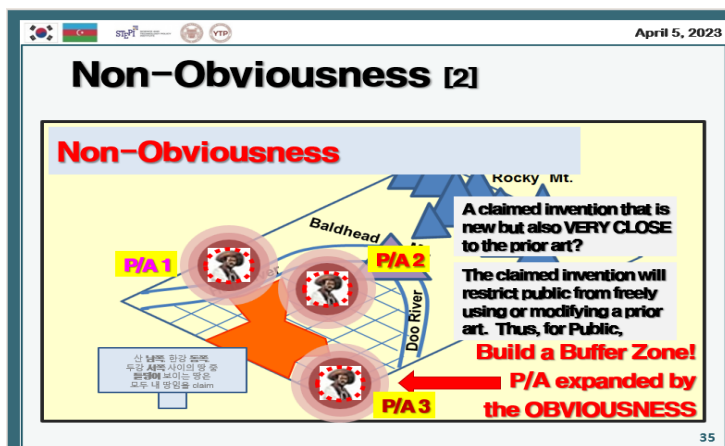
determining novelty using the claim chart runs both ways. Because most patents prefer open-ended claims, it is advisable to remember that a user must compare the claim and the prior art in the left to right direction in order to determine the novelty.

It is noted that the determination of novelty is somewhat objective or straightforward, since all that needs to be done is to make a direct comparison of the claim elements and limitations with respect to the single prior art. Accordingly, relatively speaking, there is not much room for the fact finder in determining the novelty requirement. Of course, terms such as “objective” and “straightforward” are the terms used to compare the novelty determination with the non-obviousness requirement, which is explained below.

2.3.2.3 Non-Obviousness Requirement

Even if a claim satisfies the novelty requirement, i.e. when at least one claim element or at least one claim limitation is not disclosed by a single prior art, it does not mean that the claim is patentable. To be patentable, a claim must also not be obvious. Why, then, does the patent system require non-obviousness in addition to novelty? The answer is a little complex compared to the novelty requirement. The assumption is that the public has the ability to improve or modify knowledge in the public domain. Accordingly, even when an invention claimed by an inventor has not become known to the public and has not been disclosed by a single prior art, i.e. the invention is novel, there is a chance that the public will be able to modify the existing prior art and easily arrive at the claimed invention of the inventor. Thus, there is no need to protect a inventor’s invention that is obvious. The net result is that even a novel claim cannot be patented when the claim is obvious in view of the prior art, and this result can be represented graphically as shown in [Figure 3-11]

[Figure 3-11] Diagram Showing the Essence of the Non-Obviousness Requirement, i.e. This Non-Obviousness Requirement Sets up a Buffer zone Around the Prior Art, Thereby Denying Patents to An Invention Which is Obvious to the Public



Source: Author's elaboration.

The diagram in [Figure 3-11] explains the essence of the non-obviousness requirement. When a claim includes within its scope at least a portion of the prior art, the claim is then not novel and cannot be patented. Conversely, when the claim does not include within its scope any part of the prior art, the claim is then novel and satisfies the novelty requirement.

However, when a claim does not include within its scope any part of the prior art, but is very close to the prior art, that claim may then become obvious and, therefore, cannot be patented. Thus, qualitatively speaking, it may be said that this non-obviousness requirement is building a buffer zone around the prior art. For the inventor to obtain a patent, his or her patent claim must not include any portion of the prior art and, in addition, must maintain a “certain distance” from the prior art.

It is appreciated that the all-element rule is not used to determine non-obviousness. Rather, determining non-obviousness requires one to ask whether or not the difference between the claim and the prior art, when viewing the claim as a whole, would be obvious to a person possessing ordinary skills in the relevant art at the time of filing the patent application. When the answer is yes, then the claim is obvious over the prior art; whereas, when the answer is no, the claim is not obvious over the prior art.

As described in the above paragraph, the standard for determining non-obviousness is that of a person equipped with ordinary skills in the art (referred to as a POSITA). As is manifestly clear, this hypothetical person, or POSITA, is a fictitious person who is traditionally thought to have some knowledge of the relevant art. Accordingly, if a claimed invention is obvious to the POSITA, the claim is obvious. However, the POSITA has long been at best a grey area in the patent world. In addition, the US courts have issued opinions that have greatly changed the background or capacity of the POSITA.³⁵ Accordingly, when compared with determining novelty, determining non-obviousness is more objective and difficult to predict. In other words, compared to the determination of novelty, the determination of non-obviousness is a question that lacks predictability. Given the murky atmosphere surrounding the determination of obviousness, it is no wonder that determinations of obviousness invalidate many issued patents in the US as well as in other countries. (Simon, 2024, pp. 25-61.)

2.3.2.4 Pitfalls in Prior Art Search

As discussed above, the claim of an issued patent can be invalidated when the claim is found to be neither novel nor obvious. Because the basis for such a determination lies with the prior art, it may be theoretically reasonable to assume that a patent claim will remain valid when a patent applicant searches all the relevant prior art before filing the application and manipulates the scope of the claim based on the result of the prior art search. However, the above assumption is almost always wrong in the real world, because the prior art search has both practical and legal limitations.

The first reason is that there inevitably exists a certain limit to financial resources.³⁶ Any prior art search involves cost, and the quality of a prior art search is strongly dependent upon the amount of money spent and the number of man-hours involved in such a search. Any attempt to search all the databases in the world will require large amounts of time and money. In addition, such a search requires human professionals who can read the prior art, compare the patent claim with the prior art, and determine whether the claim has been

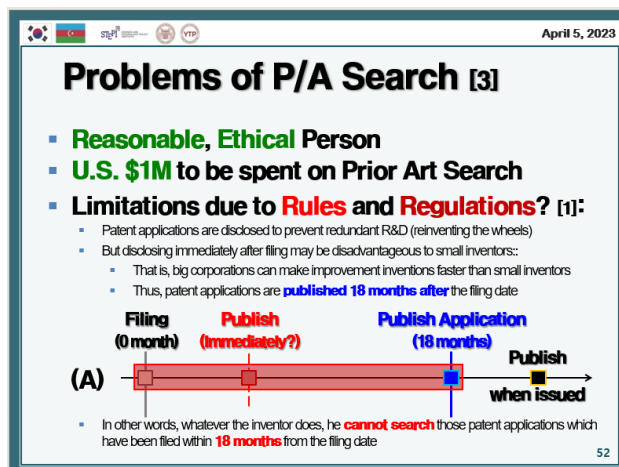
³⁵ <https://tile.loc.gov/storage-services/service/ll/usrep/usrep550/usrep550398/usrep550398.pdf> (date accessed: 10 October 2023)

³⁶ <https://www.upcounsel.com/patent-search-cost> (date accessed: 10 October 2023)

disclosed by an earlier prior art or whether the claim is obvious from the perspective of the POSITA over the prior art.

The second reason is that are various human factors which can change the determination of novelty or non-obviousness. Even if all patent law professionals were to agree that a patent claim is novel and not obvious, a contrary decision by a judge could trump the conviction of such professionals. This human-factor problem becomes more complex in determining non-obviousness. Accordingly, even when the results of a prior art search is seemingly perfect, there exist many possibilities that the final court decision will deviate from the prediction of the parties concerned. The third reason is a legal limitation in that patent applications are usually disclosed to the public not earlier than eighteen months after the date of their filing. Accordingly, when performing a prior search on October 8, 2023, a patent searcher would not be able to find any patent applications filed between April 6, 2022 and October 8, 2023.

[Figure 3-12] Diagram Explaining Patent Applications Published 18 Months After Filing



Source: Author's elaboration.

The diagram in [Figure 3-12] explains that there is a dead window between the filing date of a patent application and another date which is 18 months earlier than the date of filing. Therefore, how many resources a patent applicant may spend, this is a legal limitation that will hinder him or her in searching for the most recent prior art, such as patent applications.

It should be noted that, despite this Section (4), the author does not instruct the reader to rely on the prior art search and, therefore, not to conduct a useless prior art search. Rather, this author wishes to warn the reader that one cannot and should not rely absolutely on the results of prior search.

In addition, the unwary but fortunate inventor may be able to locate the prior art which invalidates any current claims due to a failure to satisfy the novelty requirement or non-obviousness requirement, particularly in cases where the prior art becomes known to the public more than eighteen months before the inventor intends to file the patent application. In such cases, the inventor can analyze the prior art and amend the current claim accordingly in order to satisfy the novelty and non-obviousness requirements.

2.3.2.5 Literal Infringement

To prevent or at least disincentives a competitor from copying the patented invention of a patent owner, the owner can exercise his or her right to exclude others from practicing the patented invention without first obtaining the consent of the patent owner.

However, the greater the expected profit is, the more competitors will attempt to copy the patented invention of the patent owner. A smarter competitor may attempt to patent around the scope of the patent and thereby avoid infringement.

Almost all patent systems recognize various types of patent infringements, including the following: i) direct infringement, ii) indirect infringement, iii) contributory infringement, and iv) induced infringement.³⁷ To streamline the lecture, this author has decided to focus on the simplest type of infringement, namely, the direct infringement.

Direct infringement occurs when a competitor makes, uses, offers to sell, sells, or imports a product (or service) that embodies a patented invention without obtaining the permission of the patent owner. The act of infringement does not require that the accused product (or service) be an exact copy of a product (or service) which is offered by the patent owner, provided that the accused product (or service) meets all of the elements and limitations of

³⁷ <https://www.upcounsel.com/types-of-patent-infringement> (date accessed: 10 October 2023)

a claim in the infringed patent.³⁸

The direct infringement in turn consists of two different types: literal infringement and infringement under the doctrine of equivalents, where the latter is also referred to as “infringement under the DOE” or simply “DOE infringement”. As a preview, determining a literal infringement is very similar to determining novelty, whereas determining an infringement under the DOE is quantitatively similar to determining non-obviousness.

[Figure 3-13] Example of Application of the “all-elements-rule” and “claim charts” to Determine Literal Infringement of a Patent Claim for An Accused Product

April 5, 2023

Literal Infr. – Examples [1]

■ **Claim Chart for Literal Infringement Test [1]:**

CLAIM (Elements, Limitations)	Does P/A Disclose?	
	Yes/No	Where?
A	Yes	침해제품의 processor
B	Yes	침해제품의 I/O module
C	Yes	침해제품의 controller

YES, all-element rule holds. Thus, the claim is **LITERALLY INFRINGED by the product or service.**

CLAIM (Elements, Limitations)	Does P/A Disclose?	
	Yes/No	Where?
A	Yes	침해제품의 processor
B	Yes	침해제품의 I/O module
C	Yes	침해제품의 controller
Q	NO	-

NO, the rule does not hold. Thus, the claim is **NOT Literally Infringed.**

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Source: Author's elaboration.

[Figure 3-13] shows an example of the application of the “all-elements rule” and “claim charts” to determine a literal infringement of a claim by an accused product (or service). When applied to determine a literal infringement, the all-element rule states that a claim is directly infringed when each and every (claim) element and (claim) limitation are present in the accused product (or service).

In particular, the claim is said to be “directly infringed” by the accused product (or service). As in the case of novelty, the claim is also the subject of comparison when determining a literal infringement.

³⁸ <https://bltg-ip.com/what-is-direct-vs-indirect-infringement-of-patents/> (date accessed: 10 October 2023)

A claim chart is usually a two-column table that is employed to assist comparison of a claim with respect to an accused product (or service). To this end, all (claim) elements and (claim) limitations are listed in the left-hand column of the claim chart, while the right-hand column of the chart shows whether the (claim) element and limitation of each row of that column are found in the accused product (or service).

It should be noted that determining a literal infringement using the claim chart is to compare from left to right, rather than from right to left, when a claim is open-ended.³⁹ When a claim is closed,⁴⁰ however, the direction of comparison for determining novelty using the claim chart runs both ways. Because most patents prefer the open-ended claim, it is safe to remember that the determination of novelty using the claim chart is to compare from left to right.

It should also be noted that determining literal infringement is rather objective or straightforward, because all that needs to be done is to make a literal comparison of the claim elements and limitations with respect to the accused product (or service). Accordingly, relatively speaking, there is not much room for the fact finder in determining a case of literal infringement. Of course, such terms as “objective” or “straightforward” are relative terms which are used to compare the determination of a literal infringement with the determination of an infringement under the DOE, which will be explained in detail below.

2.3.2.6 Infringement Under the DOE (Doctrine of Equivalents)

As described above, the literal infringement requires that all claim elements and limitations must be included in an accused product. Accordingly, a competitor may try to evade literal infringement by slightly modifying one or more claim elements or limitations.

[Figure 3-14] shows a few examples in which a competitor can successfully patent around (or evade) a literal infringement of a claim by slightly modifying one claim element or limitation. An exemplary claim includes three elements, an example of which may be a chopstick with an elongated body, a handle covering the distal end of the body, or a hook

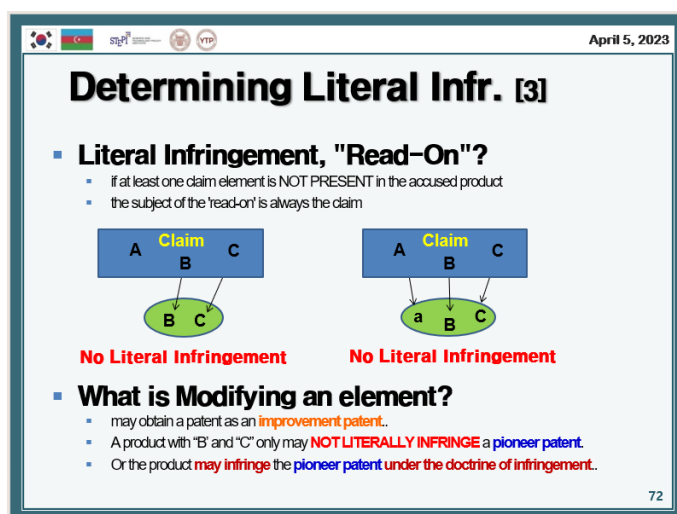
³⁹ <https://www.lsd.law/define/open-ended-claim> (date accessed: 10 October 2023)

⁴⁰ <https://www.lawinsider.com/dictionary/closed-claim> (date accessed: 10 October 2023)

coupled to a handle that allows the user to hang the chopstick on the wall.

In the left-hand panel of [Figure 3-14], a competitor makes a chopstick 'A' which is similar to the patented chopstick, but which does not include a hook. In this case, the chopstick 'A' does not literally infringe the claim, for the hook, as just one element of the claim, is not found in the competitor's chopstick 'A'.

[Figure 3-14] Examples of Evading (or Patenting Around) Literal Infringement by Slightly Modifying One Claim Element or Limitation



Source: Author's elaboration

In the right-hand panel of [Figure 3-14], a competitor makes a chopstick 'B' which is similar to the patented chopstick, but which includes a hook not positioned on the handle but on the body. Due to this slight difference, one element of the claim (i.e. the hook coupled to the handle) is not found in the competitor's chopstick 'B'. Accordingly, the chopstick 'B' does not literally infringe the claim.

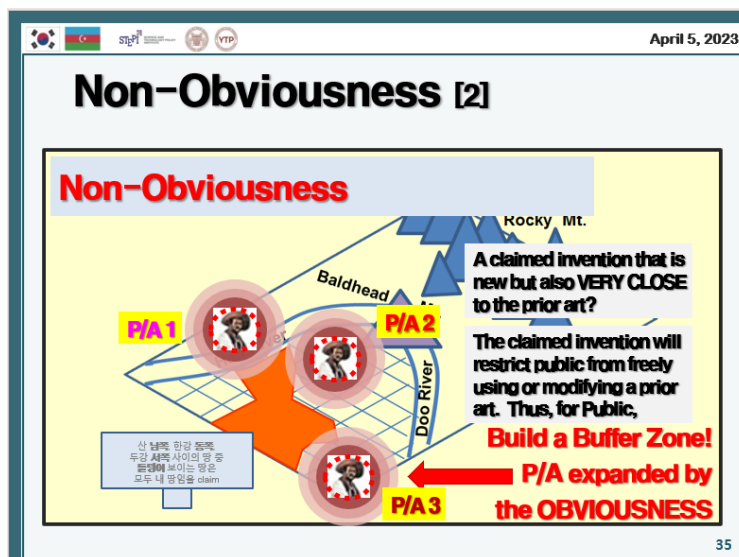
If literal infringement were the only type of direct infringement, a competitor could slightly change one element or limitation of the claim and successfully escape the accusation of infringement. This would weaken severely the extent of patent protection and, consequently, inventors would hesitate to waste their time and energy in conceiving inventions and seeking patent protection. Thus, to better protect the inventor, the patent

system provides an additional type of direct infringement, namely, infringement under the doctrine of equivalents (DOE).

The diagram in [Figure 3-14] explains the essence of infringement under the DOE. When at least a portion of an accused product overlaps with the scope of the patent, that product literally infringes the claim.

Even when an accused product does not overlap with the patent's scope, the product may infringe the claim under the DOE if the accused product falls too closely within the scope of the claim. More precisely, if a different part of the accused product performs (1) at least a substantially similar function (2) in at least a substantially similar way (3) to obtain at least a substantially similar result, the claim has been infringed under the DOE.

[Figure 3-15] Diagram Explaining the Essence of the Non-Obviousness Requirement, i.e. This Non-Obviousness Requirement Sets Up a Buffer Zone Around the Prior Art, Thereby Denying Patents to An Invention Which is Obvious to the Public



Source: Author's elaboration.

As described in the above paragraph, the standard for determining an infringement under the DOE is substantial similarity. However, determining what is substantially similar is not always clear and straightforward. Thus, when compared with literal infringement, determining infringement under the DOE is more objective and difficult to predict.

2.3.3 Selected Topics from IP Lecture 3

2.3.3.1 The Patent System Guarantees Exclusive Rights but May Not Guarantee Use Rights

A capitalistic society grants many different rights to almost every kind of personal property. For example, the owner of a personal property has a “use right,” i.e. the right to use his or her own property. Otherwise, there would be no need to purchase such a property in exchange for money.

Another important right is the “exclusive right,” i.e. the right to exclude others from using a given property without the consent of the owner. Otherwise, such a property would be a community property to be shared by the owner and others. In addition, the owner of personal property is accorded an alienable right, i.e. the right to sell or lease one’s property to others. Otherwise, the owner would have to own that property until he or she dies. Without any of the above bundle of rights, the value of the property would drop significantly.

Like the rights described above, the patent system awards or guarantees a patent owner the exclusive right. Therefore, a patent owner can exclude others from practicing a patented invention without his or her consent. Such a right is the basis of patent rights in the US⁴¹, China⁴², Korea⁴³, Azerbaijan⁴⁴, and many other countries around the world. However, almost all (if not all) patent systems do not guarantee use rights, i.e. patent systems do NOT guarantee the owner of a patent the right to practice his or her own patent. The main reason for this is that most patent systems allow a later inventor (improver) to obtain an improvement patent whose scope is a subset of the pioneer patent of an earlier pioneer inventor.

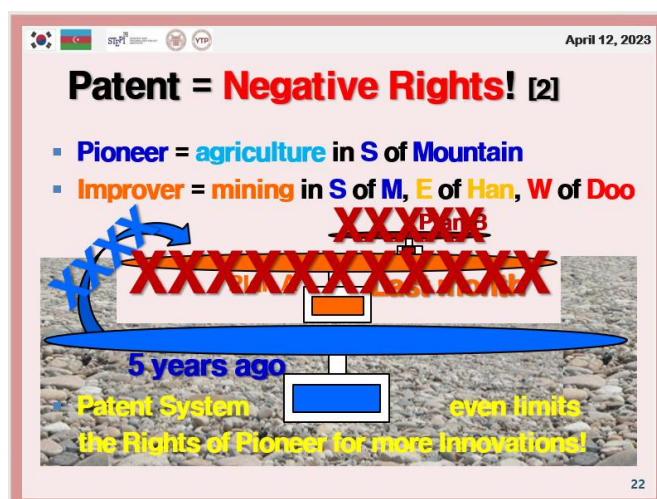
⁴¹ <https://www.bitlaw.com/patent/rights.html> (date accessed: 8 October 2023)

⁴² <https://www.wipo.int/patent-judicial-guide/en/full-guide/china> (date accessed: 8 October 2023)

⁴³ https://content.next.westlaw.com/0-501-7335?__lrTS=20210601170608873&transitionType=Default&contextData=%28sc.Default%29 (date accessed: 8 October 2023)

⁴⁴ <https://wipolex-res.wipo.int/edocs/lexdocs/laws/en/az/az010en.pdf> (date accessed: 8 October 2023)

[Figure 3-16] Diagram Explaining the Overlapping and Blocking Nature of a Pioneer Patent and an Improvement Patent



Source: Author's elaboration.

The diagram in [Figure 3-16] explains the overlapping and blocking nature of a pioneer patent of an earlier inventor (i.e. a pioneer) and an improvement patent of a later inventor (i.e. an improver).

Let's assume a situation in which an improver cannot obtain a patent for an improvement invention which is a subset of the scope of a pioneer invention. A pioneer may obtain a patent which recites an invention with a lower economic value.

When an improver conceives an improvement invention with a higher economic value, the improver knows that he or she cannot obtain a patent for that improvement invention if it does not satisfy the novelty and/or non-obviousness requirements. As a result, improvers are not incentivized to invest their time, energy, and money in finding improvement inventions, which in turn leads to a market failure in that the public is not able to enjoy improvement inventions with higher economic values.

For this reason, the patent system awards improvers with opportunities to obtain improvement patents for those improvement inventions which display patentable features that are absent from the corresponding pioneer patents. However, there is a catch. As discussed above, the scope of an improvement patent is a subset of a pioneer patent.

Therefore, when an improver practices his improvement patent, the improver literally infringes the pioneer patent. As a result, the improver of the improvement patent cannot practice his or her own patent.

It is appreciated that the patent system does not even allow a pioneer to practice his or her improvement patent, as the improver is the owner of the improvement patent and can exclude the pioneer from practicing the improvement patent.

2.3.3.2 TRIZ to Inventing Processes

TRIZ (in Russian, теория решения изобретательских задач) is referred to as “the theory of solving inventive problems”. TRIZ was created by Genrikh Saulovich Altshuller, an engineer, inventor, and writer from Azerbaijan.

Due to its inherent nature about creating creativity, it seems very worthwhile to include TRIZ in this material and to apply numerous TRIZ principles to the process of inventing in view of the patent system. To this end, this author first summarized the forty basic inventive principles of TRIZ.⁴⁵

1) TRIZ Principle No. 1 = Segmentation:

- 1-1. Divide an object into independent parts.
- 1-2. Make an object easy to disassemble.
- 1-3. Increase the degree of fragmentation or segmentation.

Segmentation consists in splitting a complex task into a number of simpler tasks, each of which is easier to solve, using a variety of tools ranging from very simple but effective ones such as Pareto analysis, to very complex ones like Quality Function Deployment.

2) TRIZ Principle No. 2 = Taking Out:

- 2-1. Separate an interfering part or property from an object or single out the only necessary part (or property) of an object.

⁴⁵ <https://www.quality-assurance-solutions.com/Triz-Inventive-Principles-1.html> (Date accessed: 8 October 2023)

2-2. Perform a value analysis of each process step.

3) TRIZ Principle No. 3 = Local Quality:

3-1. Change an object's structure from uniform to non-uniform, e.g. change an external environment (or external influence) from uniform to non-uniform.

3-2. Make each part of an object function in the conditions most suitable for its operation.

3-3. Make each part of an object fulfill a different and useful function.

For example, a particular part can be improved by singling out a specific object (product, service, processes, activities or tools), and then improving it to better provide its function, or place it in an environment so that it is optimally useful.

4) TRIZ Principle No. 4 = Asymmetry:

4-1. Change the shape of an object from symmetrical to asymmetrical.

4-2. If an object is asymmetrical, increase its degree of asymmetry.

Symmetric solutions might be more appealing to the human eye but, occasionally, asymmetric solutions might provide better functionality.

5) TRIZ Principle No. 5 = Merging:

5-1. Bring identical or similar objects closer together (or merge them); assemble identical or similar parts to perform parallel operations.

5-2. Make operations contiguous or parallel; bring them together in time.

The principle of merging or combining means bringing together things which exist or happen at the same time or in the same place, e.g. when designing QMS, several tools can be combined into a powerful method. One element of the QMS is corrective action.

6) TRIZ Principle No. 6 = Universality:

Make a part or object perform multiple functions; eliminate the need for other parts.

This inventive principle is used where objects can perform multiple functions. It is particularly useful when an object can be eliminated by having another object perform the same function.

7) TRIZ Principle No. 7 = Nesting:

7-1. Place one object inside another; place each object, in turn, inside another, and so forth.

7-2. Make one part pass through a cavity in the other.

“Nesting” means putting one thing inside another. In management systems, PDCA loops are nested in such a way that there is a strategic PDCA loop with a one-year period and an operational PDCA loop with a one week/month period.

8) TRIZ Principle No. 8 = Anti-Weight:

8-1. To compensate for the weight of an object, merge it with other objects that provide lift.

8-2. To compensate for the weight of an object, make it interact with the environment (e.g. use aerodynamics, hydrodynamics, buoyancy and other forces).

Use the force field analysis to identify which force causes a shift in an undesirable direction, then apply a compensating force to steer the system in the opposite direction.

9) TRIZ Principle No. 9 = Preliminary Anti-Action:

9-1. If it is necessary to perform an action with both harmful and useful effects, this action must be replaced with countermeasures to control the harmful effects.

9-2. Beforehand, create stresses in an object that will oppose the known undesirable working stresses afterwards.

Preliminary anti-action is a risk avoidance tool that seeks to eliminate or diminish any possibility of risk by using preliminary countermeasures. Failure Modes and Effects Analysis

(FMEA) is a systematic method of evaluating a process in order to ascertain where and how it might fail and to assess the possibility of implementing preventive actions.

10) TRIZ Principle No. 10 = Preliminary Action:

10-1. Perform, before it is needed, the required change of an object (either fully or partially).

10-2. Pre-arrange objects so that they can come into action from the most convenient place and without delaying their delivery.

The principle of 'preliminary action' means performing preparatory activities to speed up the execution of planned tasks.

For example, (1) prepare a maintenance program before starting to sell products, (2) prepare a RISK analysis; identify and perform necessary preventive actions to reduce risks before starting new processes, and take (3) preventive action from a corrective action, and so on.

11) TRIZ Principle No. 11 = Beforehand Cushioning:

Prepare emergency means beforehand to compensate for the relatively low reliability of an object.

This inventive principle is used to prepare things ahead of an event so as to increase reliability. To increase the reliability of a system, use a redundancy of system elements.

12) TRIZ Principle No. 12 = Equipotentiality:

In a potential field, limit changes in position (e.g. change the operating conditions to eliminate the need to raise or lower objects in a gravity field).

This principle of "equipotentiality" in quality management refers to the equal treatment of all customers. It means forming communication channels at all levels of management including internally, and between the customer and the supplier.

In a company, the organization takes care to eliminate the need to transfer operations from one management level to another.

13) TRIZ Principle No. 13 = The Other Way Around:

13-1. Invert the action(s) used to solve the problem (e.g. instead of cooling an object, heat it).

13-2. Make movable parts (or the external environment) fixed, and fixed parts movable.

13-3. Turn the object (or process) “upside down.”

The inventive ‘other way round’ principle means doing things in the opposite way to normal practice (examples of inverted actions).

For example, use the KANBAN system. Instead of pushing a product on to the market, let the market pull the product from your company. Design, produce and sell products as a kit and have customers assemble the products themselves.

14) TRIZ Principle No. 14 = Spheroidality - Curvature:

14-1. Instead of using rectilinear parts, surfaces or forms, use curvilinear ones; move from flat surfaces to spherical ones; from parts shaped as a cube (parallelepiped) to ball-shaped structures.

14-2. Use rollers, balls, spirals, and domes.

14-3. Go from linear to rotary motion, use centrifugal forces.

This inventive principle in the field of quality management means performing activities in such a way as to enable smooth operations requiring less time and effort at lower cost. Examples include ergonomic desk/work-station design, quality circles, or the circular work cell.

15) TRIZ Principle No. 15 = Dynamics:

15-1. Allow (or design) the characteristics of an object, external environment or process to change to the most optimal characteristics, or find an optimal operating condition.

15-2. Divide an object into parts capable of movement relative to each other.

15-3. If an object (or process) is rigid or inflexible, make it movable or adaptive.

This inventive principle means involves the creation of systems that are able to adapt to external and internal changes. In terms of quality management, this principle can be used to create a system of interconnected processes capable of quickly responding to all requirements.

For example, the ability to create, on the fly, custom-customer shipping labels, to implement an in-house design process capable of accommodating any customer requirement, or to use CAD for design.

16) TRIZ Principle No. 16 = Partial/Excessive Actions:

If an objective is hard to achieve 100% using a given solution or method, the problem may be considerably easier to solve by using “slightly less” or “slightly more” of the same method. When it is not possible to achieve perfection, we can consider doing things at less than or even more than 100%. Using Pareto analysis, we can prioritize issues and decide to solve a few vital options while ignoring many trivial ones. Examples may include a procedure for corrective actions according to the ISO 9001 requirements, sampling, auditing, or even a management review.

17) TRIZ Principle No. 17 = Another Dimension:

17-1. Move an object in a two- or three-dimensional space.

17-2. Use a multi-story arrangement of objects instead of a single-story one.

17-3. Tilt or re-orient the object, lay it on its side.

17-4. Use 'another side' of a given area.

When proven tools and techniques cannot solve problems, one can try to use another dimension, for example, transform the functional structure of an organization to a more process-oriented organization; or one can redistribute responsibility or authority, for example, split the authority for project management between the project manager and the department manager.

18) TRIZ Principle No. 18 = Mechanical Vibration:

18-1. Cause an object to oscillate or vibrate.

18-2. Use piezoelectric vibrators instead of mechanical ones.

18-3. Increase the object's frequency (even up to the ultrasonic level).

18-4. Use the object's resonant frequency.

18-5. Use combined ultrasonic and electromagnetic field oscillations.

This inventive principle implies a way of improving system and process performance by increasing the PDCA loop frequency. By varying the frequency, and the amount and content of information in a loop, it is possible to create different effects.

19) TRIZ Principle No. 19 = Periodic Action:

19-1. Instead of continuous action, use periodic, pulsating actions.

19-2. If an action is already periodic, change its magnitude or frequency.

19-3. Use the pauses between impulses to perform a different action.

This inventive principle entails the use of periodic actions instead of continuous vibration. In quality management, use an alternate PDCA cycle to bring about incremental improvements, and a PDSA cycle to achieve quantum leap improvements.

20) TRIZ Principle No. 20 = Continuity of Useful Action:

20-1. Carry on working continuously; make all the parts of an object work at full load, all the time.

20-2. Eliminate all idle or intermittent actions or work.

The inventive principle of 'continuity of useful action' refers to the maintenance of a useful function or the elimination of idle time, or to eliminating or shortening the duration of harmful functions.

21) TRIZ Principle No. 21 = Skipping:

Conduct a given process or certain stages (e.g. destructible, harmful or hazardous operations) at high speed.

Skipping or rushing through involves conducting a hazardous or costly operation at high speed in order to minimize harmful effects. For example, any conflict that causes a delay (a harmful function) will cause a loss of revenue (an effect). For that reason, conflicts should be resolved as quickly as possible.

22) TRIZ Principle No. 22 = Convert Harm to Benefit:

22-1. “Blessing in disguise” or “Turning lemons into lemonade.”

22-2. Use harmful factors (particularly, harmful effects of the environment or surroundings) to achieve a positive effect.

22-3. Eliminate the primary harmful action by adding it to another harmful action to resolve the problem.

22-4. Amplify a harmful factor to such a degree that it is no longer harmful.

“Blessing in Disguise” means converting a harmful function/effect into a useful function or effect. For example, we can run a process at the maximum process setting and discover the created defects from this possible setting, after which we can address these possible defects. For example, we can create documentation that shows these defects.

Another example is the customer corrective action. During this process, one can determine the root cause of the corrective action and then correct the root cause. Then, one can take these actions and apply them to other products to prevent possible issues.

23) TRIZ Principle No. 23 = Feedback:

23-1. Introduce feedback (referring back, cross-checking) to improve a process or action.

23-2. If feedback is already used, change its magnitude or influence.

Feedback information refers to information about the process results that is sent to the

process owner. This allows the process owner to adjust its current and future operations in order to achieve the desired objective.

24) TRIZ Principle No. 24 = Intermediary:

24-1. Use an intermediary carrier article or intermediary process.

24-2. Merge one object temporarily with another (i.e. one that can be easily removed).

This 'intermediary' refers to the use of an intermediate carrier between two objects, systems or persons. For example, a consultant provides unbiased professional or expert advice in a particular area such as quality management, security, safety, environment, etc. In another example, the QMS Auditor is a person appointed and authorized to examine documents and records, processes and other elements of a quality system in order to determine their conformity with the requirements.

25) TRIZ Principle No. 25 = Self-Service:

The inventive principle 'self-service' implies that either the customer or the product serves itself or uses waste resources.

26) TRIZ Principle No. 26 = Copying:

The inventive principle 'copying' implies the use of cheaper, disposable and inexpensive copies as a substitute for expensive, fragile or difficult-to-replace objects. It may involve replacing an object or process with optical copies.

27) TRIZ Principle No. 27 = Cheap Short-Life Object:

The inventive principle of 'cheap, short-life objects' implies the replacement of an expensive object, system or process with inexpensive, short-life objects, systems or processes while compromising certain qualities such as durability and maintainability.

Examples may include the use of disposable tools when these would be cheaper than the cost of cleaning, sharpening or maintaining non-disposable tools (i.e. medical surgery instruments), or the use of CAD and CAM tools to design and evaluate the design and use

computers to assist with all the operations involved in a manufacturing process.

28) TRIZ Principle No. 28 = Mechanics Substitution:

This principle entails replacing an existing concept, system, process or object with another kind which allows easier implementation of the required function with better characteristics.

29) TRIZ Principle No. 29 = Pneumatics and Hydraulics:

29-1. The inventive principle if ‘pneumatics and hydraulics’ refers to the use of the intangible parts of an object or system instead of the tangible ones.

29-2. For example, the flexibility of an organization with regard to the interface between that organization and its target customers. Front office personnel should be competent in communicating with customers and be able to solve problems quickly. The Kano diagram can be used to determine the ‘customer excitement’ requirements.

30) TRIZ Principle No. 30 = Flexible Shells and Thin Films:

The inventive principle ‘thin films’ refers to the use of thin structures on the borders between systems, processes and objects with special characteristics such as low cost, low space, low weight, and high flexibility. They can be used to separate, isolate, filter and protect.

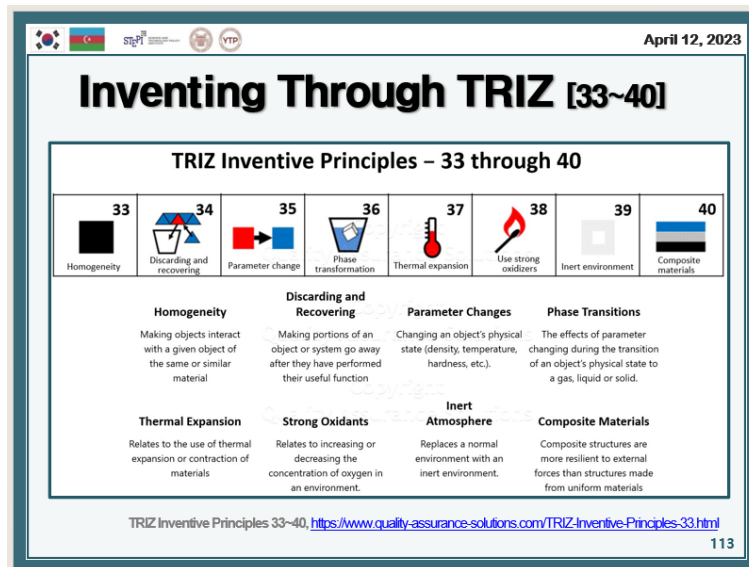
31) TRIZ Principle No. 31 = Porous Material:

The inventive principle ‘porous materials’ refers to the act of making an object or system porous or the addition of porous elements. If the object or system is already porous, the pores can be used to introduce a useful function into the organization. Porous materials allow some substances to pass through and block out others.

32) TRIZ Principle No. 32 = Color Changes:

In quality management, use color or a change of color to draw attention to changes in status that require instant action, such as the Andon system, Kanban, etc.

[Figure 3-17] TRIZ Principles Nos. 33 to 40



Source: Author's elaboration

To simplify, TRIZ Principles Nos. 33 to 40 have been summarized in a figure, where [Figure 3-17] shows the remaining TRIZ Principles Nos. 33 to 40.⁴⁶

2.3.3.3 Examples of applying TRIZ to the inventive processes

Certain TRIZ principles are actually very helpful in streamlining the brain of an inventor, as they allow the inventor to think about modifications of or alternatives to a problem he or she aims to solve. However, many TRIZ principles cannot really help an inventor to satisfy the novelty or non-obviousness requirement. In addition, many TRIZ principles may not be valuable to a competitor who strives to patent around (or avoid) literal infringement or infringement under the DOE.

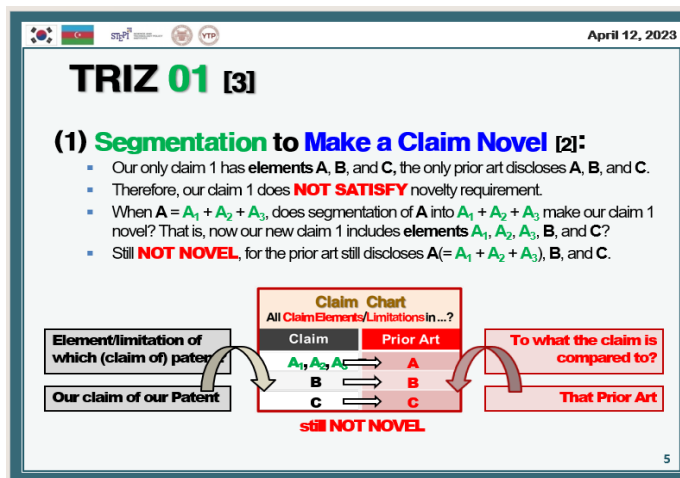
⁴⁶ TRIZ Inventive Principles, <https://www.quality-assurance-solutions.com/Triz-Inventive-Principles-1.html> (date accessed: 8 October 2023)

1) TRIZ Principle No. 1 does not convert a non-novel invention to a novel invention

TRIZ Principle No. 1 teaches to segregate an element of an existing product in order to come up with a solution to a problem. For example, a product includes A, B, and C, but a prior art has disclosed A, B, and C. The inventor can attempt to segment A into A1, A2, and A3, and thereby find the solution to the problem more easily.

Although this segmentation may help to create a new solution to the problem, the new solution unfortunately cannot satisfy the novelty requirement.

[Figure 3-18] TRIZ Principle No. 1 Does Not Affect the Determination of Novelty



Source: Author's elaboration.

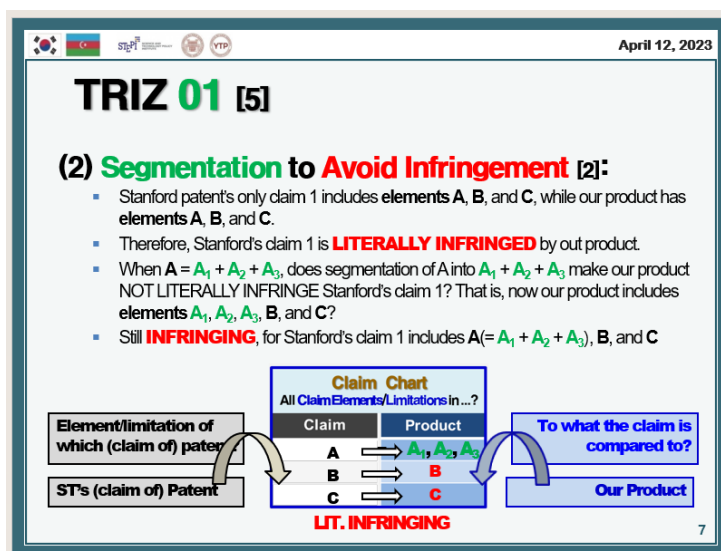
The diagram in [Figure 3-18] shows why TRIZ Principle No. 1 does not and cannot convert an old, non-novel invention into a new, novel invention. As shown in the figure, the new invention includes all sub-elements, regardless of whether the element A can be segmented into 3, 4 or 5 sub-elements. Therefore, the all-elements rule still holds and the claim is still not novel.

In other words, the all-elements rule still applies after the segmentation of an element of a non-novel claim. Therefore, the new invention, which includes A1, A2, A3, B, and C, has anyway been disclosed by the prior art, which includes A (= A1 + A2 + A3), B, C. As a result, the new invention cannot be patented because the segmented claim still cannot satisfy the novelty requirement over the prior art.

2) TRIZ Principle No. 1 cannot convert literally infringing products to non-infringing products

A competitor may apply the TRIZ Principle No. 1 in order to patent around the patent of a patent owner. Let's assume that claim no. 1 of the patent recites a product, including A, B, and C, and that the competitor's current product 'X' also includes A, B, and C. By segmenting A into $A_1 + A_2 + A_3$, the competitor can try to avoid a literal infringement.

[Figure 3-19] TRIZ Principle No. 1 Does Not Affect the Determination of a Literal Infringement



Source: Author's elaboration.

[Figure 3-19] contains a diagram explaining why TRIZ Principle No. 1 does not and cannot convert a literally infringing product into a non-infringing product. As shown in the figure, the new product of the competitor includes all such sub-elements, regardless of whether the element A of the infringing product can be segmented into 3, 4 or 5 sub-elements.

In other words, the all-elements rule still applies after such segmentation of a competitor's product which has infringed the claim of the patent owner. Therefore, all elements of the claim (A, B, and C) of the patent are still present in the new product of the competitor, which includes $A (= A_1 + A_2 + A_3), B$, and C. Therefore, the new product cannot avoid a literal infringement.

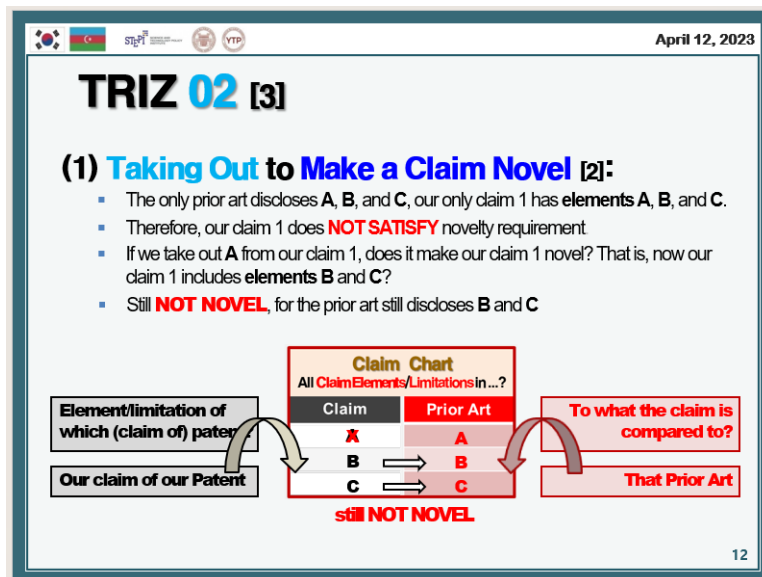
Due to such reasons, these examples (1) and (2) clearly demonstrate that none of the TRIZ inventive principles can assist an inventor in converting their non-novel claim into a novel claim, and also that they cannot assist competitors in converting a literally infringing product into a non-infringing product.

3) TRIZ Principle No. 2 does not convert a non-novel invention to a novel invention

TRIZ Principle No. 2 teaches to take out (or eliminate) at least one element of an existing product in order to come up with a solution to a problem. For example, when a product includes elements A, B, and C, one can attempt to take 'A' out of the existing product. In this way, one can easily find the solution to the problem.

Although this taking out or elimination of an element may help to create a new solution to a problem, the solution unfortunately cannot satisfy the novelty requirement.

[Figure 3-20] TRIZ Principle No. 2 Can Affect the Determination of Novelty



Source: Author's elaboration.

The diagram in [Figure 3-20] shows why TRIZ Principle No. 2 cannot convert a non-novel invention into a novel invention. As shown in the figure, an old invention includes elements A, B, and C, while a new solution takes out element A. Therefore, the new solution includes

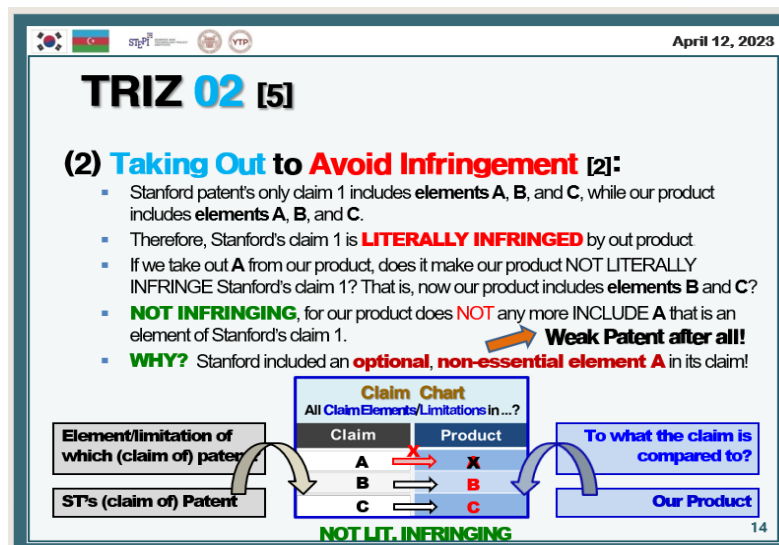
only B and C. However, B and C have still been disclosed by a prior art. As such, the all-elements rule still applies.

In other words, the all-elements rule still applies after the taking out or elimination of an element of a non-novel claim. Therefore, the new invention, which includes B and C, has been anyway disclosed by the prior art, which includes A, B, and C. As a result, the new claim cannot be patented because the claim with one element taken out still cannot satisfy the novelty requirement over the prior art.

4) TRIZ Principle No. 2 can convert a literally infringing product into a non-infringing product

A competitor may apply the TRIZ Principle No. 2 in order to patent around the patent of a patent owner. Let's assume that claim 1 of the patent recites a product including elements A, B, and C, and that the competitor's current product 'X' also includes A, B, and C. By eliminating A from the product, the competitor can try to avoid literal infringement.

[Figure 3-21] TRIZ Principle No. 1 Can Affect the Determination of A Literal Infringement



Source: Author's elaboration.

The diagram in [Figure 3-21] explains why TRIZ Principle No. 2 can convert a literally infringing product into a non-infringing product. As shown in the figure, the new product of

the competitor no longer includes A, and his or her new product only includes B and C.

In this case, claim element A is no longer present in the competitor's product, and the all-element rule no longer holds. Therefore, the competitor's new product can successfully evade a literal infringement of the patent owner's claim.

2.3.3.4 Inventing, TRIZ Principles, Patenting, and Lessons

As proven in the real world, various TRIZ inventive principles are extremely helpful tools in providing new and creative perspectives for the inventor. With these principles, inventors may easily find certain features which they had previously ignored or to which they did not give due attention.

However, the above examples clearly show that none of the TRIZ inventive principles can help inventors to convert their non-novel claims into novel claims, and that they cannot help competitors to convert a literally infringing product into a non-infringing product.

In other words, inventing can be one thing, while patenting is another. As a result, the renders have to be aware of these pros and cons and be very careful in applying such TRIZ principles to both the inventing processes and the patenting processes.

2.3.4 Selected Topics from IP Lecture 4

2.3.4.1 Claim Structure and Its Parts

A claim is structured as a (claim) preamble, a transition phrase, and a (claim) body. In the US, the transition phrase includes the terms "comprises," "consists of," and "essentially consists of." When a claim employs "comprises" as the transition phrase, it is called an "open-ended claim," whereas it is called a "closed claim" when it uses "consists of" as the transition phrase, and a "partially closed claim" when it uses "essentially consists of."

Claim Preamble

The claim preamble is the beginning part of a claim. The preamble typically describes the technological or legal category or usage of the claimed invention. It is appreciated that the scope of the claim is determined by the elements and/or limitations included in the body of

the claim. Accordingly, the preamble does NOT usually affect a claim's scope.

Of course, there are exceptions to this rule. Sometimes the words in the preamble are construed as the elements or limitations which restrict the scope of the claim. When a part of the preamble assigns minimal meaning or importance to a claim element or limitation, that portion of the preamble is deemed to limit the claim's scope. However, when a part of the preamble is necessary in construing the claimed invention, that portion may be deemed to limit the claim's scope.

In giving their opinions, the US courts have repeated these exceptions using different words. For example,

- ① A term in a preamble is a claim limitation when it gives "life and meaning" to the claim.

⁴⁷

- ② A term in a preamble is a claim limitation that limits the claim's scope when it is necessary for the claim's construction. ⁴⁸

- ③ A term in a preamble used to be a claim limitation if it describes the intended use or purpose of the claimed invention. ⁴⁹

2.3.4.2 Claim Transition Phrase

As explained above, the transition phrase of US patents includes "comprises," "consists of," and "essentially consists of." When a claim employs "comprises" as the transition phrase, it is called an "open-ended claim," whereas it is called a "closed claim" when it uses "consists of" as the transition phrase, and a "partially closed claim" when it uses "essentially consists of."

1) Open-Ended Claims and Their Pros/Cons

An open-ended claim with "comprises" as its transition phrase is characterized as having the broadest scope. The reason for this is that, as long as an accused product includes all of the

⁴⁷ Catalina Marketing Int'l v. Coolsavings.com, Inc., 289 F.3d 801, 810 (Fed. Cir. 2002)

⁴⁸ Vaupel Textilmaschinen KG v. Meccanica Euro Italia S.P.A., 944 F.2d 870, 880 (Fed. Cir. 1991)

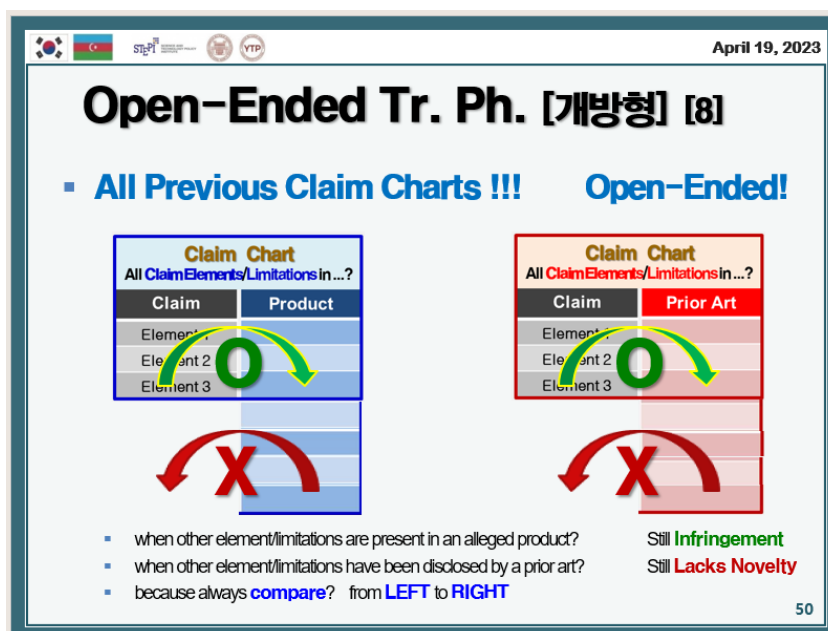
⁴⁹ Intirtool, Ltd. v. Texar Corp., 369 F.3d 1289, 1296 (Fed. Cir. 2004)

claim elements and claim limitations, it literally infringes the claim when the claim employs “comprises” as its transition phrase.

[Figure 3-22] includes a table that explains this feature. For example, even if the accused product includes other parts which are not found in the claim, the product still literally infringes the claim in as far as all the claim elements and limitations are found in the product. (See the green and red arrows in the left-hand panel of the figure.) In this context, the open-ended claim has the broadest scope.

Therefore, the open-ended claim can be most beneficial in preventing or suppressing an infringement, since it may be extremely difficult for a competitor to produce a non-infringing product which may not include at least one element or limitation of the claim.

[Figure 3-22] Broadest Scope and Pros and Cons of an Open-Ended Claim



Source: Author's elaboration.

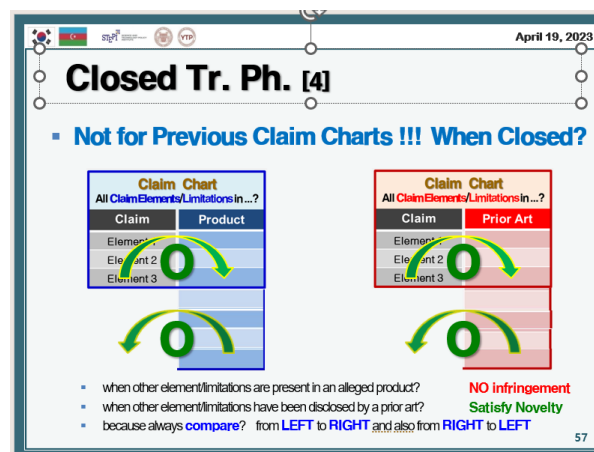
However, the open-ended claim has its own drawbacks for the same reason. When a prior art discloses all the elements and limitations of the claim, the claim is not novel. Even when the prior art discloses other elements which are not part of the claim, that claim is still not novel. In this context, the open-ended claim with the broadest scope is most prone to failure to satisfy the novelty requirement. This feature is characterized by the green and red arrows in the left-hand panel of the figure.

2) Closed Claims and Their Pros/Cons

A closed claim employing “consists of” as its transition phrase is characterized as having the narrowest scope. This is because the scope of such a claim is limited to that which is identical to the claim itself. Therefore, when a prior art discloses all the claim elements and limitations but also discloses an additional element, the claim is novel over that prior art when the claim employs “consists of” as its transition phrase.

Figure 1-20 includes a table explaining this feature. As shown in the right-hand panel, when a prior art discloses other parts which are not included in the claim, the prior art does not qualify in denying the novelty of the claim as far as the claim employs “consists of” as its transition phrase. (See the green and red arrows in the right-hand panel of the figure.) In this context, the closed claim has the narrowest scope and is most beneficial in proving the novelty (and non-obviousness) of the claim.

[Figure 3-23] Narrowest Scope and Pros and Cons of a Closed Claim



Source: Author's elaboration.

However, the closed claim similarly has its own drawbacks for the same reason. When a competitor sells a product which is similar to the claimed invention but also includes at least one part which is not a claim element or limitation, the product does not literally infringe the claim. In this context, the closed claim with the narrowest scope is most vulnerable to competitor's patenting around. (See the green and red arrows in the left-hand panel of the figure.)

2.3.4.3 Claim Body

The claim body is where the claim elements are recited along with the claim limitations. In general, each claim must consist of a single sentence that ends with a period (.). Accordingly, the claim body is a very complex sentence that may include many sub-sentences and phrases. To this end, the claim body has to be written according to the various grammatical rules governing claims. For the sake of simplification, such details are to be omitted here. Readers who are interested can find them in TCKL Lecture 5, dated April 19, 2023.

2.3.5 Selected Topics from IP Lecture 5

2.3.5.1 Why Claim Construction is Required

A claim recites numerous elements and limitations, all of which determine the scope of that claim. Accordingly, the claim defines the metes and bounds of its exclusivity. For this reason, a patent owner and an accused infringer always fight about the right meanings of the claim itself, the claim elements and the claim limitations in the court room day in and day out. The patent owner always argues that his patent is valid and has been infringed, and presents his version of the claim's meaning to the court. Conversely, the accused infringer always argues that the claim is invalid or, even when the claim is valid, that he did not infringe the claim, and presents his version of the claim's meaning to the court. The court then has to decide which of the conflicting versions embodies the claim's true meaning. However, this decision is very difficult to make because the claim has to be compared with the prior art, as well as with the accused product or service, so various experts come and give their opinions about the technical meaning of the claim elements and limitations and the like. Accordingly, interpreting the claim properly is not an easy task. Due to this reason, the US courts have

developed their own rules for the interpretation of claims, and such rules are conveniently referred to as the “claim construction.”

2.3.5.2 Canons of Claim Construction

The US courts have delivered many opinions about claim construction, and there can be many different canons. However, the US patent community refers to the following six canons.⁵⁰

The 1st canon asserts that a claim must be interpreted such that its claim elements or claim limitations are presumed to have an ordinary and customary (i.e. dictionary) meaning. This author has selected the following case to explain this canon:

- Sample case: *G.E. Lighting Solutions v. AgiLight*, 750 F.3d 1304 (Fed. Cir. 2014)⁵¹

The 2nd canon states that a patent applicant can be his or her own lexicographer. That is, a patent applicant can provide a special definition of a certain term. In this case, the meaning selected by the applicant trumps the dictionary meaning of that term. The author has selected the following case to explain this canon:

- Sample case: *X2Y Attenuators v. ITC and Intel*, 757 F.3d 1358 (Fed. Cir. 2014)⁵²

The 3rd canon asserts that a claim must be interpreted to have a consistent meaning. The author has selected the following case to explain this canon:

- Sample case: *Markem-Imaje Corp. v. Zipher Ltd.*, 657 F.3d 1293 (Fed. Cir. 2011)⁵³

The 4th canon states that a claim must be interpreted such that its scope includes the applicant’s preferred embodiment. Accordingly, when a patent applicant explains that a certain embodiment is his preferred one, all elements and limitations included in the embodiment can be imported into the claim, thereby greatly narrowing the scope of the claim. The author has selected the following case to explain this canon:

⁵⁰ https://patentlyo.com/patent/2004/11/the_canons_of_c.html (Date accessed: 8 October 2023).

⁵¹ *G.E. Lighting Solutions v. AgiLight*, 750 F.3d 1304 (Fed. Cir. 2014), <https://cafc.uscourts.gov/5-01-2014-13-1267-ge-lighting-solutions-llc-v-agilight-inc-opinion-13-1267-opinion-4-28-2014-1/> (Date accessed: 8 October 2023).

⁵² *X2Y Attenuators v. ITC and Intel*, 757 F.3d 1358 (Fed. Cir. 2014), <https://fedcircuitblog.com/wp-content/uploads/2019/09/PLAINTIFF-APPELLANT-X2Y-ATTENUATORS-LLC%E2%80%99S-OPENING-BRIEF.pdf> (Date accessed: 8 October 2023).

⁵³ *Markem-Imaje Corp. v. Zipher Ltd.*, 657 F.3d 1293 (Fed. Cir. 2011), <https://law.justia.com/cases/federal/appellate-courts/cafc/13-1032/13-1032-2012-12-12.html> (Date accessed: 8 October 2023).

- Sample case: *Trading Tech. Int'l v. eSpeed, Inc.*, 598 F.3d 1340 (Fed. Cir. 2010) ⁵⁴

The US courts have also adopted the “doctrine of claim differentiation,” which states that “a patent applicant shall not draft two claims to have the same scope” and, therefore, that “each claim shall have a different scope.” Accordingly, a certain word or phrase recited in an independent claim must have a wider scope than the same word or phrase recited in the dependent claim. The caveat here is that this doctrine will be trumped when other canons of claim construction yield a different interpretation of the claim.

The 5th canon states that a claim must be interpreted in view of a patent specification, but that no claim limitation can be imported into the claim. In other words, one must refer to the patent specification to interpret the claim, but one cannot import any word or phrase of the specification into the claim. The author has selected the following case to explain this canon:

- Sample case: *Unique Concepts, Inc. v. Brown*, 939 F.2d 158 (Fed. Cir. 1991) ⁵⁵

For the sake of simplification, such details are omitted here. Readers who are interested in such details can find them in TCKL Lecture 5 dated May 3, 2023.

2.4 Conclusion

Technology transfer and commercialization are crucial to growing the national economy. In particular, technology transfer and the commercialization of home-grown technologies and their intellectual properties can bring in many benefits, as well as helping to secure sustainable growth of the national economy. In this regard, Azerbaijan’s central government needs to set up and fine-tune its technology and IP management policies.

In doing so, it is strongly recommended that the government take account of how to develop and nurture the human resources required for such technology transfer and commercialization.

⁵⁴ *Trading Tech. Int'l v. eSpeed, Inc.*, 598 F.3d 1340 (Fed. Cir. 2010), <https://caselaw.findlaw.com/court/us-federal-circuit/1509115.html> (Date accessed: 8 October 2023).

⁵⁵ *Unique Concepts, Inc. v. Brown*, 939 F.2d 158 (Fed. Cir. 1991), <https://casetext.com/case/unique-concepts-inc-v-brown-2> (Date accessed: 8 Oct, 2023.)

It is vital to understand that technological innovations spring up out of nowhere and cannot be automatically protected under the IP laws. To the contrary, professionals equipped with diverse expertise have to be involved in creating technological innovations, protecting such innovations and IPs and, thereafter, transferring or commercializing the technologies.

As discussed along with [Figure 3-4] to [Figure 3-6], technological innovations, inventions, and patents, and their transfer or commercialization, require a complex IP ecosystem comprising upstream, midstream, and downstream elements. Each such layer of the IP ecosystem is interrelated to and interacts with the others. In addition, each layer of the IP ecosystem requires professionals who are equipped with a wealth of expertise and come from diverse backgrounds.

It should be remembered that it is the innovators who can create innovations and inventions, that it is the legal professionals who can protect innovations and inventions under IP laws, and that it is the technological or business experts who can commercialize or transfer such innovations, inventions or patents. Thus, Azerbaijan's government should set up IP management policies that take account of how to recruit and then properly nurture a pool of home-grown professionals.

In short, Azerbaijan will need to recruit professionals capable of carrying out the difficult process of setting up a domestic IP ecosystem, and the government will need to accelerate such a process by helping - directly and indirectly - to educate and transmit the basic knowledge to such professionals.

This chapter provides an exemplary pool of knowledge which can be used to educate and nurture the professionals who will be responsible for setting up each layer of the IP ecosystem. In the virtuous IP ecosystem, technically competitive and commercially valuable technologies can be proactively transferred to Azerbaijan's domestic industries, and they can also be proactively commercialized by such professionals or by Azerbaijan's industries.

3. Startup Programs

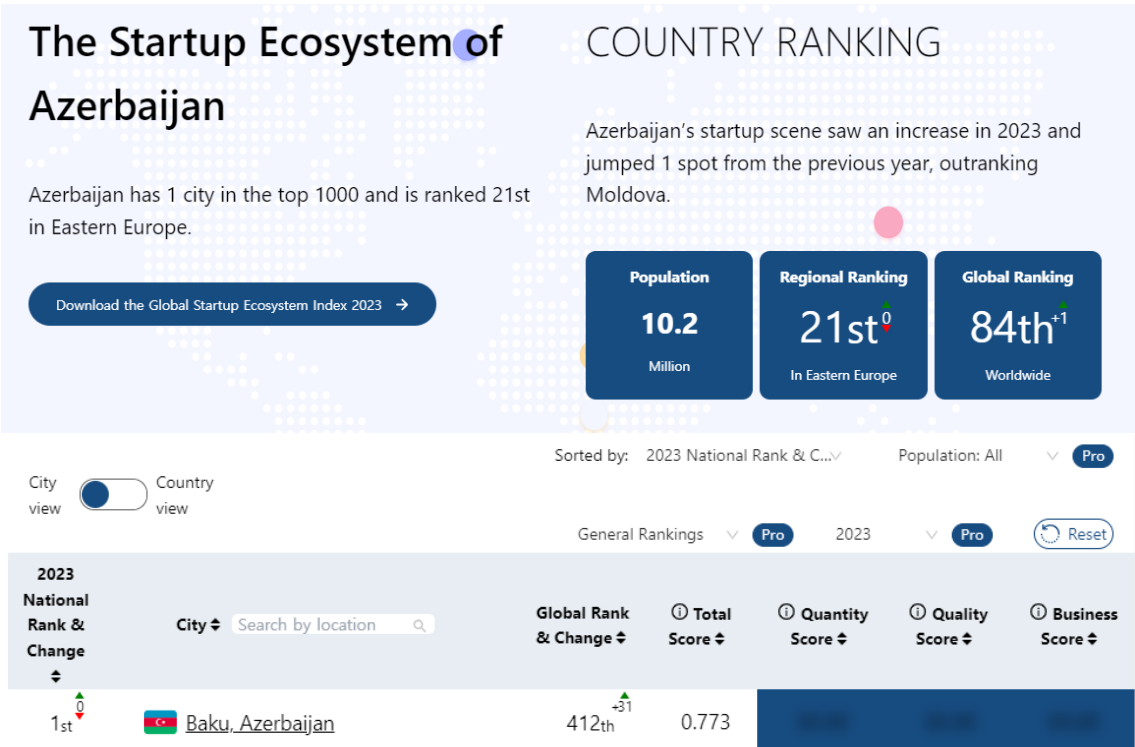
3.1 Startup Incubation Program in Azerbaijan

3.1.1 The status of startup promotion in Azerbaijan

The startup and incubator support programs in Azerbaijan are led by the government, and the budget plan is also set by the government, although the total budget plan is not reported officially.

According to the global startup ecosystem benchmarking site, Startup Blink, Azerbaijan has only one city (Baku) in the top 1,000 cities, and and 84th globally.⁵⁶

[Figure 3-24] Startup Ecosystem in Azerbaijan



Source: Startup blink (2023).

⁵⁶ <https://www.startupblink.com/startup-ecosystem/azerbaijan> (Date accessed: 8 October 2023).

According to KOTRA (2018),⁵⁷ the central government announced the Strategic Road Maps for the National Economy and the Main Economic Sectors, which aims to develop the non-oil industry and foster small and medium-sized enterprises.

Situated at the crossroads of Eastern Europe and Western Asia, Azerbaijan is working on making the country attractive to entrepreneurs and the founders of startups, and on transforming and digitizing its economy to reduce its dependency on oil exports.

The Agency for Small and Medium Business Development of Azerbaijan was established in December 2017 as a way to foster small and medium-sized enterprises. The ASAN Centre⁵⁸ has established the business centre to provide education, consulting, financial, and administrative services to small and medium-sized enterprises. In areas where there is no ASAN Centre, the business centre is operated by the regional department under the Ministry of Economy.

These notable ecosystem stakeholders have helped shape the narrative of the startup scene. The State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan (ASAN) has been an important initiator of ecosystem-related projects, and is considered a global leader in terms of digitalization and smart cities.

One of those projects is the Azerbaijan 500 ASAN Startup Program, which aims to innovate and develop the country's budding startup ecosystem through online events. Moreover, ASAN partnered with the Asian Development Bank between 2019 and 2023 to boost private sector development, raise public sector efficiency, improve infrastructure, and strengthen human capital. ASAN also launched INNOLAND, an incubator, accelerator, and research centre. Likewise, the National Academy of Sciences established the High-Tech Park as a science and technology park with an environment that enables and is conducive to research and development projects.

The private sector supports the ecosystem by supplementing the need for funding,

⁵⁷ https://dream.kotra.or.kr/kotranews/cms/news/actionKotraBoardDetail.do?SITE_NO=3&MENU_ID=180&CONTENTS_NO=1&bbsGbn=243&bbsSn=243&pNttSn=171731 (Date accessed: 8 October 2023).

⁵⁸ An administrative agency that issues notarized and administrative documents to the same unit as the district office in Korea (Date accessed: 8 October 2023).

networking, and accelerators with such initiatives as the Yeni Fikir LLC and SUP VC. A startup studio in Azerbaijan, the Social Innovation Lab (Sil.), has dedicated itself to creating the first unicorn in Azerbaijan by 2025 and to helping fifty startups raise funding every year. This is an ambitious goal for a country with a relatively small market, and local startups have yet to get close to receiving high valuations. More initiatives like these will be needed to help grow entrepreneurship, increase networking opportunities between national and international startup developers, and encourage private investment.

[Figure 3-25] ASAN Xidmet (ASAN Service)



Source: CASPIAN NEWS (2023).

In August 2018, the Entrepreneurial Development Fund,⁴ an organization under the Ministry of Economy, was made independent and allocated a budget of 1.1 billion Azerbaijan Manat (about \$645 million) to promote the development of private companies in the non-oil industry. Funds range from entrepreneurs' education to funding for projects, and priority support industries are known to include the agricultural, tourism, information technology and communications/manufacturing sectors.

In March 2018, the Youth Fund was established to finance startups, and the tax law was amended to exempt companies with a startup certificate from income tax for three years. In addition, the Ministry of Transportation and Information and Communication held a startup competition jointly with the Heidar Aliyev Foundation.

Azerbaijan's 2030 strategy aims to make the country a regional leader in innovation and entrepreneurship

p. The strategy includes a multi-faceted approach to support the development of startups and technology providers, comprising tax breaks, grants, and access to mentorship and training among other initiatives. To be exact, the strategy maintains a clear articulation of the ecosystem: “The ecosystem to be built in the new era (melding appropriate incentives, incubation and acceleration centres, financing models based on public-private cooperation, venture capital, crowdfunding institutions, etc.) should be aimed at increasing the technological capacities of the economy”.

Through this commitment to the innovation ecosystem, an extensive series of programs has been initiated, ranging from e-government services to startup programs. Such initiatives vouch for an aggressive start to building a technology and entrepreneurship ecosystem in the broader Caucasus region. The section below covers only a few of these programs, including ASAN Finance, SIMA and fiscal incentives for technology entrepreneurs and developers.

ASAN Finance is a valuable resource for the citizens of Azerbaijan. It is an e-government project that provides a single window through which locals can access financial services from a variety of providers. The project was launched in 2017 by the State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan.

The service provides a variety of solutions, including loan application, access to insurance and investment products, and financial education. ASAN Finance has been a success in Azerbaijan, with the project processing over one million financial transactions in 2021 alone. ASAN Finance has also been praised by international organizations such as the World Bank and the Asian Development Bank.

SIMA electronic signature and KYC product is a digital identity and signature solution that provides a secure and convenient way for citizens and businesses to conduct online transactions. The solution is based on the use of a hardware security module (HSM) to generate and store digital signatures, and a mobile app to manage and use the signatures. SIMA solutions facilitate access to public services such as filing taxes, applying for a passport,

and registering a business. Opening bank accounts, applying for loans, making payments, and signing contracts are now fully automated through the SIMA online services.

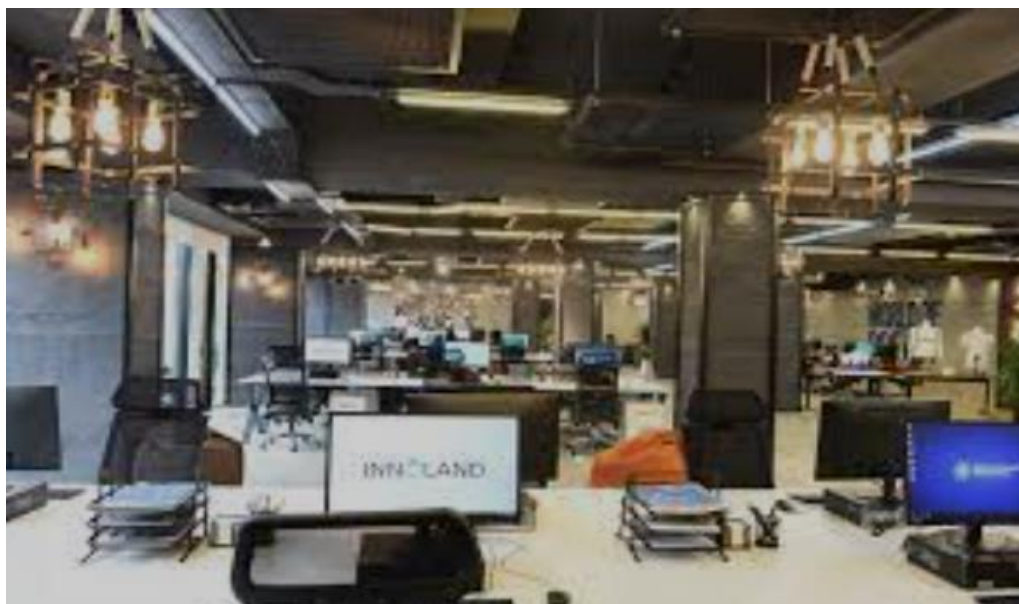
Businesses operating in industrial and technology parks (ITPs) can enjoy tax breaks for ten years. Starting from 2023, companies involved in system integration, software preparation, and development activities may also be considered as ITP residents and enjoy the tax benefits provided by law - even if they are not physically located within an ITP.

The tax breaks include:

- A 0% income tax rate for the monthly incomes of employees up to AZN8,000 and a 5% rate for incomes above that.
- Exemption from business profit tax.
- Exemption from the 10% withholding tax on dividends.
- Exemption from property and land taxes.
- Exemption from social taxes for non-nationals.

The SME Agency (SMBDA) offers a startup certificate that grants a three-year exemption from income tax, and also provides access to a generous fiscal support scheme for innovating startups.

[Figure 3-26] INNOLAND



Source: INNOLAND (2023)⁵⁹

As fostering a startup ecosystem is a high priority, the INNOLAND Incubation and Acceleration Centre is a versatile startup development program that is run within the State Agency for Public Service and Social Innovations under the President of the Republic of Azerbaijan. The program includes incubation and acceleration cohorts, a training centre, and a co-working space. State-owned ABB, the largest bank in Azerbaijan, has launched an entrepreneurship program, ABB Innovation, for pre-seed startups. Sabah Labs is another innovation centre that focuses on building startups and training software programrs under the Educational Institute.

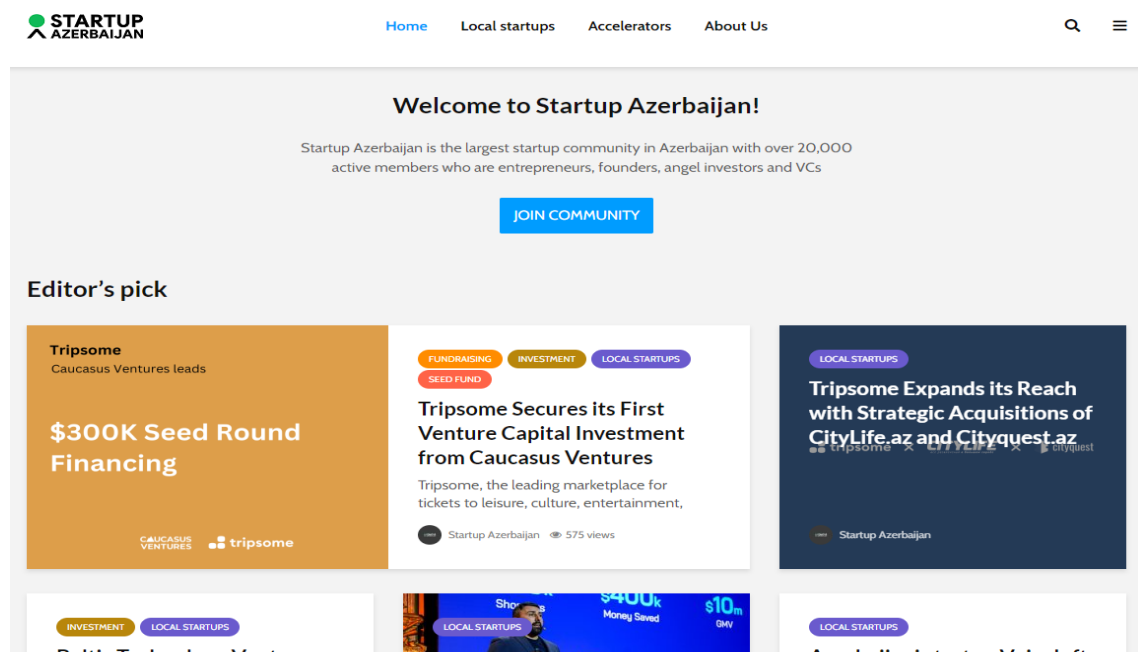
NextSTEP, SUP.VC, and Gamepons are other major initiatives shaping the startup landscape. Technest, Tech Academy, and ABB Tech are also nurturing the talent ecosystem by running rigorous training and scholarship programs for coding. The Agency for Innovation and Digital Development (IDDA) has been working on a series of venture capital funding schemes as well as on startup venture studios. Caucasus Ventures is now officially in action as the first startup VC in the country. The Central Bank of the Republic of Azerbaijan will soon introduce

⁵⁹ <https://innoland.az/en/news/14> (Date accessed: 8 October 2023)

a sandbox that will facilitate the compliance process for registered fintech startups.

Startup Azerbaijan: An organization established by the Ministry of Transportation and Information and Communication, it conducts projects such as encouraging startups and strengthening links with startups. Please visit <http://www.startupazerbaijan>.

[Figure 3-27] Startup Azerbaijan



Source: Startup Azerbaijan (2023)⁶⁰

Azerbaijan Startup EcoSystem: A site that collects all information about the startup ecology for young entrepreneurs. www.startupazerbaijan.

- Next Step Innovation Centre (www.icnextstep.com): An organization that provides funding for the product concepts and prototype development of founders' startups, and startups that have completed the training provided by the organization. It has an office in downtown Baku that provides office space for selected prospective startups

⁶⁰ <https://www.startupazerbaijan.az/about/> (Date accessed: 8 October 2023)

- Social Innovation Lab SIL (Social Innovation Lab SIL): A research institute created to train fifty tech startup companies annually by 2025, provide founder training, and offer overseas market expansion services. www.socialinnovationlab.az
- SUP (www.sup.az): Intensive training programs for startups, including product invention, marketing practice, etc. as part of a 3-month course.

Support programs for local telecommunications startups

- Barama (www.barama.az): Startup support business for the telecommunications operator Azercell Telecom.
- Bakcell AppLab (www.applab.bakcell.com): Telecommunications operator Bakcell helps startup founders to develop mobile apps.

Startup programs in universities: Azerbaijan National Oil University and ADA University, etc. have startup support centres.

As mentioned above, there are various government-led startup and incubator programs, but due to the poor corporate environment they are still often operated in a demonstrative manner. It has been shown that very few businesses have been started through these programs. The local conditions that hinder the development of startups are outlined below.

- (Closed economy) Azerbaijan's geographical proximity to Russia and Turkey and its small market make it less attractive for foreign companies to enter. The private sector lacks capital liquidity and has a closed economic structure in which only a small number of companies have concentrated capital
- (Brain drain) If one faces powerful vested interests and has no background in a society focused on blood ties and delays, it is difficult to be incorporated into vested interests, so talented people tend to leave the country.
- (Big business-oriented economy) It is difficult for new startup companies to emerge due to the fact that some large companies are owned by high-ranking officials by tacitly tolerating the ownership of the company.

- (Taxation burden) Tax audits of general private companies are frequent and taxation is heavy. They often run their businesses by voice.
- (Education system) The overall quality of education is low due to the low salary levels of faculty and staff and the poor educational facilities

According to Mr. Vadim, director of Boss (www.boss.az), the largest job posting site in the country, it is not easy for foreigners to start a business there due to the poor business environment, such as the closed economic structure and language barriers. However, given the recent development of the tourism industry and the good image of Korea, he mentioned that tourism and trade could be considered promising areas for startups in the short term.

- (Tourism) Demand for local travel services is on the rise as more Korean tourists visit the three “Kafkaz” countries.
- (Trade) Demand for Korean products is steady due to a good level of awareness of Korea, and imports of fruits and vegetables from Azerbaijan are also promising

The poor business environment delays the introduction of advanced management know-how and services, which means that business service and/or product that are already common in other countries can become new and profitable business items in Azerbaijan. If there is sufficient market research and preparation, it is worth trying to start a business on the spot.

Household goods stores such as MINISO have recently been established and are operating successfully, and a proxy driver is operating at the suggestion of an international student who studied in Korea. Recently, Korean dramas began to air during prime time (9 p.m.) on Hezar Dalga TV, which is expected to have the effect of boosting exports of related Korean Wave products.

3.1.2 Onsite discovery interview on entrepreneurship in Azerbaijan

In October 2022, a local market survey with major universities and startup support organizations was conducted in Azerbaijan for approximately one week. The results of the main interview are as follows:

In the case of Azerbaijan, there is next to no government R&D funding available to create a separate fund to support startups in universities or to promote research and development, which is the seed base of startups.

Since mid-2010, some universities have been seeking to revitalize startups by providing education in the information and communication field, which does not require much initial capital for startups. However, since no funds are available for follow-up investment or support after completing entrepreneurship training, it is difficult to provide practical entrepreneurship care beyond the operation of educational programs.

[Figure 3-28] On-site Discovery Interview in Azerbaijan



Major universities in Azerbaijan usually operate startup education centred on undergraduate centres and hope to create a startup ecosystem that takes into account the industrial characteristics of Azerbaijan, but they are having difficulty in creating a fund to support this, so they hope to establish a startup incubation system in connection with overseas accelerators such as the United States and venture capital.

However, unlike major countries such as the United States, South Korea, and Europe, which have secured intellectual property and patent pools in advance to protect startups' business items and secure differentiation, Azerbaijan still does not manage and secure their intellectual property through research and development in the field is recognized as a public property characteristic, so there are few motivating factors to actively secure intellectual properties in Azerbaijan.

In addition, there are universities in Azerbaijan that apply for patents, but it is difficult to ascertain what patents have been applied for by which universities, making it difficult to obtain accurate statistics on how many patents are applied for by universities.

The results of local interviews with major universities and startup-related institutions in Azerbaijan are as follows below.

3.1.2.1 University startups

Baku State University runs an in-school innovation centre and encourages research based on science and technology, but the number of R&D tasks is small and the need for intellectual property rights, such as patents, is barely recognized. In relation to startups, support for undergraduate students is being provided mainly by innovation centres, and it has been reported that some ideas with the potential to solve local problems in Azerbaijan have been funded by oil-related companies such as BP in the UK and carried out on a project-by-project basis.

[Figure 3-29] “Azercell” at ADA University



Source: ADA University (2023)⁶¹

In conjunction with the KOICA ODA project, ADA University is promoting cooperation with Soongsil University in Korea for the operation of IT joint degree programs and startups. Although a project is being carried out to foster and support startups with an innovation centre in the university, the education-oriented startup support system has been maintained because there are insufficient funds to operate startup programs. Azerbaijan government is trying to expand cooperation with Korea and Italy to develop startup innovation capabilities, but the operation of practical incubation programs for startups has been temporarily suspended due to a lack of professors specializing in startups and the

⁶¹ <https://www.ada.edu.az/en/news/580-ada-students-take-on-developing-innovative-ideas-under-the-idea-incubation-programs-of-azercell> (Date accessed: 8 October 2023)

difficulty of raising follow-up funds⁶². Although some telecommunications companies are interested in supporting startups, they hope to cooperate with countries and institutions around the world with which they do not yet have a significant investment or cooperative relationship, and they only have a fund support system that can support the creation of practical startups beyond entrepreneurship education. Recently, Azercell launched the “Ideas Incubation Program” initiated by Azercell Telecom LLC in cooperation with the ADA University Foundation.

The University of Economy operated a boot camp type of program in 2022 in relation to startup development, resulting in the discovery of about twenty-five teams. Although the startup incubation program was developed for these teams through follow-up education and linkage with international funds such as ABB⁶³, there is still a limit to the support for early seed investment. However, ABB supports the operation of the largest startup incubation centre, fund, and hackathon-style startup education program in Azerbaijan in terms of size. Unlike other universities, the University of Economy operates a technology commercialization centre that can apply for and support patents, and about sixteen patents are applied for annually, of which around 50% are actually registered. In the field of cooperation with Korea, they are trying to run joint

degree programs with Kyunghee University and Woosong University, and have established cooperative relations with universities in Hungary and Greece.

3.1.2.2 Other startup support organizations

Sabah Hub was launched two years ago, led by Azerbaijan's Ministry of Science and Technology. It provides course of education through partnerships with Europe's largest i-Business schools, and continues to cooperate in the field of education by utilizing experts from Stanford and MIT after i-Business. Its education program is focused on entrepreneurship and mainly links peer-to-peer IT-related education, design education, and

⁶² Related programs reference site: inc.unec.edu.az/en/unec-incubation-centre/ (Date accessed: 8 October 2023)

⁶³ ABB (Azerbaijan Bank of Business): Azerbaijan Development Bank. It is a large bank that ranks first or second in Azerbaijan. 51% of their funds are government funds, while the remaining 49% are from major shareholders such as other banks and PEs such as BlackRock. It plays a similar role to the Korea Development Bank (reference site: abbinnovation.az/en) (Date accessed: 8 October 2023)

business education in the field of education. It provides mentoring to tenant companies through separate contracts with Stanford-related educators.

Regarding pitching, there are events such as “Baku Investment Day.” As Azerbaijan’s startup culture is still in its early stages, Sabah Hub has established an incubation institution and has also established a separate academy for young entrepreneurship education program. The mission of institutional operation is to foster human resources → create infrastructure → upgrade fund support, and the size of the target fund is about US\$ 25 million.

However, the fund for creating and spreading the Azerbaijani startup ecosystem is still small and in poor condition, so Azerbaijan is trying to create the Sabah Fund jointly with local large companies and overseas institutions such as PASHA Holdings, SOCAR, the Asian Development Bank, and the European Development Bank. MVP operates mainly in the ICT field, but there are a few deep-tech startups.

In Azerbaijan, deep-tech is judged to have potential in the fields of agriculture, energy, education, and medicine. Azerbaijan regards the medical field as a promising field for investment or startups because it costs only about US\$ 150 million per year for countries such as Turkey, Iran, Germany, and South Korea, which believe that it is possible to enter the deep-tech company market in the medical field.

Medical education is progressing well in Azerbaijan, but practice is weak. Doctors also believe that their salaries are not very high, so there will be improvements in that area and business opportunities for startups.

Regarding the global startup support ecosystem in Azerbaijan, early-stage companies can also be supported by four countries, namely, Kazakhstan, Estonia, Uzbekistan, and Turkey, and after that, they are trying to attract Dubai. Upon request from these countries, occupancy and support are approved after six months of verification. The number of VCs in the capital, Baku, is inaccurate, but Estonia, Kazakhstan, and some U.S. government funds also have some linked VCs. And those VCs has a co-partnership with other startup support organizations such as INNOLAND.

However, Azerbaijan's related industries are not large, so they are cooperative rather than

competitive. Azerbaijan's representative startup company is B2B, but it is not big and is trying to create a unicorn company within five years. As for the size of the market, it is judged that unicorn-class companies can be created only by linking activities in nearby regions and the United States.

Currently, startups in Kazakhstan and Uzbekistan are linked to Turkey and Europe, whereas those in Estonia come to Azerbaijan due to the insufficient laws related to blockchain. So those countries are currently considering plans to cooperate with industry-academia universities related to transfer technology.

From the perspective of technology commercialization, HTP experts provide mentors to Sabah, and cooperation with Sabah Hub is also being planned, just as the Innovation Agency is proceeding with the cooperative initiative in terms of TTCP. As part of the cooperation between the HTP and Sabah, the two institutions hope to request the operation of programs such as joint education and training or incubation online for five to six months.

Regarding Sabah's incubation portfolio, we can see that there are three research-based (personal idea-based) startups out of about eighty startups, while the remaining seventy-seven are IT application development-oriented startups. There are few patent-based startups and few patent professionals in Azerbaijan, so we believe that platforms such as the HTP's TTCP will be helpful in improving this area. The Innovation and Digital Development Agency, as an institution that was established for the purpose of creating a technology-based startup ecosystem, supports various organizations, is responsible for incubating innovative technologies, and aims to revitalize technology-based startups in the private sector.

Regarding the purpose of its establishment, the IDDA aims to transform Azerbaijan into a country that is faithful to technology-based entrepreneurship and to create a digital government. As an organization whose main purpose is to promote the innovation ecosystem and digital development, it aims to cultivate talented professionals through the creation of an innovation ecosystem, develop technology companies, support investment ecosystems, spread entrepreneurship, and establish “Abscheron Valley” (infrastructure related to shared offices for startups).

As digital development consists in creating digital government, society, infrastructure, businesses, and ecosystems are collaborating with various countries. Under Azerbaijan's long-term vision and system, the purpose-specific mission is to create a venture fund, provide education, support the establishment of public and private companies, establish a role-oriented implementation structure between stakeholders, and create a result-oriented and sustainable ecosystem. A few of well-known investors are the GP, and funds are created in a matching form or institutions participate in LPs and total funding amount is about \$30MM AUM for AZ investors with 3,000 technology based researchers. Usually, when entering the investment phase, an approximately 5% stake is secured at the most, and the market value of the companies involved is valued at approximately \$200,000. 100 new companies have prototypes for their businesses, and 3 innovation spaces in the region to be examined and managed for supporting 30+ early state of startups.

There are five main areas of activity, as follows:

The first includes investment ecosystem support, defined as "Business Angels Club" support, matching fund development, and VC fund creation support.

The second includes technology talent development to strengthen technology capabilities, peer-to-peer tech skills education, entrepreneurship education, investor education, media education, STEM Olympiads, internships and job (up) matching between investors in technology companies and members of the ecosystem, mentorship programs, and innovation champion education.

The third is focused on technology enterprise development, support for university innovation, the development of enterprise innovation, program incubation, acceleration programs, and venture studio support.

The fourth includes promotion of the technology ecosystem, including ecosystem matching, inter-organizational roadshows, branding, positioning, technical media support, vertical community wisdom, events and networking support, and IDDA Award support.

The fifth and final area of activity is focused on developing innovation spaces, configuring

technology ecosystems, supporting the creation of technology parks, and supporting the development and support of relocation programs. Eight universities are participating in this innovation development project.

INNOLAND is another government-led startup support organization. Having moved its current location in 2018, it is operated within a structure similar to that of Korea's startup-supported shared office. Most of the startup companies that have moved into INNOLAND are startups in the ICT field, and none of the companies has a separate lab. It is open 24 hours a day with government support. Accelerators that support seed funds in residential facilities, and funds can be invested at the pre-seed level.

Elsewhere in Azerbaijan there are many incubation spaces that are similar to the shared offices in Korea, so it is necessary to strengthen the financial support for technology-based startups moving into these facilities in order to create a technology innovative startup ecosystem centred on specific regions, including major metropolitan cities.

3.1.3 South Korea's technology startup education and promotion system

South Korea's R&D investment budget amounted to about KRW 89.04 trillion as of 2019, of which the government's R&D investment has continued to increase each year, from KRW 20.5 trillion (23.0% of the total R&D budget) in 2019 to KRW 24.1 trillion in 2020, KRW 27.2 trillion in 2021, KRW 29.8 trillion in 2022, and KRW 31.1 trillion in 2023.

However, in terms of performance, the success rate of commercialization in South Korea is around 20%, which is significantly lower than in the United Kingdom (70.7%), the United States (69.3%), and Japan (54.1%), ranking 27th worldwide. In particular, in the case of universities, research productivity - which represents the technology transfer revenue per billion won of total university research expenditure in South Korea - was just 1.41% as of 2019, which is less than one-third of the 4.66% recorded in the United States (2017).

The situation in the U.S. was not much different ten years ago, with around US\$7 billion (8 trillion won) worth of funds being invested in basic research by the National Science Foundation (NSF) in 2010 during the Obama administration, which invested even more federal funding, i.e. about \$172 billion (180 trillion won) per year in R&D, and yet the

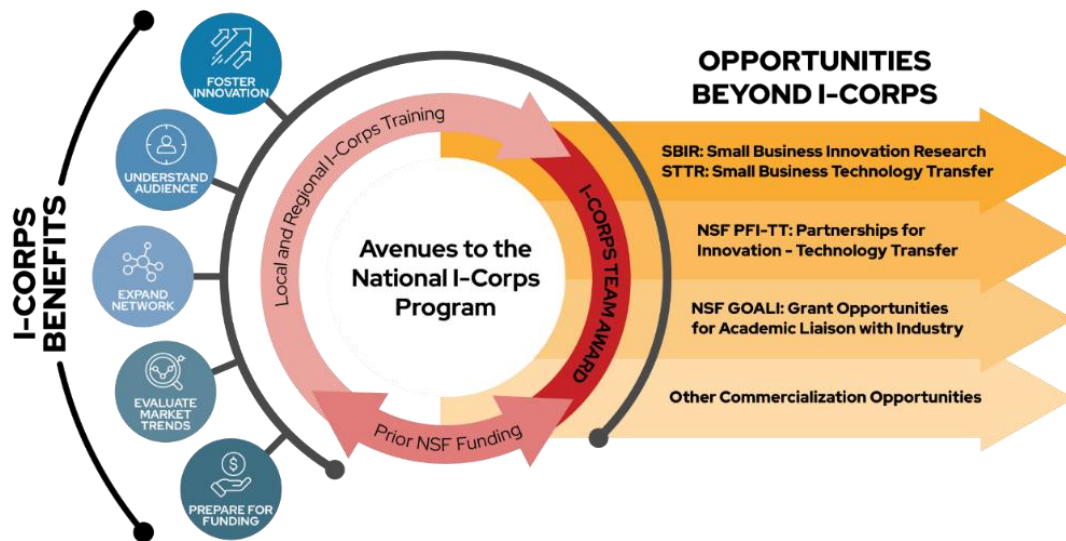
technology transfer rate remained low and only a few technology startups were established.⁶⁴

In this regard, Professor Kim Jong-sung (2017) of Boston University introduced Professor Pisano's Science Business (2006)⁶⁵ of Harvard Business School as a potential solution to controlling the gap between low R&D productivity and technology commercialization in the United States. Professor Pisano argued that after years of research by biotechnology companies, a fundamental difference had emerged between science and technology and business, and asserted that scientists needed to educate themselves about how to develop their businesses, and that business would improve only when scientists actively sought out and cooperated with experts in diverse other fields through a market exploration process (Customer Discovery) of "broad research," while moving away from their previous method of collaborating with colleagues in the same field on "deep research." In response, the Obama administration has, from 2012, targeted about 400, i.e. about 1% per year out of about 40,000 research heads at universities and state-funded institutes, and provided them with governmental project support. NSF I-Corps develop and support a "L2M (Lab to Market)" type education and training programs that selects three researchers and enables them to transform research that conducts "deep research" through market verification into a "broad research" method. The National Science Foundation's I-Corps (Innovation-Corps) Education and Training Program succeeded in establishing 1,036 startup teams as public technology-based startups through a cumulative training course involving about 2,241 prospective startups from 2012 to 2020.

⁶⁴ Jim Chung(2019), "Why NSF I-Corps", George Washington University TTT program.

⁶⁵ Pisano, Gary P.(2006), Science Business: The Promise, the Reality, and the Future of Biotech", Harvard Business Review Press, 11 January 2006.

[Figure 3-30] I-Corps Benefits



Source: University of Wisconsin-Madison (2023)⁶⁶

The output of these technology-type R&D investments leads to the creation of both economic and social outcomes. The issue of losing is also becoming a serious matter for Korea.

In relation to this, the Korean government designated KAIST and POSTECH, specialized universities in science and technology, as pilot business colleges under the supervision of the Ministry of Science and ICT since 2015, and has conducted NSF I-Corps-type technology startup education since 2023, and it is reported that it has been generating outstanding results in terms of startup rate, survival rate of startups, job creation, and investment attraction.

Therefore, this report shares the trials and errors that Korea's NSF I-Corps education and training courses experienced in the process of customer discovery in Korea, centred on the Korea Entrepreneurship Foundation (KoEF), the National Research Foundation(NRF), and the Korea Innovation Centre Washington(KIC D.C.) in the US, which were the actual program

⁶⁶ NSF I-Corps & University of Wisconsin-Madison (2023), <https://teo.wisc.edu/nsf-i-corps/> (accessed date: 10 October 2023)

operators and/or I-Corps regional governance.

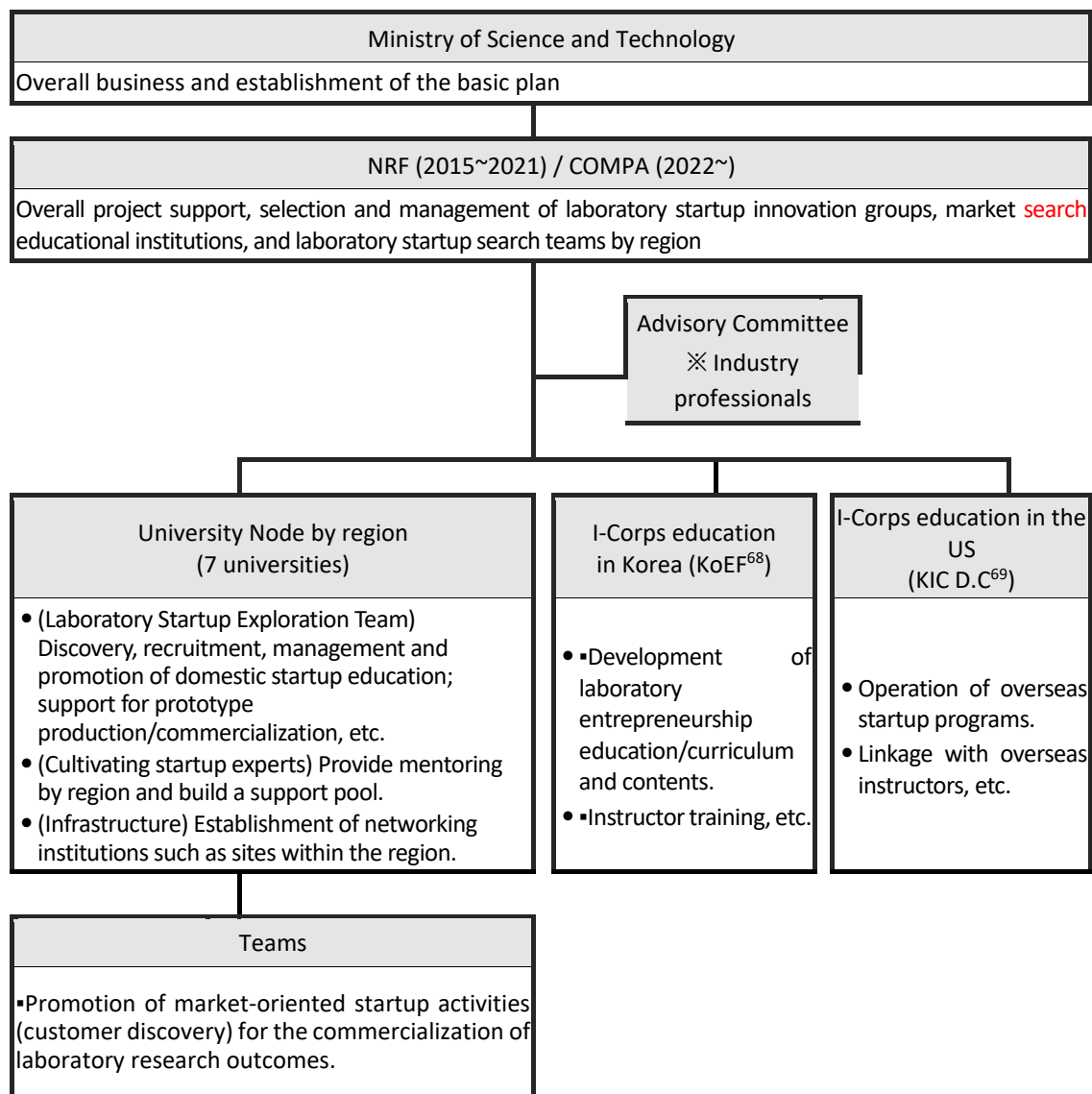
This particular governance structure will be referred to by Azerbaijan's relevant startup incubation centre. Further, this type of innovative startup support ecosystem project and governance structure could be used in reorganizing Azerbaijan's currently weak economic structure, in which energy exports account for 90% of all exports, 30% of GDP, and 75% of fiscal revenues⁶⁷, and could also contribute to fostering the non-oil sector in the long term.

3.1.3.1 Structure of the Korea I-Corps program for R&D Startups

The governmental department responsible for promoting the I-Corps project in Korea is the Ministry of Science, Technology and Information and Communication. Regarding the operation of domestic education programs, the Korea Youth Entrepreneurship Foundation (KoEF) is in charge of domestic education, while, in the case of overseas education, the KIC D.C operates overseas education courses in connection with George Washington University and UC Berkely University in the United States.

⁶⁷ Sooyoung Hwang (2023), "National Credit Rating Report", Export-Import Bank of Korea, Overseas Economic Research Institute, 27 February 2023., p. 6.

[Figure 3-31] Structure of the Korea I-Corps Program, NRF (2022)

Source: NRF & GS SHIM (2022)⁷⁰⁶⁸ KoEF (Korea Entrepreneurship Foundation): <http://www.koef.or.kr/> (Date accessed: 10 October 2023)⁶⁹ KIC D.C (Korea Innovation Centre Washington D.C): <https://www.facebook.com/KoreaInnovationCentreDC/> (Date accessed: 10 October 2023)⁷⁰ NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Program," NRF, January 2022., p. 103.

The second phase of the Korean I-Corps project, which started in 2021, consists of two phases. During the first to second phase, it selects seven regional laboratory startup innovation groups (Node University) and market discovery educational institutions through phase evaluation.

[Table 3-2] Korea I-Corps Program Support Plan by Year

Year	Phase I		Phase II		
	1 st year	2 nd year	3 rd year	4 th year	5 th year
Period	03.2021-02.2202	03.2022-02.2023	03.2023-02.2024	03.2024-02.2025	03.2025-02.2026

Source: NRF & GS SHIM (2022)⁷¹

The roles played by each major entity in operating the business are as follows:

- University's Lab to Startup Support Team: Startup team recruitment and incubation.
- Instructor: Training teams use the I-Corps methodology and lead "lessons learned" sessions.
- Mentor: Teams coach and manage an office hour program.

If we look at them in detail, they are organized with the following roles and responsibilities.

⁷¹ NRF & GS SHIM (2022), "Policy research of Korean I-Corps Programs", NRF, Jan., 2022., pp.104

[Table 3-3] Korea I-Corps Program's Major Entities and Their Roles

	Group	Role	Roles and Responsibilities
Stakeholders	Node Universities	Startup Incubation and support	Discover and nurture a startup search team (intensive education, pivoting, etc.). Establish a mentor pool for each region and support mentoring. Establish a site within the region and discover overseas market discovery educational institutions.
	KoEF	Education (Local)	Develop a domestic entrepreneurship education curriculum, domestic common startup education, and basic and practical startup education for domestic startup search teams. Discover, nurture and support instructors.
	KIC D.C	Education (Global)	Conduct basic and practical startup education for overseas startup search teams. After overseas training, support additional mentoring and incubation by overseas startup experts. Linkage with international instructors.
Professionals	Mentors	Scale-up (Pivoting)	Pivot and supplement the directions for startup business items. Assist with discovering customers and share startup know-how.
	Instructors	I-Corps methodology training	Customer Discovery Methodology Training Train the startup search team (enhance their entrepreneurial mindset, etc.)

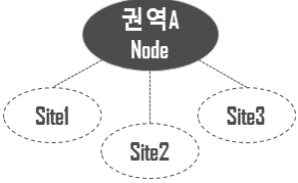
Source: NRF & GS SHIM(2022)⁷²

In addition, the base university, called the Node University, discovers and nurtures technology startup teams for other universities within the region, and conducts education and team discovery activities for preliminary technology startup teams as an accelerated course as a part of the mini I-Corps program.

⁷² NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Program", NRF, January 2022., pp.107.

[Table 3-4] Role of Node and Site University

<Structure >



	(Node) University	(Site) University
Condition	Regional university as a business incubation agency.	Belongs to the Node University of the region.
Major role	Recruitment of technology-based startups for the region. Startup incubation program (around 5 months).	Recruitment of technology-based startups and introduction of the teams to the Node University. Mini I-Corps program (2-5 weeks).
Follow-up	Follow-up for excellent teams with other startup programs.	Follow-up for excellent teams that enter the I-Corps program.

Source: NRF(2022)⁷³

3.1.3.2 Structure of startup governance and the I-Corps program for promoting technology entrepreneurship in Azerbaijan

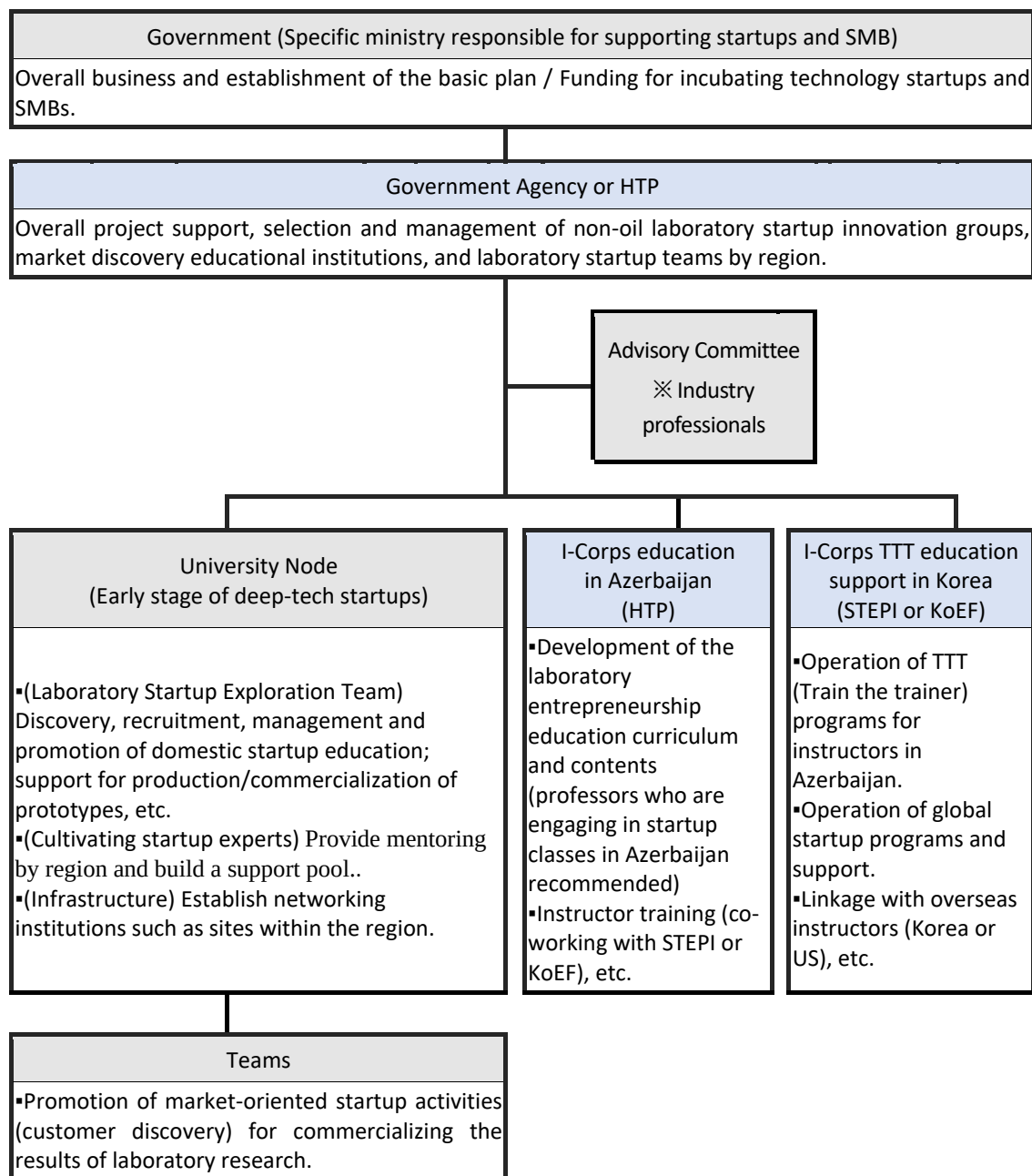
As previously discussed, in order to revitalize technology startups in the non-oil industry, one of Azerbaijan's major government ministries that is currently in charge of startup support or playing the same role as the Ministry of Science and ICT in South Korea needs to be designated as a "ministry for startup governance."

In this regard, it will be necessary to expand the role of the HTP or reorganize its organizational structure in order to create a technology startup ecosystem centred on universities and major national research institutes as agents of governance for practical technology startups.

In addition, after the system of governance is established, it will be necessary to establish a sustainable and ongoing cooperation system with South Korean institutions such as STEPI, which promotes the transfer of TTT education and training programs in Korea.

⁷³ NRF & GS SHIM (2022), "Policy research of the Korean I-Corps Program", NRF, January 2022., pp.108.

[Figure 3-32] Recommended Structure of the I-Corps Program in Azerbaijan



Source: NRF & GS SHIM(2022)

3.1.3.3 Status of the HTP and startups

The HTP used to be an umbrella organization of the ANAS (Azerbaijan National Academy of Sciences, founded in 1945), but its functions were separated from ANAS, which was considered a top institution, throughout 2022 and 2023. ANAS conducts major research in the scientific and social sciences of Azerbaijan and is a major national research organization that comprehensively coordinates its activities and is granted relevant status under the Presidential Decree of 2001. ANAS employs 10,000 people (4,939 science workers, 664 doctors of science, and 2,028 doctors of philosophy) in total.

The HTP's activities and programs related to startups include improving the Intellectual Property Protection Act (enforced in 2018 by Presidential Decree), applying the results of educational and scientific research in various high-tech and innovative fields through entrepreneurship, promoting innovation initiatives in high-tech fields (startup competitions, etc.), IP funds, innovation funds, startup incubation centres for the establishment and expansion of high technology park. HTP and/or relevant startup organization need to support the startup ecosystem by complementing research and technology transfer centres and other specialized service centre

3.1.3.4 Azerbaijan's technology IP resources for Lab-to-Market (L2M)

Startups

Azerbaijan's foundation for revitalizing startups in the non-oil industry is very weak due to certain structural problems in society.

In particular, the holding of intellectual property rights and the technology transfer ecosystem in various engineering fields, which may be considered the "technology seed" of innovative technology startups in the non-oil industry, is extremely low even in universities and other innovative public research institutes due to the widespread negative perception of technology privatization due to Azerbaijan's socialist characteristics and bureaucracy.

Based on data from 107 countries from the Korea National Statistical Office (2022), the trend of patent application and registration in Azerbaijan and neighboring countries for the last three years (2017-2019) is presented in the Table below:

[Table 3-5] Current Trend of Patent Application and Registration in Azerbaijan

Country	2017		2018		2019		Remarks (WW Ranking)
	Patents Applied for	Patents registered	Patents Applied for	Patents registered	Patents Applied for	Patents registered	
Azerbaijan	226	67	171	64	167	102	71
Armenia	110	74	105	100	116	100	83
Georgia	232	206	260	133	197	115	67
Turkey	8,555	1,900	7,466	2,882	8,088	1,941	20
Iran	16,259	4,151	12,823	3,367	12,147	2,769	15
Russia	36,883	34,254	37,957	35,774	35,511	34,008	7
Republic of Korea	204,775	120,662	209,992	119,012	218,975	125,661	4

Source: Korea Statistics (2022) ⁷⁴

The figure for Azerbaijan is based on three-year cumulative patent applications. There is a low trend of intellectual property applications among its neighboring countries including Armenia (71st), Georgia (67th), Turkey (20th), Iran (15th), Russia (7th), and Armenia (83rd).

Looking at this as the average annual growth rate (CAGR), i.e. 14.04% over the past three years, it is necessary to provide government-level policy support and raise awareness of the importance of intellectual property rights as “the seeds” of technology transfer, commercialization, and technology startup.

In other words, in Azerbaijan, the government's technology startup governance, institutional improvement, and the technology base, which can be practically utilized in the non-oil industry. And Azerbaijan has to enforce to non-oil based technology field to develop other industry and invest technology oriented startup and financial investment to create social impact.

In addition, to improve awareness, it is necessary to create a technology-linked innovative

⁷⁴ Korea Statistics (2022), IRP Report, National Statistics Portal, “Patent Application and Registration in Major Countries” Status, Statistics Korea, June 2022.
https://kosis.kr/statHtml/statHtml.do?orgId=101&tblId=DT_XNN0104&vw_cd=MT_RTITLE&list_id=101_001_NN_NN_TT_A_N_N_TT_10&seqNo=&lang_mode=ko&language=kor&obj_var_id=&itm_id=&conn_path=MT_RTITLE (Date accessed: 12 November 2023).

startup ecosystem that allows the practical use of intellectual property rights to be secured. In addition, the Korean I-Corps program - which can be evaluated as successful in creating jobs and attracting investment by introducing NSF I-Corps from around the world, increasing the practical use of R&D, and expanding Lab To Market-type technology startups - can be considered a major precedent for the various experiences and achievements experienced in the process of Korean I-Corps program over the past eight years.

Therefore, NSF I-Corps, influences to develop a representative technology startup program in Korea, and the similar program for Azerbaijan will help create an innovative startup ecosystem in the non-oil industry from the perspective of outcomes, which may require the training of instructors (TTT) centred on major startup incubators and governance.

3.1.4 Overview of the I-Corps TTT program for Azerbaijan

The NSF I-Corps and Lean Startup curricula promote early-stage startup techniques which identify customers, validate value propositions, and build successful business models for commercialization. The I-Corps at the national node university in the US is normally open to faculty, students, staff, and regional science and technology entrepreneurs with innovations or inventions that have commercial potential.

The I-Corps Team's program aims to develop and nurture a national innovation network, and to support researchers interested in entrepreneurial education and mentoring, with the goal of reducing the time it takes to bring technologies from the laboratory to the marketplace.

I-Corps provides real-world, immersive instruction that allows scientists and engineers to evaluate a technology's commercial potential, a critical first step towards successful translation into products, processes and services that will benefit society.

This starts with the formation of a team. Each team is composed of three to five members and includes a technical leader, an entrepreneurial leader, and an industry mentor.

Each team must first submit an executive summary via the application portal. The executive summary must include information about each team member, the proposed technology,

and the commercial application. Next, the team is invited to an interview with I-Corps staff. A subset of the team is then selected to participate in the National I-Corps training program and submit an I-Corps Teams proposal.

In addition, all of the NSF I-Corps programs are led by one or two lead instructors and several TTT trained instructors at national I-Corps node universities in US.

[Figure 3-33] GWU's I-Corps and TTT and Mentoring Programs Through Office Hours



Source: GWU (2023)⁷⁵

In Korea, eight Korean instructors have been selected every year since 2017, while George Washington University (GWU), one of the NSF's node universities, has supported the TTT (Train the Trainers) curriculum for Korean Instructors. They have also experienced GWU's milestone-based mentoring programs, which provide one on one. TTT team support for learning mentors-in-residence guide teams through the customer discovery journey — meeting innovators before they even have a finalized idea, and continuing with them throughout commercialization, as part of a long-term relationship.

Mentors provide consistent oversight and direction, evaluating progress milestones and assisting with the interpretation of market signals. They help the teams to develop strong, validated business models, minimum viable products, and go-to-market strategies,

⁷⁵ GWU (2023), "Train the trainer program for KoEF I-Corps Instructors" presentation slide, Entrepreneurship Centre of GWU, p. 7.

preparing them for acceleration and investment.

Using this milestone-based foundation, GWU-like node universities in the US have developed programs at the local, regional and international levels, working with partners to maximize the reach of our mentorship offerings while building integrated early-stage innovation ecosystems.

As of the end of 2022, about forty-eight Korean instructors had completed the TTT curriculum of GWU, they were mainly selected from among the following categories:

- ① professors at universities;
- ② instructors specializing in startups;
- ③ holders of professional qualifications, including patent attorneys, lawyers, accountants, etc.; and
- ④ highly experienced people with sufficient business (non-oil industry) experience for Azerbaijan.

In the case of Azerbaijan, TTT program managers of the HTP and startup professionals from the related agencies participating in this project have to localize and apply the I-Corps programs. And they have to be involving and Implementing in the train the training programs, through this, the establishment of a system that can foster innovative startup prospective entrepreneurs in the non-oil field is promoted

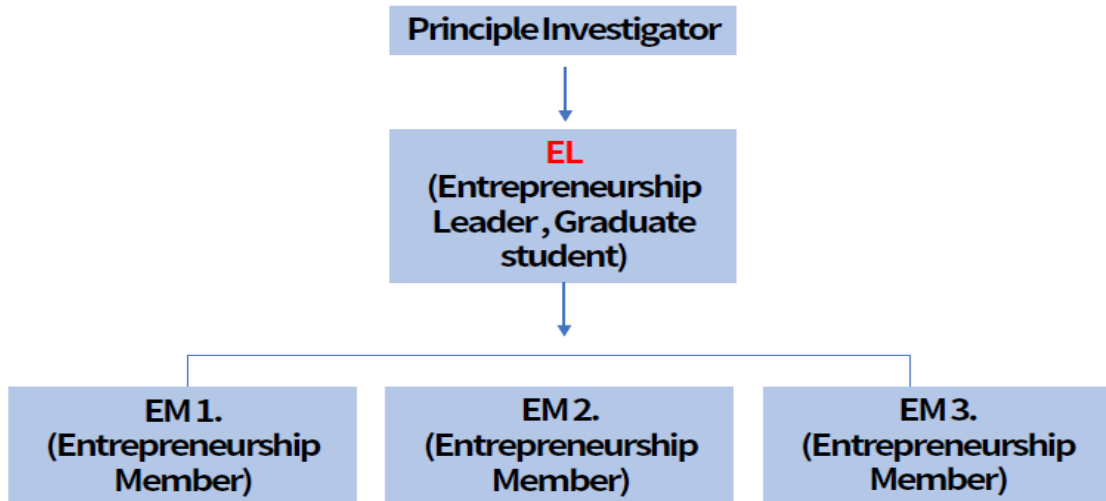
The training course is organized both online and offline, and the training course is conducted to carry out team projects related with virtual business items, so that Azerbaijani faculty or TTT invited professionals can conduct the process of customer discovery themselves

The Azerbaijani faculty and/or professionals participating in the TTT education and training course include one EL (Entrepreneurship Leader) and multiple EMs (Entrepreneurship Members).

However, the Azerbaijan I-Corps program for the actual student programs has one PI (Principle Investigator) and one PM (Project Manager, optional), along with an EL and an EM,

as in South Korea.

[Figure 3-34] Structure of the I-Corps Team in Korea



Source: Shim (2022)⁷⁶

As regards the TTT education and training course, a total of four Korean instructors have been appointed to run it in 2024. One of the four Korean instructors will be assigned the role of lead instructor and will manage the TTT curriculum overall, while the other three will be assigned the role of providing feedback to the team in the form of thematic lectures and in office hours during the I-Corps program.

The TTT training course is largely composed of four parts:

(Online real-time) Lecture-type Education (including an orientation curriculum): This encompasses theoretical education on the core methodology of the I-Corps education and training; understanding the I-Corps curriculum; establishing a business hypothesis; value proposition canvas; customer discovery; BM canvas; and beyond the canvas, etc. However, the I-Corps educational video contents produced by the Korea Youth Entrepreneurship Foundation and the YouTube lecture contents released by Professor Steve Blank (Stanford University)⁷⁷ can be used as auxiliary educational contents free of charge to help you run

⁷⁶ NRF & GS SHIM (2022), "Policy research of Korean I-Corps Programs", NRF, Jan., 2022., pp.110

⁷⁷ <https://www.youtube.com/watch?v=rqYmsrjJH3g&list=PLysM940HUUsAutvEc9JdcejACk-la0b9> (Date accessed: 10 October

lectures and understand the curriculum.

Customer Discovery (repeated weekly): This includes weekly in-depth interviews with real customers based on the customer discovery methodology learned from the lecture. Guidance is provided to ensure that customer discovery is carried out in groups of at least two people (one questioner and one recorder).

Office Hours (repeated weekly): The instructor provides feedback on the response of the target customer acquired through customer discovery. However, Office Hours is conducted using a separate ppt template developed for the I-Corps training course, rather than a method in which learners arbitrarily organize and present the customer discovery results.

DeBrief (Conducted immediately after Office Hours): The participants discuss the customer discovery and team improvements conducted by the instructors with or without the involvement of the I-Corps TTT team.

In South Korea's TTT education and training course, 137 teams are selected in advance every year, but only 90% of the teams will be invited to attend the national I-Corps program (10% of the teams will be discontinued). The BootCamp for the selected teams is conducted as a collective training program, as shown in the following figure.

After BootCamp, a gap of about one week is allowed, after which online video lecture materials produced for pre-education purposes are self-studied and executed individually.

In addition to the lecture-type education, the next basic curriculum allows one to understand the overall I-Corps education and training process through about twenty actual customer discoveries and Office Hours.

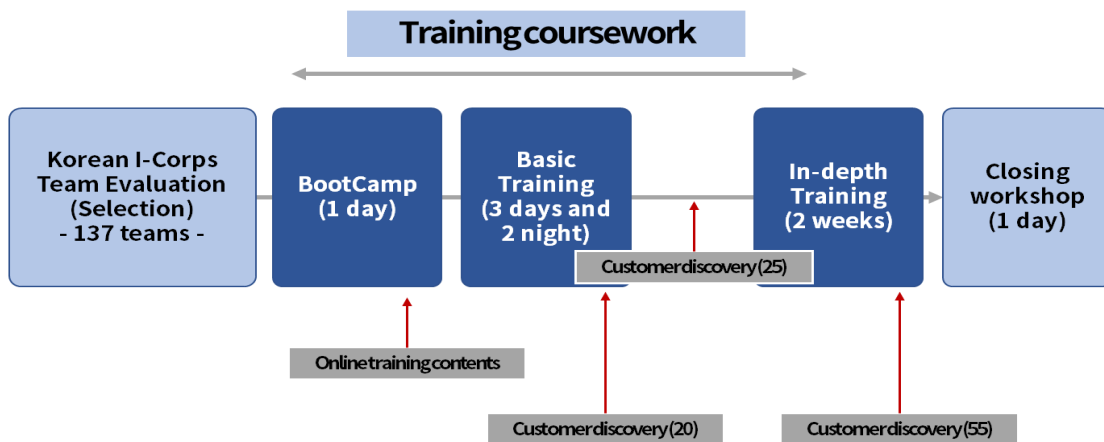
After the basic curriculum has been completed, the task is to leave a gap of about one to two weeks and then conduct an additional 25 customer discoveries.

The advanced curriculum requires about 55 customer discoveries to be made over about 2 weeks, and office hours are conducted once per week.

In the final week, the final presentation and evaluation is conducted, and the subsequent prototype development costs are differentially covered according to the results of the evaluation (financial support of 20 million to 40 million won is provided to each team depending on the results of the evaluation).

[Figure 3-35] I-Corps Teams' Customer Discovery in Korea

- **Total number of customer discovery: 100 times (1 on 1 in-depth interview)**



Source: Shim (2023)⁷⁸

In the case of the Azerbaijan I-Corp education training program, the prospective founder program recommends that the programs be operated to conduct about 100 in-depth interviews (customer discovery) for the market verification of business hypotheses.

- Basic training course: Conduct 20 times of customer discovery
- Period of gap between basic and practical training: 25 customer discoveries
- Practice training session: 55 times of customer discoveries
- Totally, the teams have to complete a total of 100 customer discoveries

⁷⁸ Shin(2023). "Introduction of the Korean I-Corps Training Program: Structure and Philosophy," GS SHIM, p. 25.

[Figure 3-36] Team Mission for Customer Discovery

• **Why teams have to do 100 times of customer discovery interviews**



Source: Shim (2023)⁷⁹

However, in the case of the TTT curriculum, the program consists in conducting about forty interviews as a team, rather than one hundred interviews, in consideration of the characteristics of the teacher education.

- Online training: 10 customer discoveries per week (3 to 4 weeks). Lecture-type training will be conducted at the beginning of the curriculum, and customer discovery tasks will be assigned after the second week of training.
- Final offline assessment (local in Azerbaijan): Final presentation assessment, including the results of customer discovery up to the last week. Regarding the presentation evaluation, a separate evaluation index used in the I-Corps process in Korea will be provided as a reference.
- 30 to 50 customer discoveries: The TTT curriculum also recommends that you pursue more than 30 customer discoveries once insights begin to be inferred from the market.

In the case of TTT training, the weekly customer discovery is conducted in the same way as feedback is received on the business hypothesis verification and customer discovery performance of the Korean instructors through "Office Hours."

⁷⁹ Shin(2023). "Introduction of the Korean I-Corps Training Program: Structure and Philosophy", GS SHIM, p. 26.

[Figure 3-37] PPT Template for Office Hours

KoEF 한국청년기업가정신센터

고객 인터뷰 기록(Customer Experiment Card) - 1/3

인터뷰 대상자(Interviewee)	
누구인가? (Who)	조직/상황 (Organization/Location)
왜 인터뷰를 하는가? (Why this interview)	고객 유형 (Customer Type)

인터뷰 상세(Interview Detail)		
질문 - 테스트 실험: 가설 (Test Experiments: Hypothesis)	답변 - 고객 경험: 행동 (Customer Experiences: Behavior)	확안한 내용/사실(패턴) (Insight)

Source: KoEF (2023)⁸⁰

In addition, in the case of TTT education, office hours are conducted according to the same PPT templates used in the actual I-Corps curriculum. The PPT templates used in Office Hours are provided with slightly different contents each week.

TTT Education and Training is due to be implemented in 2024, and will be conducted online for a total of four to five weeks. However, in the last week, the final presentation evaluation will be conducted in Azerbaijan and a certificate of completion of the curriculum will be issued at the relevant ceremony.

⁸⁰ KoEF (2023), “Korean I-Corps Team Presentation Template for Office Hourse”, KoEF, p. 7.

3.1.5 Planning of TTT programs (tentative) for Azerbaijan in 2024

To implement TTT education and training in Azerbaijan, the following programs will be operated under this project in 2024.

<1st Week>

Type of programs: Online training

Time to go: Sep. 2024

Lecture time: approx. 3 hours

Date	Type of training	Description	Instructor
Week 1	Lecture 1	Introduction to the Korean I-Corps TTT Program	Lead instructor
	Lecture 2	Understanding the BM Canvas	Instructor A
	Lecture 3	Business Hypothesis and Customer Segments	Lead instructor
	Assignment	Introduction of the PPT Template (Assignment for Week 1: Who, What, Why)	Lead instructor

<2nd Week>

Type of programs: Online training

Time to go: Sep. 2024

Lecture time: approx. 3 to 4 hours

Office hours: YES

Date	Type of training	Descriptions	Instructor
Week 2	Lecture 1	Understanding of Customer Discovery	Instructor A
	Office hours	Team Presentation and Feedback from the Instructors (15 minutes per team)	All instructors (4 instructors)
	Assignment	Introduction of the PPT Template (Assignment for Week 2: Customer Discovery, 10 per team)	Lead instructor
	Debriefing	Instructor Debriefing	All instructors (4 instructors)

<3rd Week>

Type of programs: Online training

Time to go: Sep. 2024

Lecture time: approx. 3 to 4 hours

Office hours: YES

Date	Type of training	Descriptions	Instructor
Week 3	Lecture 1	Beyond the Canvas	Instructor B
	Office hours	(Team Presentation and Feedback from the Instructors (20 minute per team)	All instructors (4 instructors)
	Assignment	Introduction of the PPT Template (Assignment for Week 3: Customer Discovery, 10 per team)	Lead instructor
	Debriefing	Instructor Debriefing	All instructors (4 instructors)

<4th Week>

Type of programs: Online training

Time to go: Sept. 2024

Lecture time: approx. 3 to 4 hours

Office hours: YES

Date	Type of training	Descriptions	Instructor
Week 4	Lecture 1	Understanding of Revenue Stream (Case study)	Instructor C
	Office hours	Team Presentation and Feedback from the Instructors (20 minutes per team)	All instructors (4 instructors)
	Assignment	Introduction of the PPT Template (Assignment for Week 4: Customer Discovery, 20 per team)	Lead instructor
	Debriefing	Instructor Debriefing	All instructors (4 instructors)

<5th Week – Final Week>

Type of programs: Offline Ceremony

Time to go: Oct. 2024

Lecture time: approx. 2 hours

Office hours: NO

Date	Type of training	Descriptions	Instructor
Week 5	Final presentation	Team Presentation (10 minutes per team)	All instructors (4 instructors)
	Lecture 1	How to Do IR Pitching	Lead Instructor
	Ceremony	TTT Certificate	All instructors (4 instructors)

3.1.6 Why Korean I-Corps for Azerbaijan?

In Korea, a total of 685 teams have completed the course between 2015, when the Korean I-Corps project was launched, and 2023, when 125 teams were selected.

Regarding the startup rate reported in 2022, 186 teams, or 42.7% of the 435 teams that completed the training in the years 2015 to 2021, established startup companies, which is close to the 50% rate of the NSF I-Corps in the United States. Some 122 companies, or about 65.5% of the startups, are still active and have shown an excellent performance rate, with about 70% of them achieving a three-year survival rate and about 44.5% of them achieving a five-year survival rate. In addition to job creation, 1,003 new employees were created, and it is analyzed that an average of 5.4 people were employed for each company started (Gyeongsoo Shim, 2022).

In order to compare the performance of the I-Corps programs in South Korea and the United States, a study based on major statistical data disclosed by the two countries is necessary. In this regard, every two years the U.S. NSF presents the two-year performance of the U.S. I-Corps education and training programs, divided into three tracks, i.e. NSF I-Corps, NIH I-Corps, and DoE I-Corps, which can be used to compare and study the major performance achievements of the two countries' programs.

[Figure 3-38] NSF I-Corps – Biennial Report



Source: NSF(2023)

Looking at the types of performance creation from 2017 to 2020, which are covered by the report of the NSF (2021), a direct comparison of the respective performance achievements of the I-Corps in the U.S. and that in South Korea shows that a total of 270 teams completed the education and training over four years, of which 132, or about 48.9%, started their businesses. And, among the startup companies, 103, or about 78%, were still in business at the end of 2022.

Among them, a total of 24 teams attracted more than USD \$ 50,000 in investment as an average, i.e. 12 teams between 2017 and 2018 and 12 teams between 2019 and 2020.

In addition, one of the major changes between 2017 and 2020 was the dispatch of Korean instructors to George Washington University, one of the major node universities in the NSF I-Corps program in the U.S., to promote TTT (Train the Trainer) education to localize the Korean I-Corps program.

In the case of Korea I-Corps instructors, about eight instructors were nurtured every year up to 2022 under an agreement with George Washington University in the United States; and, as of 2022, a total of 48 professors had been trained to teach Korea's technology startup

teams using the NSF I-Corps methodology.

As previously discussed, the report presents different performance of the teams from the indicators used by major businesses related to startups in Korea, and the related contents are as follows:

- First, it presents indicators such as startup rate, startup company survival rate, sales, and job creation, all of which are commonly used in Korea, but in the case of the United States, unlike Korea, the I-Corps startup team's investment attraction performance is managed as a major achievement, and the follow-up investment attraction performance is investigated through a multi-year follow-up survey that reflects the characteristics of the early startup company.
- Second, for startups that were established after the education and training courses, a few of teams have engaged in government funding and/or private IR programs, and the most famous government led funding programs like TIPS look very helpful to get another private funds from AC/VCs. And those funds played the role like NSF's STTR and SBIR⁸¹ in US.

[Figure 3-39] I-Corps Performance Report (2017-2020, Korea)

270 I-Corps teams		132 companies		103 companies		24 companies
Teams trained		Startup foundation		Survival rate		Success of IR funding (over USD\$ 50,000)
Completion rate (%)		48.9%		78.0%		23.3% (8.8% out of total startups)

⁸¹ SBIR: The Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs are highly competitive programs that encourage domestic small businesses to engage in Federal Research/Research and Development (R/R&D) with the potential for commercialization. Through their competitive awards-based programs, SBIR and STTR enable small businesses to explore their technological potential and provide the incentive to profit from commercialization. By including qualified small businesses in the nation's R&D arena, high-tech innovation is stimulated, and the United States gains entrepreneurial spirit as it meets its specific research and development needs.

Source: GS SHIM (2023)⁸²

[Table 3-6] Comparison of NSF I-Corps vs. Korea I-Corps programs

	NSF I-Corps	NIH I-Corps	DoE I-Corps	Korean I-Corps
Teams ^(a)	1102	144	75	275
Startups ^(b)	593	143	8	132
Budgets ^(c)	\$133,444,206	\$10,080,000	\$10,525,000	\$22,481,960
VC Funds ^(d)	\$260,791,253	\$61,533,568	\$64,414,361	\$36,124,606
% of startups established ^{(e)*} (against teams)	53.8%	99.3%	10.7%	48.0%
No. of startups by budget ^{(f)**} (Per USD \$ 1 Millions)	4.5	14.2	0.8	5.9
VC Funds ^{(g)***} (against budgets)	2x	6.1x	6.1x	1.6x
Public Funds	\$147,406,659	\$60,033,568	\$13,378,324	\$2,119,740
SBIR/STTR	\$96,493,958	\$26,599,791	\$183,000	-
Other federal funding [non-SBIR/STTR]	\$40,264,700	\$33,258,777	N/A	-
State funding	\$10,648,001	\$175,000	\$0	-
Private Funds	\$113,384,594	\$1,500,000	\$51,036,037	\$34,004,867
Venture capital	\$99,055,020	\$130,000	\$0	-
Angel or individual investor	\$9,038,330	\$0	N/A	-
Private industry, corporate funding, or others	\$5,291,244	\$1,370,000	\$47,002,037	-

*^(e)=(^(b)/^(a))×100**^(f)=(^(b)/^(c))×1Million***^(g)=^(d)/^(c)Source: NRF&GS SHIM(2022)⁸³

In particular, NSF I-Corps startups are increasingly attracting follow-up funds and pushing for business advancement through NSF SBIR rather than through the private sector, reflecting the U.S. government's strategy for sustaining the innovation ecosystem through government financial support amid a global crisis such as COVID-19.

⁸² Shim, G.S. (2023), "A Study on the performance efficiency comparison of graduate student technology startups between Korea and the US", Journal of Industry Academic Cooperation Policy, Vol.3, No.2, pp.13.

⁸³ NRF & GS SHIM (2022), "Policy Research of the Korean I-Corps Program", NRF, Jan 2022, pp.48.

Small Business Innovation Research (SBIR) is a leading scientific startup model that was introduced during the Reagan administration in 1982 and has lasted for the past forty plus years. The SBIR funding system requires 2.5% of all R&D budgets of the central government to be used annually to support startups, and since 2017, this percentage has been recognized for its effectiveness and raised to 3.2%. As of 2018, eleven federal agencies were participating in the system.

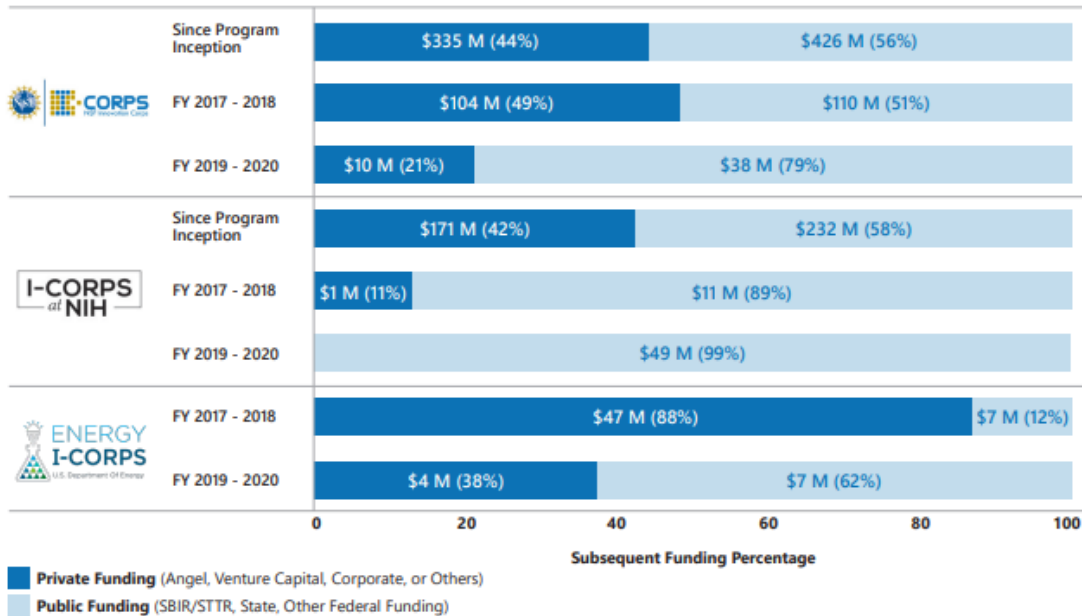
To effectively play the role of a seed fund for technology-based startups in managing programs directly in accordance with the guidelines presented to Congress, the following "3-No Operation Method" is used.

- ① Not a loan (no need to redeem R&D results if they fail).
- ② No strings attached.
- ③ No equity density (no requirement for equity in the company due to investment).

3.1.7 Report on the performance of the Korean I-Corps program

Regarding the necessity of a follow-up investigation, the need for a closer follow-up investigation and research on the performance of the Korean I-Corps program emerged with regard to the transfer of the Korean I-Corps program to Azerbaijan. Based on these findings, it is judged that successful planning and transfer of the program to Azerbaijan will be possible based on the growth process and trials and errors of the Korean I-Corps education and training program.

[Figure 3-40] NSF I-Corps – Distribution by Type of Funding (2017-2020)



Source: NSF(2021)⁸⁴

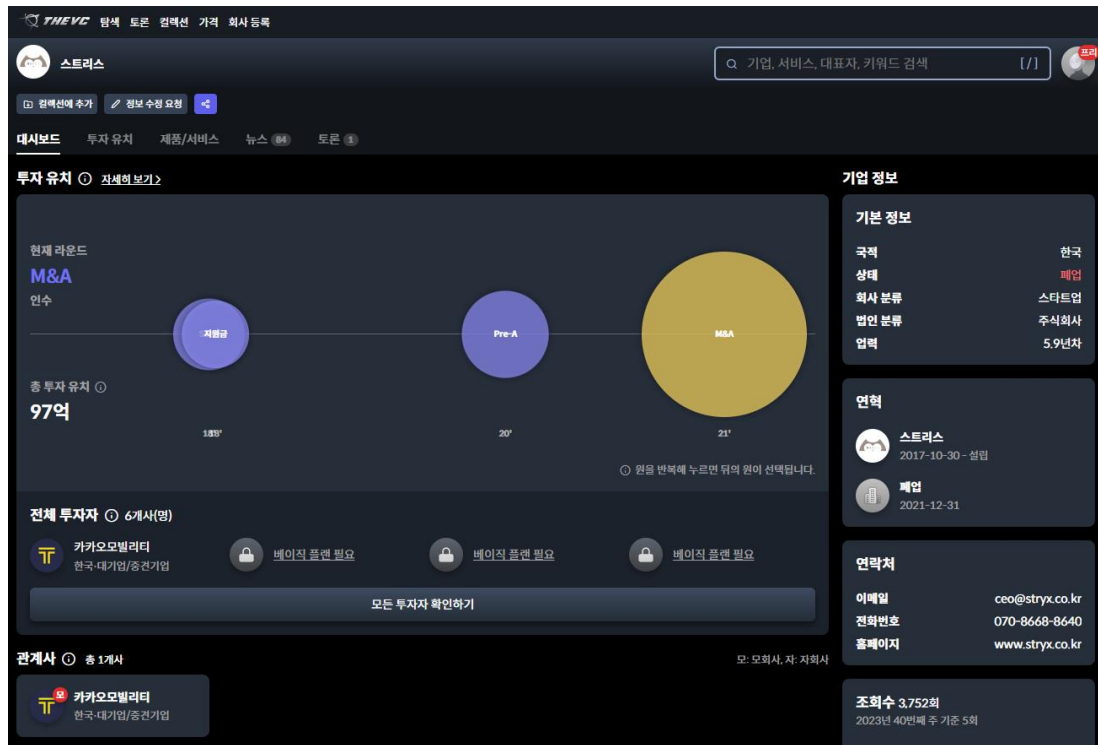
Therefore, a follow-up study on Korean I-Corps' performance in 2023 was conducted, and the period 2017 to 2020 was studied so as to enable a direct comparison with the U.S. I-Corps program. Research is also being conducted with the focus on attraction performance (Gyeongsoo Shim, 2023).

In the years 2017 to 2020, the budget for the Korean I-Corps project was KRW 3.7 billion in 2017, KRW 5.5 billion in 2018, KRW 6.9 billion in 2019, and KRW 7.4 billion KRW in 2020, while the number of L2M-type laboratory startup exploration teams benefitting from the education and training continued to increase (Lee Won-cheol, 2021). Thus, the total budget for the four-year period was KRW 23.5 billion.

Based on The VC site, the total amount of investment for the teams which have founded through the Korean I-Corps project from 2017 to 2020 are estimated in 8.5 billion KRW of government funding program and about 216.3 billion KRW of private investment.

⁸⁴ NSF (2021), "NSF Innovation Corps – Biennial Report, 2021., pp.19.

[Figure 3-41] The VC site – History of Startup Funding

Source: The VC⁸⁵

As shown in the table below, private investment attraction accounted for about 97.8% (KRW 157.7 billion) of total investment between 2017 and 2018, whereas government subsidies centred on TIPS accounted for about 2.2%. Between 2019 and 2020, the amount of private investment attraction fell slightly to 58.54 billion won, or about 92.1% of the total amount of investment attraction.

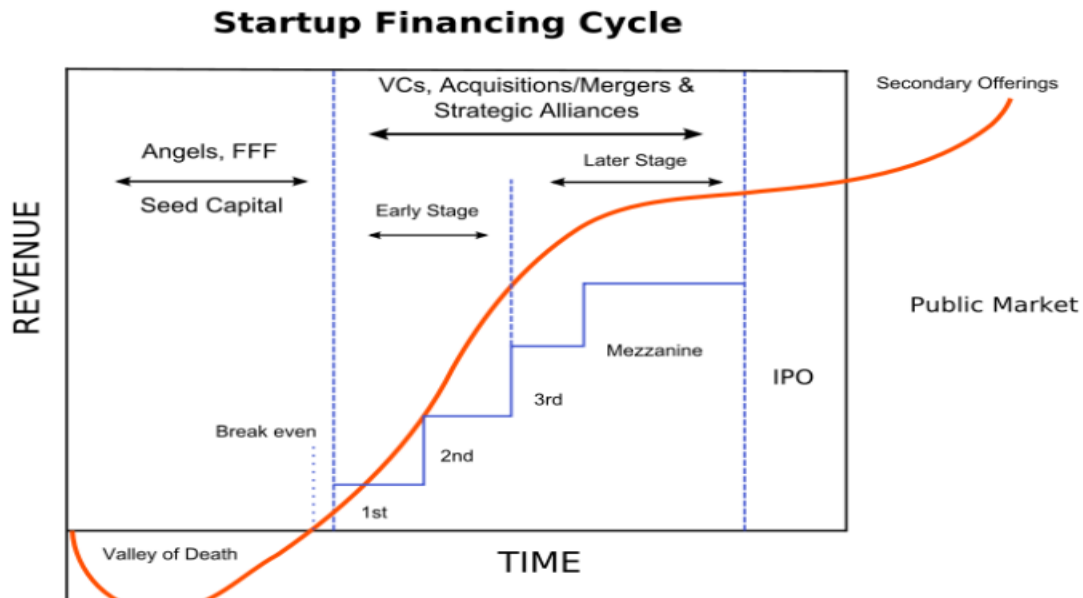
[Table 3-7] Results of Fund Raising by Korean I-Corps Teams (2017- 2020)

Type of fund	Private fund	Public fund
FY 2017-2018	1,577.7 Billion Won (97.8%)	35 Billion Won (2.2%)
FY 2019-2020	585.4 Billion Won (92.1%)	50 Billion Won (7.9%)

Source: GS Shim(2023)⁸⁶⁸⁵ <http://thevc.kr> (Date accessed: 15 November 2023)⁸⁶ GS SHIM (2023), "Comparative study of the performance efficiency of graduate student technology startups between Korea and

If we divide the Korean I-Corps teams based on the stage of the funding cycles of startups that were launched successfully through the VC⁸⁷ website and examine the distribution of the startups, we can see that there are various cases in about seven stages, i.e. Seed → Pre-A → Series A → Series B → Series C → Pre-IPO → M&A, as shown in the table below.

[Figure 3-42] The Startup Funding (financing) Cycle



Source: Deal Room (2023)⁸⁸

In particular, between 2017-2018 and 2019-2020, twelve companies each attracted at least 70 million won in investment or received subsidies from the Ministry of SMEs and Startups through government startup support projects such as TIPS⁸⁹, preliminary startup packages, and initial startup packages.

the US", Journal of Industry Academic Cooperation Policy Vol.3, No.2, pp.14.

⁸⁷ The VC: <http://thevc.kr> (accessed date: 15 November 2023)

⁸⁸ Deal Room (2023), "Startup Financing: How It Works and How To Get It", Startup Financing: How It Works & How to Get It (Guide with stats) (dealroom.net) (Last visit: Dec 25, 2023)

⁸⁹ TIPS (Tech Incubator Program for Start-up Korea): The TIPS program (private investment-led support for technology start-ups, Israeli method) is a program designed to intensively foster promising start-ups in the future by selecting start-up teams with technology items with the potential to lead the global market.

[Table 3-8] Fund Raising Stage of Korean I-Corps Startups

Funding stage	Sub-Total	Seed	Pre-A	Series A	Series B	Series C	Pre-IPO	M&A
FY 2017~2018	12	4	3	0	2	1	1	1
FY 2019~2020	12	2	4	2	0	0	0	0
Total	24	6	7	2	2	1	1	1

Source: GS SHIM (2023)⁹⁰

Among the twenty-four companies confirmed to have successfully attracted more than 70 million won in investment, eighteen companies, or about 66.7% of the total, have received more than 500 million won in government subsidies; and it has been observed that the government's step-by-step startup support project will contribute significantly to overcoming the so-called "Valley of Death," which is often represented by the gap between the market and technology that early technology-based startups often go through.

During the same period, NSF I-Corps U.S. startups attracted about \$214 million between 2017 and 2018, without much difference between private investment attraction and government subsidies. Between 2019 and 2020, the amount of investment attraction decreased significantly in the aftermath of the COVID-19 pandemic, attracting about \$48 million.

In conclusion, the total investment of surviving companies (2017-2020) as of August 2023 amounted to about KRW 224.8 billion for twenty-four companies. And during that period, Korean government has invested around KRW 23.5 billion for operating the Korean I-Corps project.

[Table 3-9] Results of Fund Raising by NSF I-Corps Teams (2017- 2020)

Type of fund	Private fund	Public fund
NSF I-Corps (Total fund raising amount)	USD \$ 335 M (44%)	USD \$ 426 M (56%)
FY 2017~2018	USD \$ 104 M (49%)	USD \$ 110 M (51%)
FY 2019-2020	USD \$ 10 M (21%)	USD \$ 38 M (79%)

Source: GS SHIM (2023)⁹¹

⁹⁰ Shim(2023), "Comparative study on the performance efficiency of graduate student technology startups between Korea and the US", Journal of Industry Academic Cooperation Policy Vol.3, No.2, pp.14.

⁹¹ Shim(2023), "Comparative study on the performance efficiency of graduate student technology startups between Korea and the US", Journal of Industry Academic Cooperation Policy Vol.3, No.2, pp.14.

At the same time, it can be said to be a very good performance, given that there is no significant difference in the input budget and the number of support teams even when directly compared to the total investment attraction of NSF I-Corps startup companies in the United States.

Of course, it is unclear whether the US NSF's report reflects all the results of cumulative investment attraction through the follow-up surveys used in this study, and there may be some limitations in the statistical surveys given the sharp decline in the investment attraction performance of startup companies in terms of both private and public subsidies between 2019 and 2020, when the COVID-19 crisis began in earnest.

Nevertheless, given the trend of the NSF I-Corps program (Gyeongsoo Shim, 2022), which usually has an annual government support budget more than six times larger than that of Korea, it can be analyzed that the rate of efficiency compared to investment through government budgets in Korea is still superior to similar startup support projects in the US.

3.1.8 Achievements and promotion of policy consultation

As in the case of Korea, promoting innovative technology-based startups will be useful in promoting policy changes and innovation in the industrial structure.

It is believed that it would be a good policy, at least in the short term, to foster professors of major Azerbaijan universities and instructors of major startup institutions by transferring the TTT (Train the Trainer) process and experience of the public technology-based market-linked startup search support project (Korean I-Corps project) currently underway in Korea.

The development of instructors who have learned the educational methodology of the public technology-based market-linked startup search support project can be fostered in a relatively short period of time.

In the first and second years (2022 and 2023), one Korean lead instructor will focus on planning the relevant I-Corps programs for Azerbaijan's potential TTT trainees, and on fostering and sourcing the Korean instructor pool for the "Train the Trainer" program in 2024.

In the third year, it is expected that Azerbaijan will be able to prepare its own I-Corps program education and operation system by directly running a training course for technology startup teams discovered in Azerbaijan in the form of co-teaching with Korean instructors.

If Azerbaijan itself fosters instructors that can teach the above education and training methodologies, it is expected that it will be possible to operate programs that can advance technology and idea startups on their own even after the end of the ODA program.

3.1.8.1 Transferral steps of the “Train the Trainer” startup program

To this end, in the first year (2022), the main education and training courses of the Korean I-Corps course were conducted online. After the pre-training sessions, the main contents of the lectures covered the background to the Korean I-Corps course, the establishment of business hypotheses, and the training of core education and training programs such as customer discovery.

Based on the online curriculum promoted in the first year, a field interview survey in Azerbaijan was conducted, the operation and management status of major universities related to intellectual property rights was identified, and the main purposes and background of the Korean I-Corps course were shared with officials from major universities and innovation institutions in the region.

In the second year, a meeting was held to discuss the contents of the first year's research report using online in relation to the “Train the Trainer” education and training programs. For the improvement of training field, the specific direction of the trainer task to be assessed in the third year and the method of improvement for the TTT programs were designed.

In the middle of the second year, offline education including the trials and errors and major achievements of Korea was conducted during the visiting and training terms in Korea, and detailed curriculum design was conducted to organize the Azerbaijan’s officials program like Korean I-Corps program.

In the second half of the second year, we plan to disclose the overall outline of these TTT programs in Azerbaijan, and finally promote the curriculum and method of operation of the

TTT programs to be promoted in the third year, along with the composition of the I-Corps instructors in Korea.

3.1.8.2 Expected Impacts of the Azerbaijan I-Corps programs

Regarding the I-Corps project in Korea, it can be evaluated as having contributed to providing an important opportunity for many universities to transform their undergraduate-centred idea startup culture into a graduate student-centred laboratory startup centred on the node university.

It can be seen that the technology of university laboratories developed through the administration of the government's R&D budget has laid the groundwork for shifting away from patents, papers, and prototype-level outputs, which can be called “research achievements”, to practical laboratory startups, which can be called “outcomes”.

Therefore, if this program is transferred to Azerbaijan, it is believed that it will provide an important opportunity for the establishment of a technology startup ecosystem in the non-oil industry.

As mentioned above, in the case of Korea, instructors trained in teaching and learning methods at major NSF I-Corps node universities in the United States are intensively fostered from the beginning of the project in order to spread the technology startup ecosystem through I-Corps.

It is possible to foster instructors who have learned the I-Corps-type education methodology in a relatively short period of time. In the first year, it will be possible to focus on fostering a pool of instructors through the “Train the Trainer” program, and in the second and third years, it will be possible to prepare Azerbaijan's own I-Corps program education and operation system by directly operating a curriculum in the form of co-teaching with Korean instructors.

As a result, if Azerbaijan fosters its own instructors who can teach the I-Corps methodology, it should be possible to operate programs that can advance technology and ideas startups on their own even after the end of the ODA program.

For the teams produced under the above curriculum, if the ODA project is advanced, it is expected that the two countries will be able to establish a cooperative system for startups in industrial areas (manufacturing, trade, culture, tourism, etc.), which are expected to be commercialized jointly by Korea and Azerbaijan.

3.1.9 Conclusion

3.1.9.1 Conclusion: It is necessary to increase the role played by universities in order to create a technological innovation startup ecosystem

In the case of Azerbaijan, there is next to no government R&D funding available to create a separate fund to support startups in universities or to promote research and development, which is the seed base of startups.

In the case of some universities, since mid-2010, this program has to be seeking to revitalize startups through education in the information and communication field, which does not require much initial capital for startups. However, since there is no funding for follow-up investment or support after completing entrepreneurship training, it is difficult to provide practical entrepreneurship care beyond the operation of educational programs.

Major universities in Azerbaijan usually operate startup education centred on the undergraduate centre and hope to create a startup ecosystem that takes into account the industrial characteristics of Azerbaijan; however, as they are having difficulty in creating a fund to support this, they hope to establish a startup incubation system in connection with overseas accelerators such as the United States and venture capital.

However, unlike major countries such as the United States, Korea, and Europe, which have established intellectual property and patent pools in advance to protect startups' business items and secure differentiation, Azerbaijan still has some of weakness of managing the output from science and technology. Although securing intellectual property through research and development in the field is recognized as a public property characteristic, there are few motivating factors to actively secure intellectual property.

In addition, there are universities in Azerbaijan that apply for patents, but it is difficult to know what patents have been applied for between universities, making it difficult to obtain accurate statistics on the number of patents applied for by universities.

3.1.9.2 Startup governance system for supporting science and technology-based entrepreneurship

As mentioned earlier, in order to revitalize technology startups in the non-oil industry, major government ministries in Azerbaijan, which play the same role as the Ministry of Science and ICT in South Korea, need to be designated as governance institutions.

In this regard, it is necessary to expand the role of the HTP under the current NAS or to reorganize the organizational structure of the HTP so as to create a technology startup ecosystem centred on universities and major research institutes in Azerbaijan as agents of governance for practical technology startups.

In addition, once the system of governance has been established, it will be necessary to establish an organic and continuous cooperation system with Korean institutions such as STEPI, which promotes the transfer of TTT education and training programs in Korea.

3.1.9.3 Using the Korean I-Corps project and experience as a benchmark

In Korea, some 685 teams have completed the course since 2015, when the Korean I-Corps project was launched, including the 125 teams selected in 2023. Regarding the startup rate reported in 2022, 186 teams, or 42.7% of the 435 teams that completed the training from 2015 to 2021, established startup companies, which is close to the 50% achieved by the NSF I-Corps in the United States.

Furthermore, 122 companies, or about 65.5% of the startups, are still active and have shown an excellent performance rate, with about 70% of them achieving a three-year survival rate and about 44.5% of them achieving a five-year survival rate. In addition to job creation, 1,003 new employees were created, and it is analyzed that an average of 5.4 people were hired per company that started.

In the case of Korea I-Corps instructors, about eight of them were nurtured every year

between 2017 and 2022 under an agreement with George Washington University in the United States, and, as of 2022, 48 professors were capable of guiding Korea's technology startup team.

Again, for a direct comparison with the U.S. I-Corps education and training programs, the performance from 2017 to 2020 is divided into two-year periods. Between 2017 and 2018, private investment attraction accounted for about 97.8%, or 157.7 billion won, of total investment, while government subsidies centred on TIPS only accounted for only about 2.2%. Between 2019 and 2020, the amount of private investment attraction fell slightly to 58.54 billion won, i.e. 92.1% of the total investment attraction.

As of August 2023, the total investment of surviving companies (2017-2020) as of August 2023 amounted to about KRW 224.8 billion for twenty-four companies. And during that period, Korean government has invested around KRW 23.5 billion for operating the Korean I-Corps project.

The total investment of the U.S. NSFI-Corps startup companies at the same time is about \$262 million (approx. 3, Even when directly compared to the size of 40.6 billion won, it can be said to be an excellent performance, given that that there was no significant difference between the input budget and the number of teams supported.

3.1.9.4 It is necessary to provide policy support to foster the non-oil industry and transfer Korea's TTT programs

Based on Korea's experience, it is believed that it will be possible to foster and promote startups in the non-oil industry in Azerbaijan. To that end, it will be necessary to shift the policy perspective towards creating several innovative startup ecosystems in Azerbaijan.

First, it is necessary to secure the technology and intellectual property rights required to foster the non-oil industry. In addition to the possession of technology and intellectual property rights require the establishment of governance and institutional improvements of the government's non-oil technology startup to utilize for those intangible assets. The government's financial investment and social awareness to increase the technology base that can be practically utilized in the non-oil industry.

In order to raise awareness of this, it is necessary to create a technology-linked innovative startup ecosystem that enables the practical use of secured intellectual property rights. The Korean I-Corps program - which can be evaluated as having successfully created jobs, attracted investment by introducing NSF I-Corps from around the world, increased the practical use of R&D, and expanded “Lab To Market” types of technology startups - has been successfully established and implemented over the last eight years. The various experiences and achievements experienced in this process can be considered a major precedent.

Therefore, it is concluded that the NSF I-Corps education and training program, the representative technology startup type program of South Korea and the United States, will be effective in creating an innovative startup ecosystem in the non-oil industry from the perspective of “outcomes”, but this will require the training of instructors (TTT) centred on major startup fostering institutions and governance systems.



CHAPTER 4

Strategic Plans for
Improving the Technology
Commercialization System
and Developing the HTP
Program

Chapter 4. Strategic Plans for Improving the Technology Commercialization System and Developing the HTP Program

1. Strategic Plan for Improving the Technology Commercialization System

1.1 R&D processes and commercialization in the world and in Azerbaijan

Around the world, countless research and development activities are carried out by universities, government institutions, and enterprises belonging to the private sector. For this reason, professions occupy an important place among the main tasks of universities, along with education and research.

Research and development projects can be carried out in many different areas, differing in terms of application and content depending on the fields of activity of the companies concerned. Sectors such as food, oil and gas, technology, infrastructure, finance and healthcare are areas of great importance for research. Apart from these areas, continuous research and development work is carried out in such fields as automobiles, textiles, chemistry, and metallurgy, as well as in service fields.

But why and for what purposes does research need the private sector, including industry?

Based on research, the answer to this question can be given briefly as follows:

- To produce new products.
- To increase product quality and standards.
- To develop new techniques that reduce costs and raise standards.

- To maintain and increase the difference between competitors by creating innovative ideas.
- To offer different solutions to existing problems.
- To develop new production technologies and models.
- To become a company in the local market and a brand in the global market.

World experience shows that companies invest significant capital in research and development activities. According to UNESCO's latest statistics, global spending on research and development recently reached a record high of almost \$1.7 trillion. Ten major countries (United States of America, China, Japan, India, Germany, South Korea, France, Great Britain, Taiwan, and Russia) account for 80% of all R&D expenditure. The European Union allocated 328 billion euros for research and development in 2021. Between 2011 and 2021, the majority of the European Union's research and development expenditure was carried out in the private sector.

As part of the UN's Sustainable Development Goals, many countries have committed to significantly increasing both their public and private research and development expenditure and their number of researchers by 2030. In addition, many countries are seeking to increase investment in both the private and public sectors by setting national targets for R&D expenditure as a share of GDP. Further, some countries determine risk-adjusted estimates of R&D expenditures, which inevitably involve capital risk because the return on investment is uncertain. Capital risk increases when more funds are allocated to research and development. There are also companies that outsource their research and development for a variety of reasons, including size and cost. Nevertheless, quite serious investments are being made in these processes.

After analyzing how research and development activities are carried out around the world, we came to the conclusion that this process cannot be carried out without systematic planning or without relying on a well-established program. In world practice, the first stage of research and development activities is considered to be observation. During the observation process, the perceived needs are checked and identified. Of course, each stage is important, but to make the determination correctly, it is extremely important to make the

right observation first of all. Therefore, fundamental research should be conducted in the observation phase. Then, the applicability of the new information obtained as a result of the observation is checked. At this stage, theoretical and experimental experiments are used. Then, the experimental development process begins. In this process, the development of the target product or service is completed, and a prototype is prepared and put into use.

How are research and development activities carried out in Azerbaijan?

The number of organizations engaged in research and development activities in our republic stood at 127 in 2020 and 134 in 2021. For the public sector, these indicators were set at 86 and 90, respectively, in 2020-21. In the private sector, four organizations carried out research and development activities in 2020, and seven in 2021. In terms of higher education institutions, these types of activities were carried out by thirty-seven organizations in both 2020 and 2021.⁹²

Research and development expenses in Azerbaijan totaled AZN 162,452,500 in 2020 and AZN 194,197,300 in 2021. In the public sector, these costs amounted to AZN 139,115,900 in 2020 and AZN 158,679,400 in 2021, while in the private sector they amounted to AZN 5,478,400 in 2020 and AZN 5,729,300 in 2021, and in universities they amounted to AZN 17,858,200 in 2020 and AZN 29,788,600 in 2021. In 2020, 64.5% of research and development expenditure in the public sector, and 58% in 2021, were spent on wages. (Source: State Statistics Committee www.stat.gov.az)

From the abovementioned statistical data, it can be seen that although there has been an increase in expenditures on research and development activities in Azerbaijan, the private sector is lagging behind in this activity. Only 3.3% of the total funds spent in 2020, and 2.9% in 2021, were allocated to the private sector. Although there has been an increase in the number of organizations engaged in research and development activities, such organizations in the private sector are in the minority (i.e. 3% and 5% respectively in 2020-21).

There may be various reasons why enterprises belonging to the private sector in Azerbaijan

⁹² www.stat.gov.az (accessed date: 15 November 2023)

carry out research and development activities within a narrow framework and with limited funds. These reasons could include the lack of a need for innovative production and services, problems in meeting with researchers, the fact that funds allocated to research are not included in budget planning or are viewed as additional costs, and the low level of education on research and development among other factors. Taking into account the above, the following measures are proposed to increase research and development activities in Azerbaijan, especially in the private sector:

- Research and development strategies should be based on competitiveness, and the competitive priority areas and research and development projects directed to these areas should be defined.
- These strategies should be created after taking into account global and future trends.
- The competitive innovation strategies of Azerbaijan at the regional level should be determined, and research and development activities in global markets should be followed.
- Basic patent research should be conducted and patent strategies defined.
- Research and development should be carried out by taking into account market needs.
- High technology areas should be targeted by research and development and activities.
- Macro and micro level action plans should be implemented.
- The strategic development departments of companies should be supported by the state.

Research and development is a creative effort carried out on a systematic basis in order to increase scientific and technical knowledge and use this knowledge in new applications, which is the initial stage before delivering a product or service to the users. These activities are also the beginning of the creation of new and advanced technologies that can provide a competitive advantage at the national and global level. There is no "taking without giving" in science. The more invested, the more results are possible. For this reason, the successful

organization of research and development projects that serve a very serious purpose can be a great impetus to the development of the innovative economy, which is one of the main strategic goals of the state.

The government in Azerbaijan is interested in the commercialization of technologies, and the measures implemented for the development of intellectual capital are becoming more and more relevant, which allows us to make optimistic predictions about the innovation ecosystem of Azerbaijan, including the commercialization of science.

All these requirements are very important for the transfer and commercialization system of indigenous technologies, because a modern system must be built to attract all stakeholders of the ecosystem to an open innovation ecosystem and to promote commercialization opportunities.

Scientific research is the source of important scientific and technological achievements. Effective links between academia and industry are needed to optimize the economic and social benefits of research and the return on public R&D investment. Knowledge flows between research institutes and industry are realized through, among other things, joint research projects, training, consultancy and contracts, the commercialization of scientific research products, personnel mobility between workplaces, and informal collaboration by researchers. These knowledge flows act as an important factor that accelerates the creation of innovation and, consequently, promotes the formation of an innovative economy. In this sense, the commercialization of technologies can be characterized as an innovation process that creates new economic value.

1.2 Strengths and weaknesses of technology commercialization

1.2.1 Strengths

The main requirements for activating the transfer and commercialization of technologies in Azerbaijan can be considered as follows:

- The legislative base has been improved.

- There has been an increase in government investment in research and development.
- The number of government technology transfer and commercialization projects is increasing.
- Awareness of intellectual property issues is accelerating and spreading.

1.2.2 Weaknesses

As a result of the research of the Korean-Azerbaijani expert group, the main weaknesses related to technology transfer and commercialization have been identified as follows:

- The legal system related to technology transfer and commercialization.
- The inactive system of commercialization of science and technology.
- The small market for these operations.
- The lack of interest in modernization and innovation.

1.3 Proposals for the efficient management of the technology transfer and science commercialization system

1.3.1 Carrying out educational reforms in TC

- It is necessary to encourage pupils and students at the secondary and higher education levels to pursue application-oriented work, to provide invention education, and to organise regular internships with industrial enterprises.
- It is necessary to enable scientific researchers and educators to teach and apply TT and TK skills.
- It is necessary to adjust the educational curricula and research topics according to meet industry requirements.

1.3.2 Mentoring services, resource assessment and business networking

It is necessary to organise competitions and provide mentoring support to teams with promising projects:

- Evaluate existing resources and format success stories by adopting progressive approaches.
- Promote science-education-business cooperation.
- Support global networks.

1.3.3 Innovation capital and its formation

- Organise activities that support the formation of enterprise capital and venture capital.
- Provide access to enterprise capital and venture capital.
- Organise incubation and acceleration services in Azerbaijan based on Korea's experience, including improvement of this field.

1.3.4 State policy and ecosystem support mechanisms

- Organise lobbying activities to achieve effective organisation of state policy in the direction of TT and TK.
- Establish long-term relationships with international markets at the state level.
- Share the experience of STEPI with Azerbaijan.

1.3.5 Strengthening activities related to intellectual property rights

- Provide patent education for the interested parties.
- Modernise the local patenting processes.
- Establish a preference for patentable research works.

1.4 News about the HTP

The activities of technology parks in Azerbaijan are regulated by the "Model Regulation on Technology Parks" approved by Decree No. 168 of the President of the Republic of Azerbaijan, dated May 15, 2014. For the formation of a sustainable and competitive economy, it is extremely important to expand the fields of innovation and high technologies based on modern scientific and technological achievements. Having a stable legal basis for

this process is considered one of the main factors. Provisions on the establishment of technoparks, and the organisation, management and development of their activities, income, the activities of their residents, land use, and other issues are reflected in the "Model Regulation on Technology Parks".

According to the Decree of President Ilham Aliyev dated December 30, 2022, some additions were made to the "Model Regulation on Technology Park". Thus, some requirements for system integration and software development have been approved. According to these requirements, in order to register as a resident of the technology park, legal and natural persons operating in the fields of system integration and (or) software development and development must be registered in the state register, must be included in tax records, and must belong to one of the small, medium or large business entities. These additions and changes are a clear example of how much importance is given to the field of information technologies, including digitization, in the operation of technoparks.

Also, based on some changes made in the Migration Code of the Republic of Azerbaijan, the Tax Code, and the Law "On Social Insurance", a number of concessions and exemptions have been determined for the residents of technology parks. Thus, according to the change made to the Migration Code, the residents of a technology park (in the case of a legal entity, those who hold the position of its head and deputies), and specialists with experience in the field of information and communication technologies recruited by that resident to work in the territory of the Republic of Azerbaijan, are not required to obtain a work permit.

According to the amendment to the Tax Code, the concessions and exemptions provided for technology park residents will also apply to individuals who have received a technology park registration certificate and operate outside the technology park as a legal entity, and to individuals who carry out entrepreneurial activities without creating a legal entity.

Therefore, according to these changes, persons who, as a resident of the technological park, perform system integration, software development and development activities outside the boundaries of the technological park, in accordance with the law, must pay property tax, industrial and technological they are exempt from paying land tax for the land they use in the park, and source tax for dividends.

Also, income tax is levied on the monthly income from wage employment of individuals working for taxpayers who perform system integration, software development and development activities outside the technology park as residents of a technology park, as follows:

- 0 percent on monthly income up to 8,000 manats for a period of three years from January 1, 2023, 5 percent on monthly income exceeding 8,000 manats;
- 5 percent on monthly income for seven years from January 1, 2026.

According to the amendments made to the Law of the Republic of Azerbaijan "On Social Insurance", the resident of the technology park (including those who carry out the activities of system integration, software preparation and development) are covered by compulsory state social insurance for a period of 10 years from January 1, 2023 .

Issuance of a confirmation document on the application of residents:

Three resident enterprises - import of construction equipment and other materials of "Baku Polimer Ishehsalat" LLC, "Proline Chemicals" LLC and "Azcosmetics Group" LLC.

All three applications were submitted to the Ministry of Science and Education of the Republic of Azerbaijan. Based on this information, the documents were submitted to the management for approval.

Since the progressive additions made to the "Model Regulation on Technology Parks" by the decree of the President of the Republic of Azerbaijan, Mr. Ilham Aliyev, dated December 30, 2022, the number of applicants for residency at the YTP has increased.

As such, Azchemicatech LLC (a producer of various medical reagents), Smart Bee LLC (a provider of various services in the field of IT and creator of innovative products), UAV Solutions LLC (a producer of UAVs and developer of various software), Trawell Group LLC (a provider of new solutions in the field of IT), 1 Soft Office LLC (a compiler of various operating programs), and Prior LLC (a facilitator of cooperation between contractors and producers in the field of agriculture) are all currently under review.

We hope that these changes to the legislation will further increase the connection between research and industry and create the basis for enterprises to carry out their research and development activities in a technopark, thereby contributing to the process of developing or improving innovative products and high technologies.

1.5 Planned works

Development and optimization of TTCP on content and technical issues.

■ User engagement

User engagement is one of the TTCP key targets and performance indicators because TTCP needs the user segment to function as a platform. Also, the classification of users through special filtering is included among the functionalities of TTCP.

■ Creating a database

Databases are formed by creating profiles in TTCP. This database innovation ecosystem can play the function of coordination, transferring experience and knowledge. This important issue from the point of view of innovation awareness is one of the main performance indicators of TTCP. Building a database can also make TTCP a rich treasure trove of innovation. By providing information about their activities with created profiles, users acquire the benefit of “self-presentation”, which gives them a wide range of privileges and opportunities to become a serious player in the ecosystem.

■ As a TTCP linking tool

As a platform connecting various ecosystem entities, TTCP also aims to have a coordinating performance. This platform is an ideal concept to change the ecosystem from a "chaotic" state to a coordinated one. With a special filtering function, TTCP categorizes users and brings together ecosystem entities that need each other. If we value the provision of coordination as the unification of goals, TTCP can assert itself as the starting point for the creation of innovations.

A virtual space where technology transfer and commercialization process is carried out

As a key performance indicator, TTCP aims to ensure the realization of the process of technology transfer and commercialization. TTCP aims to transfer and commercialize technologies with technical capability, marketability and intellectual property with real cases. Example: a research project is submitted through TTCP, its ownership is secured, the technology is evaluated, a prototype is created and made into a product and, finally, it is sold for economic gain.

■ Sustainability plan

Factors that may justify persistence:

- TTCP was established by taking into account the experience and financial support of the Republic of Korea, proven by its activities in the world, as well as local needs.
- The services offered by TTCP correspond to the needs and interests of the innovation ecosystem in Azerbaijan, including the players in the ecosystem. Therefore, setting up and running an easy-to-use platform could lead to a self-regulating process in which the service offered on the platform and its working mechanisms are provided in this form.
- TTCP is open and accessible to beneficial cooperation with all parties to the ecosystem.
- The professional activities of the team are compatible with the goals of TTCP. The team is interested in continuing this project in the form of an association. (It should be noted that the initial idea and construction works related to the establishment of the platform were put forward and started by the team in the absence of initial support. However, the originality of the claimed idea and the presence of the right team made it possible to find support in a short time.)
- Continuity plan:
 - Complete preparation of the working version by the end of 2023 and attract at least 400 users to the platform.
 - Ensure improvement works in TTCP by conducting an assessment according to the demands and needs of the users involved (1st quarter of 2024).
 - Organise online and offline events (discussions with specialized parties, seminars, forum, conference, competition, etc.).

- Evaluate the users' activities, present quarterly and annual reports to the market, and form a competitive environment between the parties (2024).
- Continue with improvement works according to the requirements of the market, including users, etc.

■ TTCP team

- The real and current professional activities of the team are compatible with the goals of TTCP. Information about the team members is given below.
- The team is interested in continuing this project in the form of an association. As is well known, TTCP was established within the framework of the "Program for the Exploitation of YTP in Azerbaijan: Implementation and Capacity Building" implemented by the Korea Science and Technology Policy Institute (STEPI), which has a leading position in the field of technology transfer and commercialization of science in the Republic of Korea. Furthermore, the members of the TTCP team are also local experts in the said project.
- The goals of TTCP are fully calculated for the development of the innovation ecosystem in Azerbaijan. IRIA is a state institution that ensures the policy and continuity of this process in Azerbaijan.
- The services offered by TTCP meet the requirements of the players in the innovation ecosystem. Also, TTCP was developed in the initial version and opened for service.
- With the support of IRIA, TTCP and its team can work more effectively in the ecosystem, especially in universities and research institutes.

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